

EEG post-hoc pairwise sweep

Dataset A: 5 conditions, 10 pairs. Dataset B: 4 ad conditions, 6 pairs, in Δ and raw space.

n = 18 · 4 s · median · ICA · participant is the inferential unit · 28 August 2026

352 tests. 0 significant under Holm, BH or BY.

8 cells reach uncorrected $p < .05$ among 320 distinct tests, against a naive chance expectation of 16.
Holm controls the family-wise error rate and is valid under any dependence. BH controls the false discovery rate and is laxer, but assumes independence or positive dependency, which these measures violate: the five relative powers sum to one.
Benjamini-Yekutieli is the FDR valid under arbitrary dependence, and on this data it is stricter than Holm.

EXPLORATORY. The confirmatory pipeline is untouched, still Holm as pre-specified, and was verified to reproduce to $1e-9$.
No pairwise cell is promoted, whatever its p or q.

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Findings

The verdict

- 352 pairwise tests. 0 significant under Holm. 0 under Benjamini-Hochberg.
- 8 cells reach uncorrected $p < .05$ among 320 distinct tests; naive chance predicts 16.
- Every nominal Dataset B cell sits on an explicit-early comparison, which is the slow-power tilt already documented as the six ICA-only exploratory hits. Localisation of a known event, not a new one.

Correction method makes no difference

- Holm controls the family-wise error rate and is valid under arbitrary dependence. BH controls the false discovery rate and is laxer, but assumes independence or positive dependency, which these measures violate: the five relative powers sum to one. Benjamini-Yekutieli restores the FDR guarantee under any dependence, and on this data it is stricter than Holm (x2.45 at $m=6$, x2.93 at $m=10$).
- BH lowers individual values (Fz theta Dataset B: Holm .166 to BH .094) but changes no verdict.
- Where one p dominates its family BH and Holm coincide exactly, so the strongest cell stays at .080; BY puts it at .196.
- Under every family definition from none to a single family of 320, all three methods give 0.

Feature reduction cannot help, and this was checked exhaustively

- The correction runs within measure, so Dataset A families are 10 tests and Dataset B families are 6, never 160.
- Dropping features removes whole families and leaves the survivors untouched: it can delete the best cell, never improve one. Dataset A stays at .368 for every subset that keeps its best measure.
- All 65,535 non-empty subsets of the 16 measures were searched under the harsher across-feature family, for all three corrections. Best adjusted p obtainable anywhere: .0799 Holm, .0799 BH, .1958 BY, all from global delta alone. None reach .05.

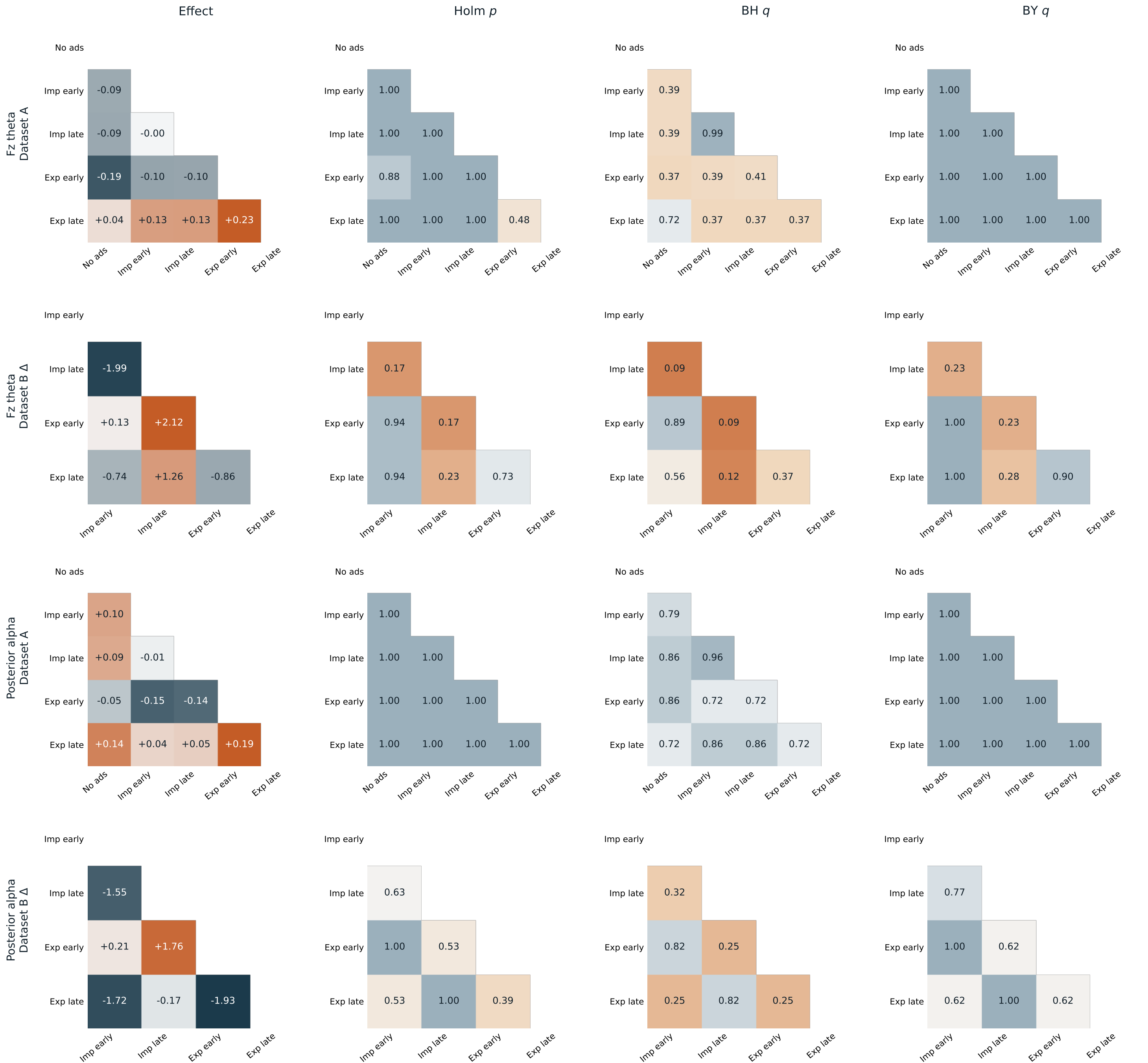
What is genuinely new: nothing dimensional

- Five conditions leave a 4-dimensional contrast space and the four planned contrasts already span it.
- Every pairwise difference is an exact linear combination of contrasts already tested, verified numerically.
- The one exception is the Delta-space timing contrast in Dataset B, tested nowhere at present, and Holm-null.

Exploratory throughout. The confirmatory pipeline is untouched, stays on Holm as pre-specified, and was verified to reproduce to $1e-9$.
Open item for Methods, not an analysis defect

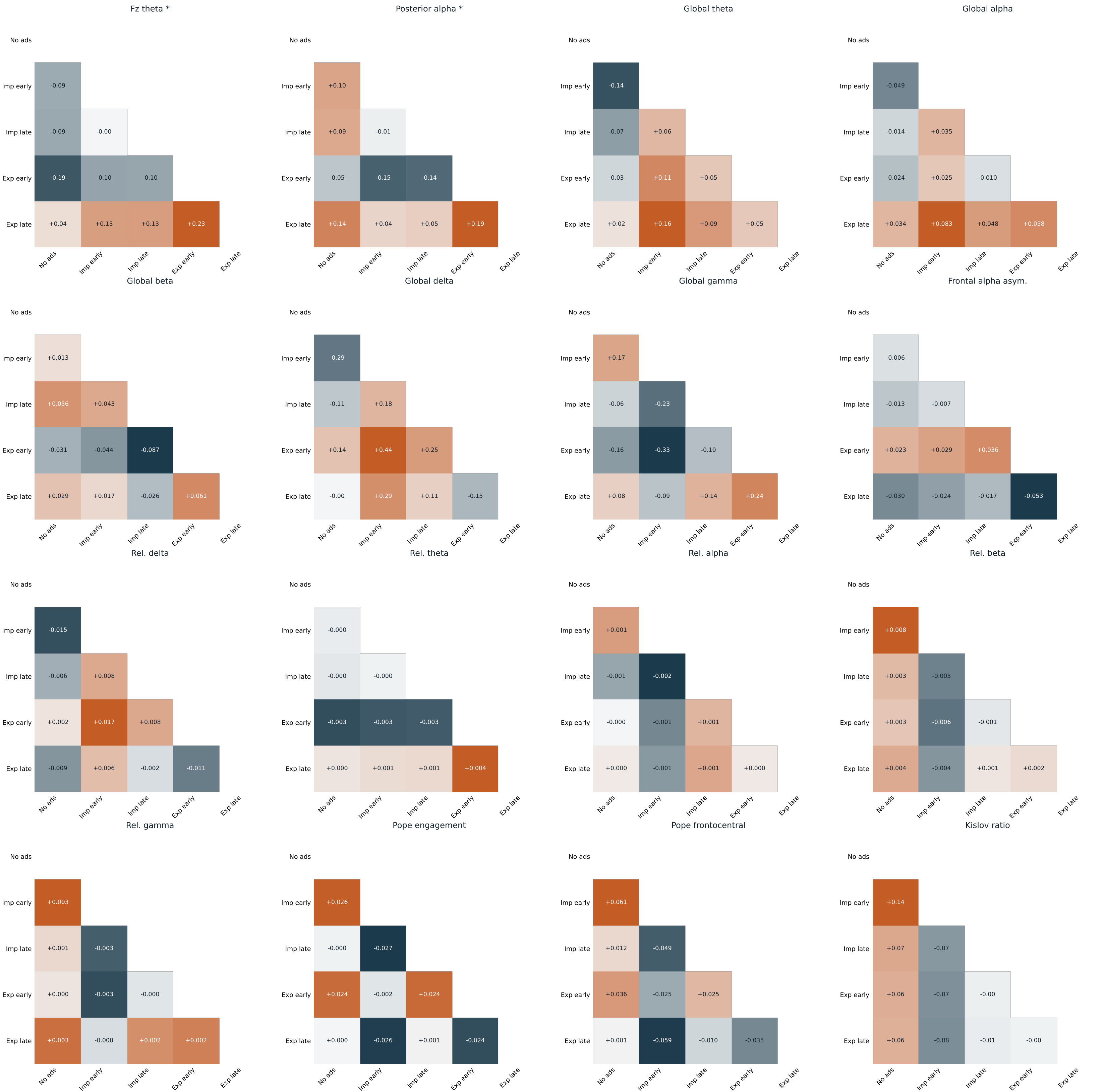
Post-hoc pairwise sweep: the two confirmatory measures

Cell = row minus column, n=18, 4 s median ICA. All three corrections run within measure within dataset. Exploratory: no cell is confirmatory, and none reaches .05 under Holm, BH or BY.



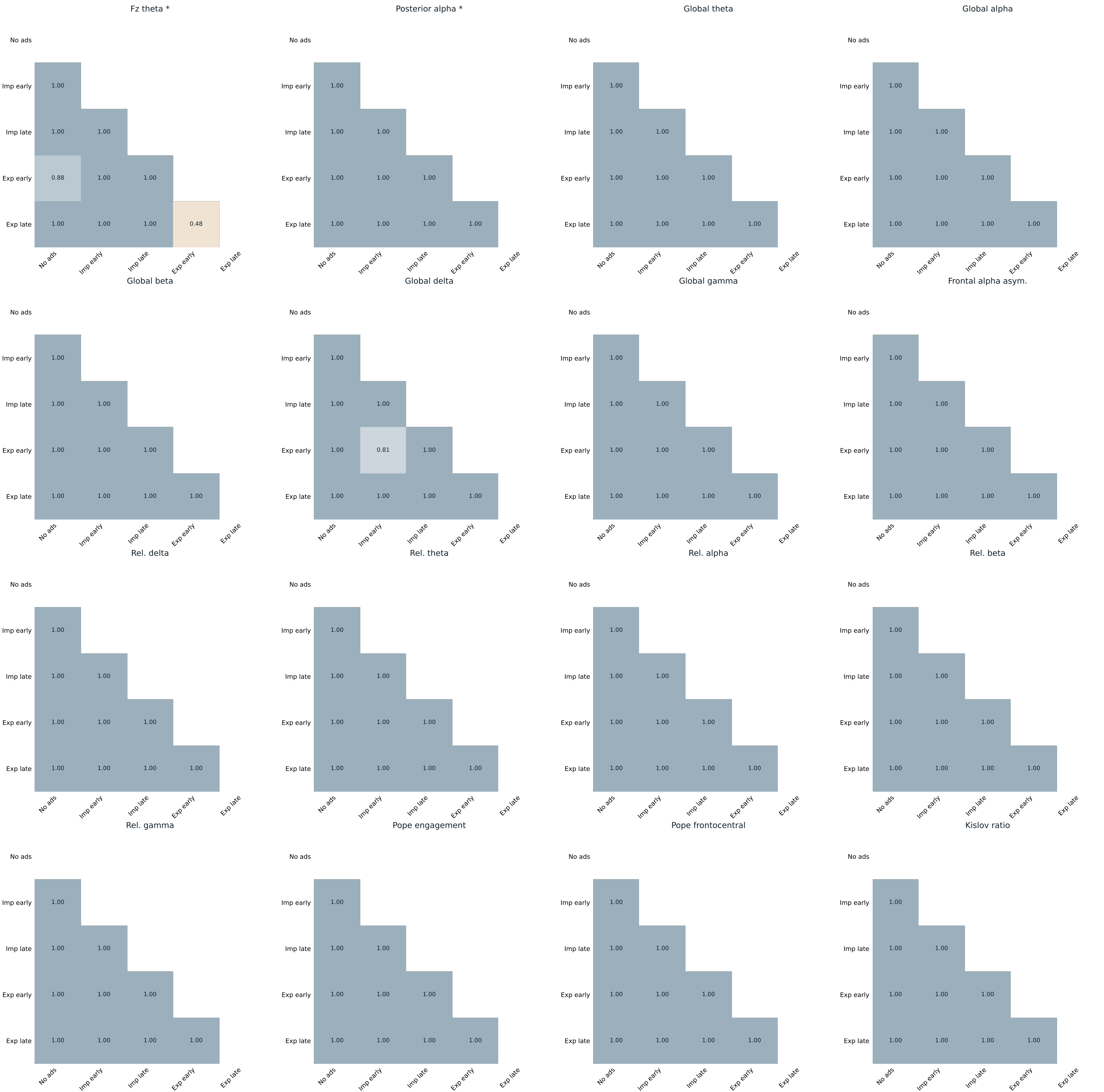
Dataset A pairwise effects (row minus column)

All 10 condition pairs, 16 measures, n=18, 4 s median ICA. Correction runs within measure across the 10 pairs. Each panel is scaled to its own range. Exploratory.



Dataset A pairwise Holm p

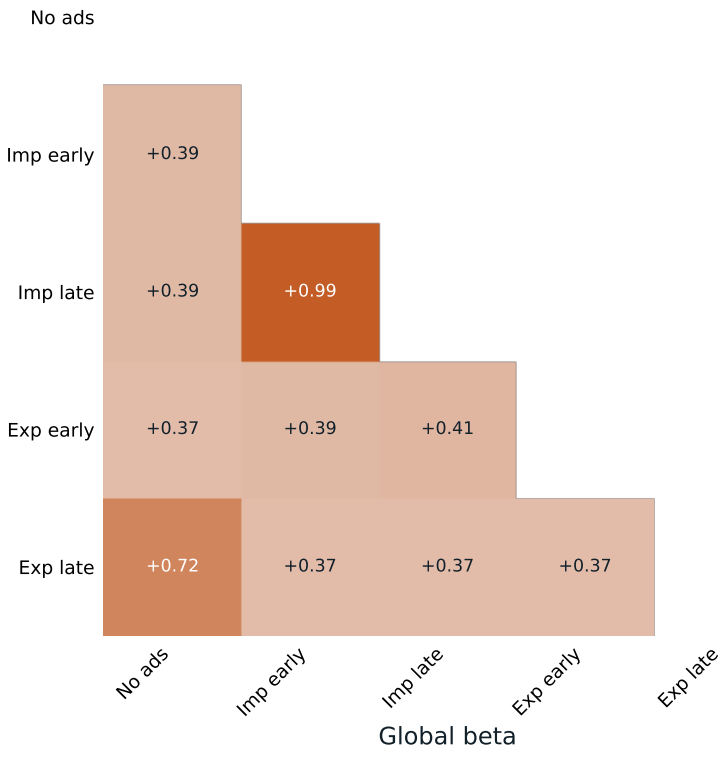
All 10 condition pairs, 16 measures, n=18, 4 s median ICA. Correction runs within measure across the 10 pairs. No cell reaches .05.



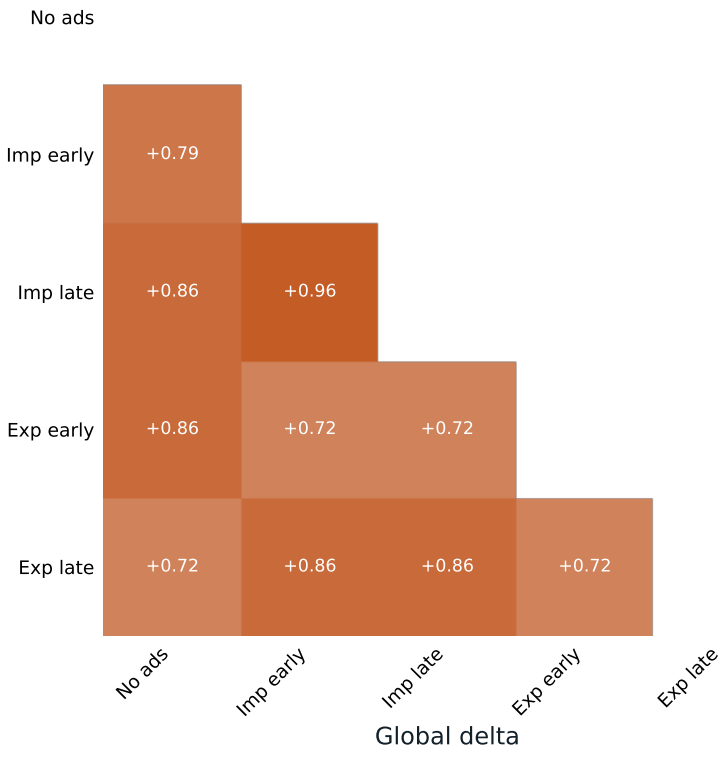
Dataset A pairwise Benjamini-Hochberg q

All 10 condition pairs, 16 measures, n=18, 4 s median ICA. Correction runs within measure across the 10 pairs. No cell reaches .05.

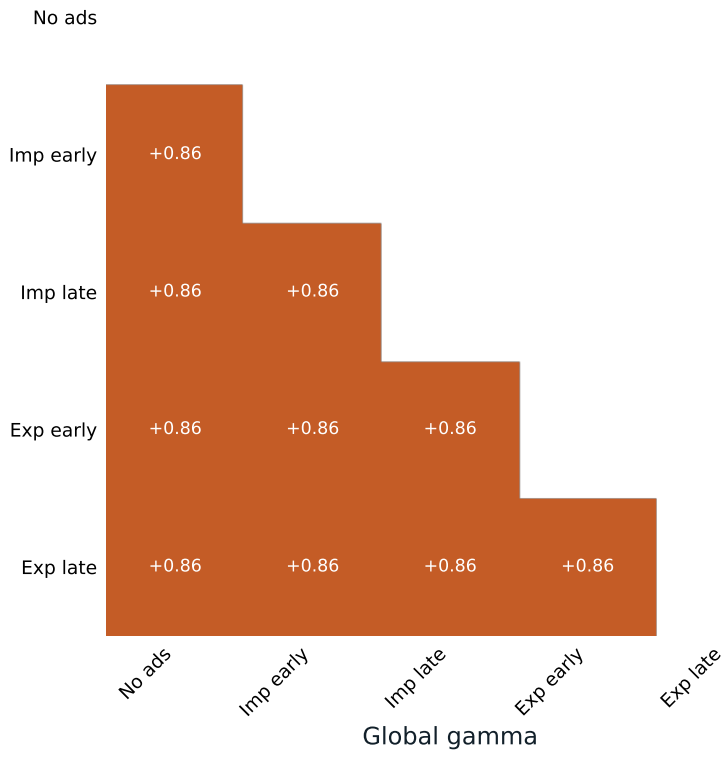
Fz theta *



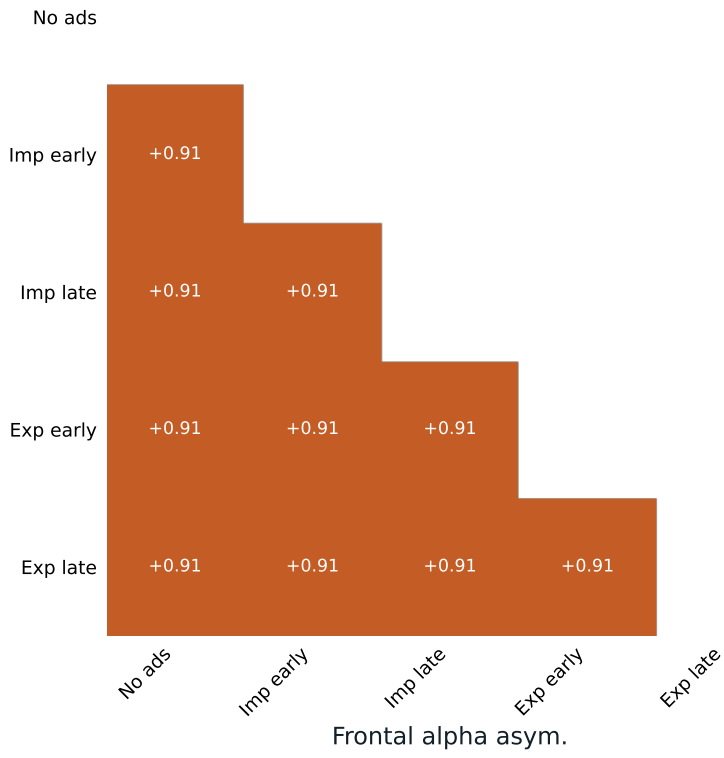
Posterior alpha *



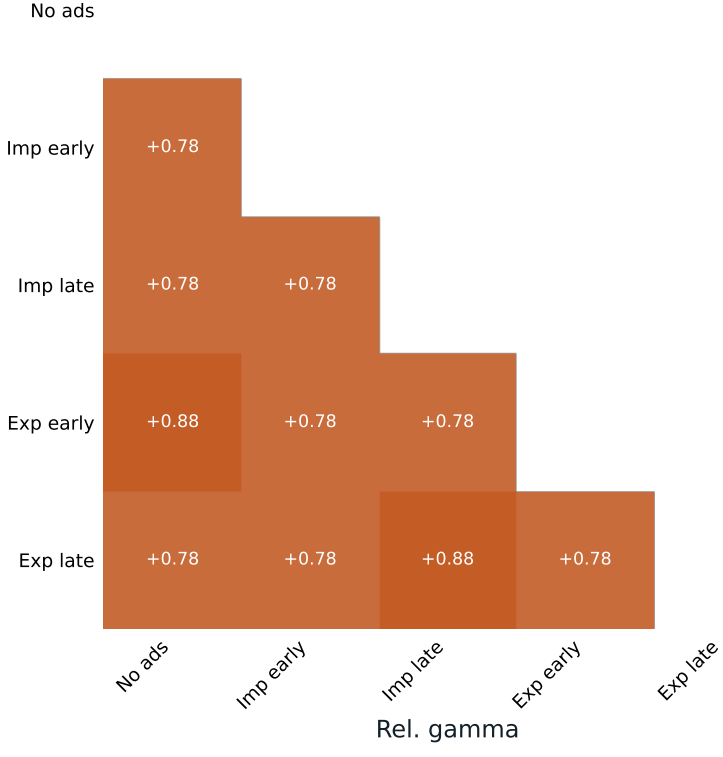
Global theta



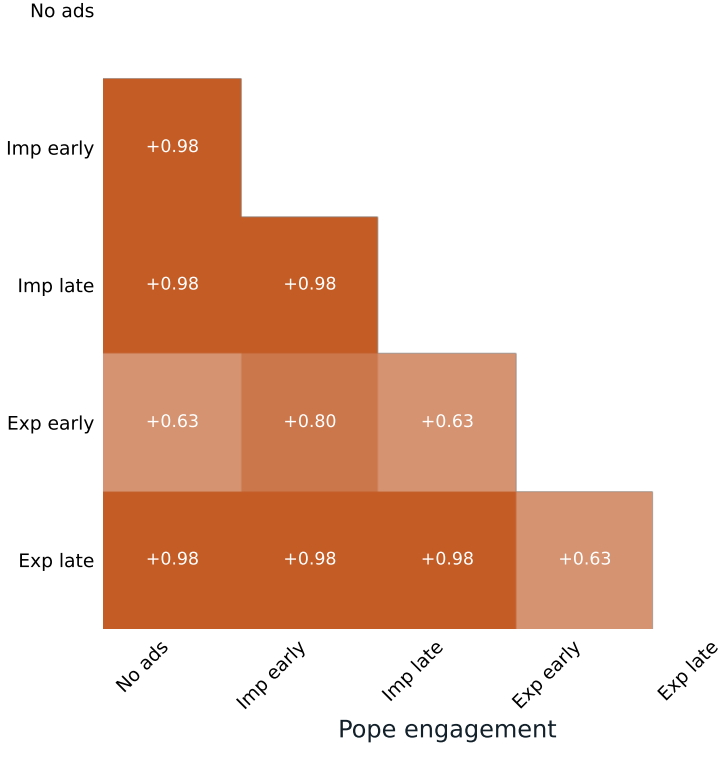
Global alpha



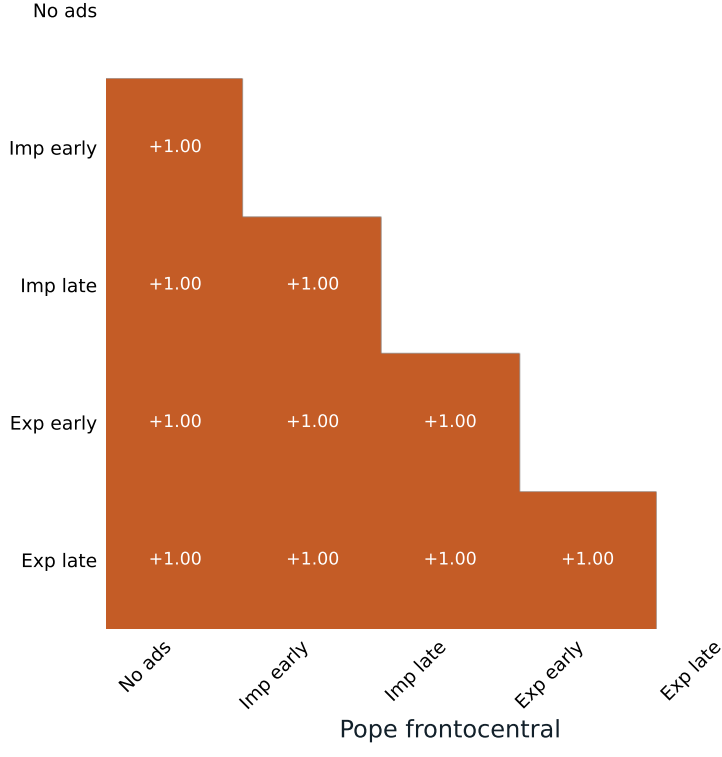
Rel. delta



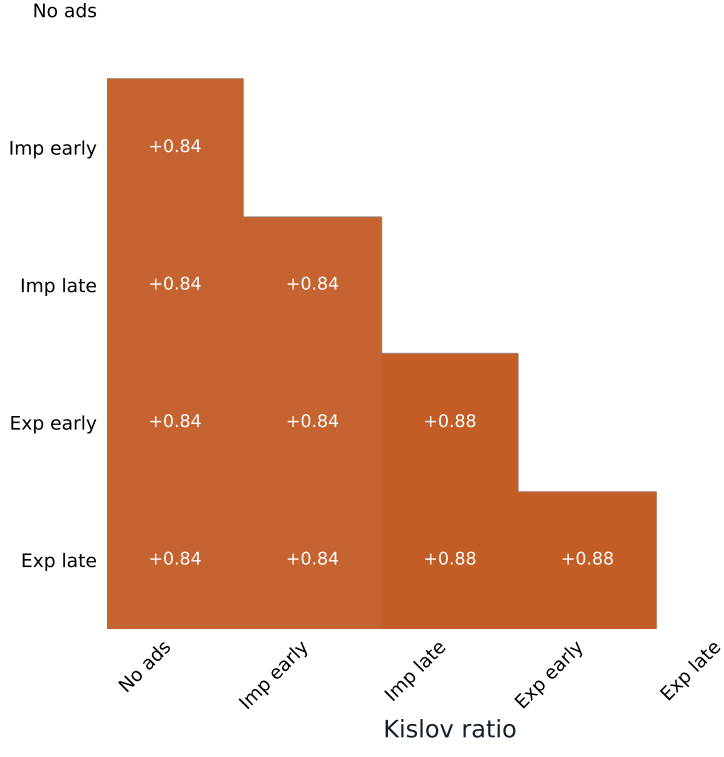
Rel. theta



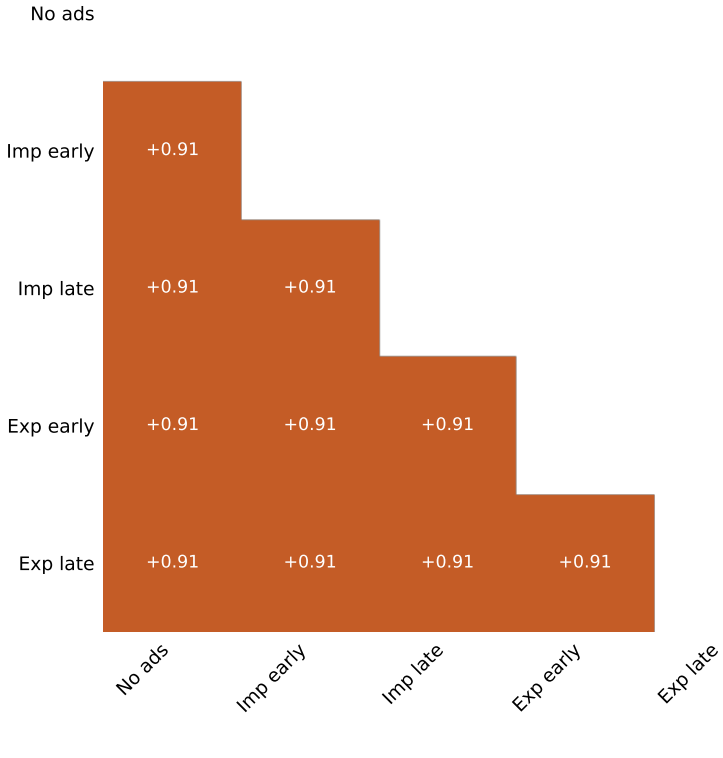
Rel. alpha



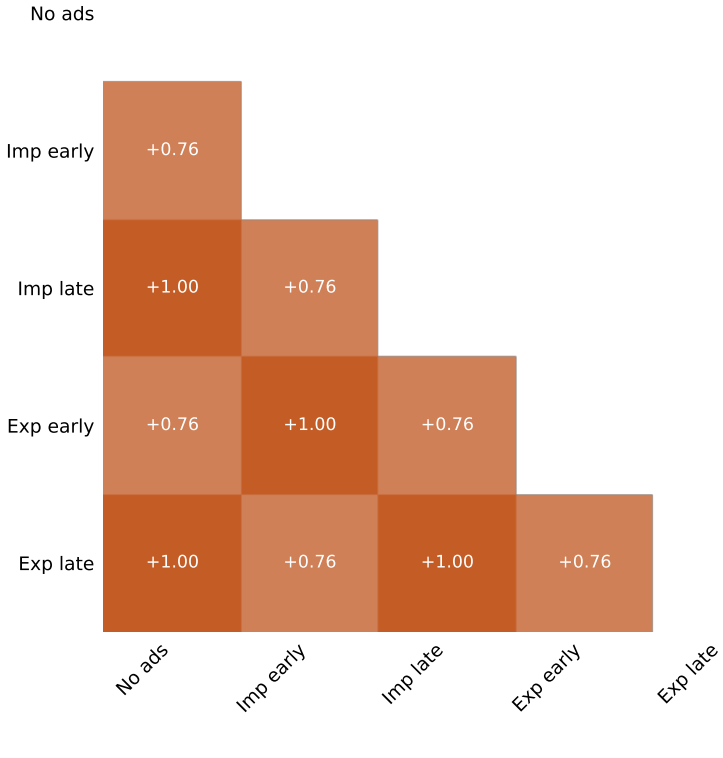
Rel. beta



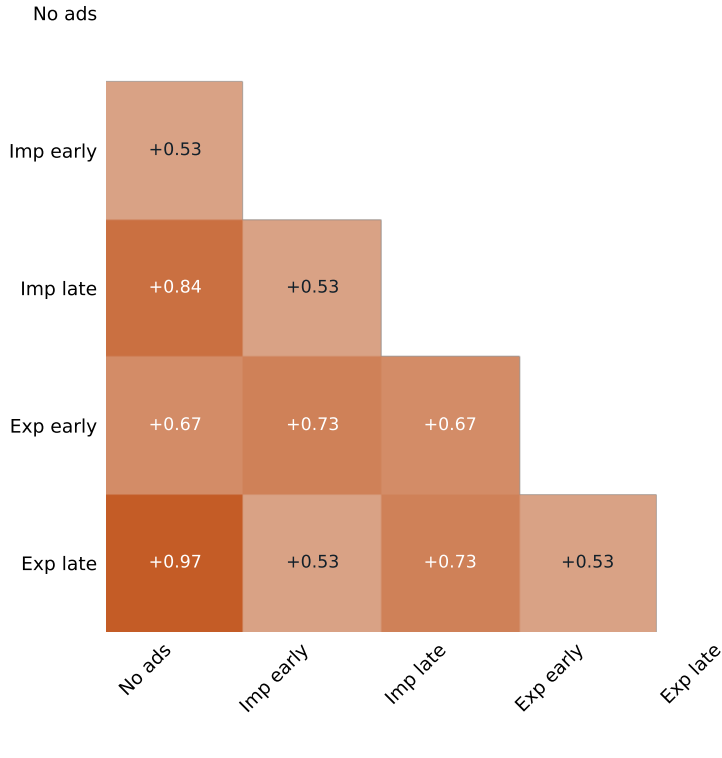
Rel. gamma



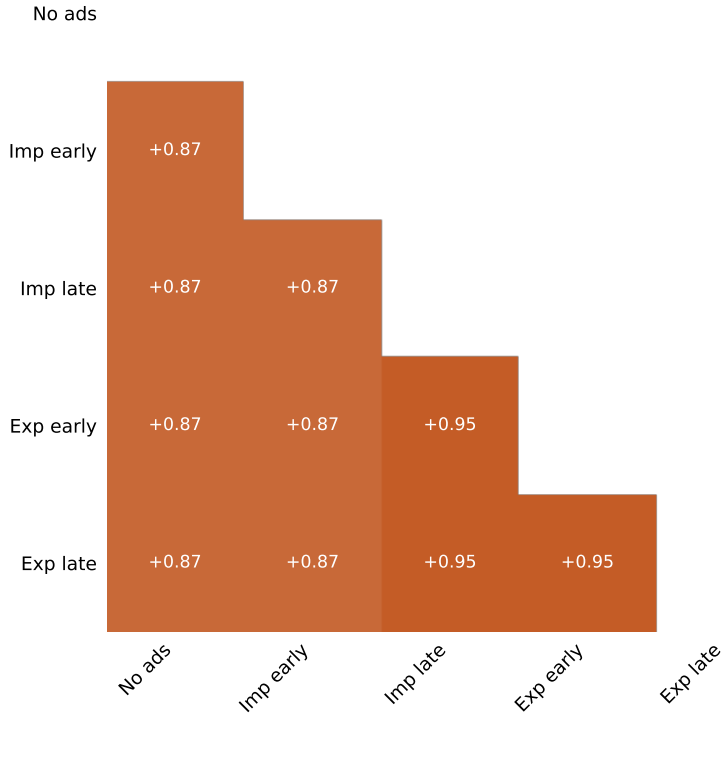
Pope engagement



Pope frontocentral

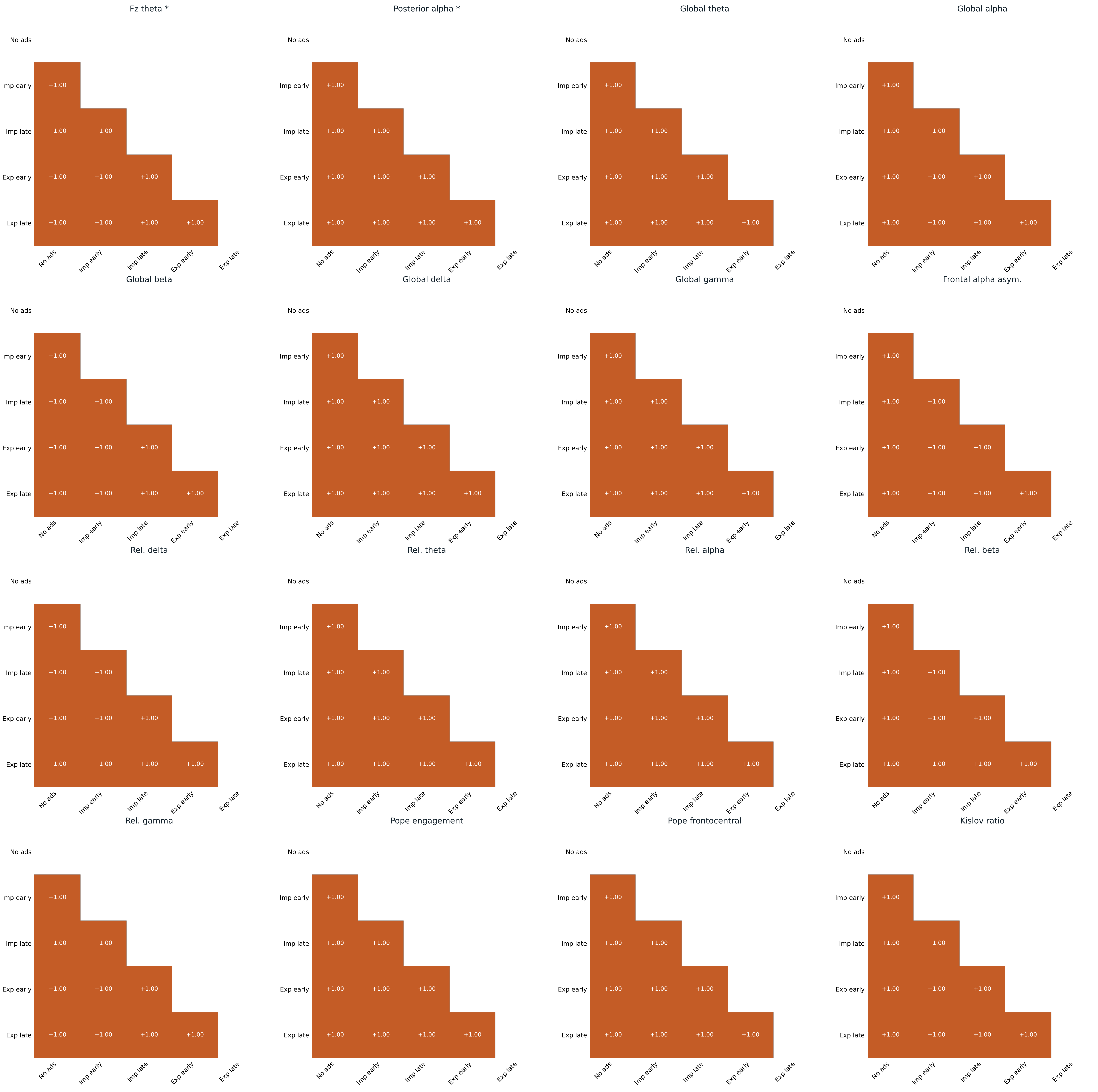


Kislov ratio



Dataset A pairwise Benjamini-Yekutieli q

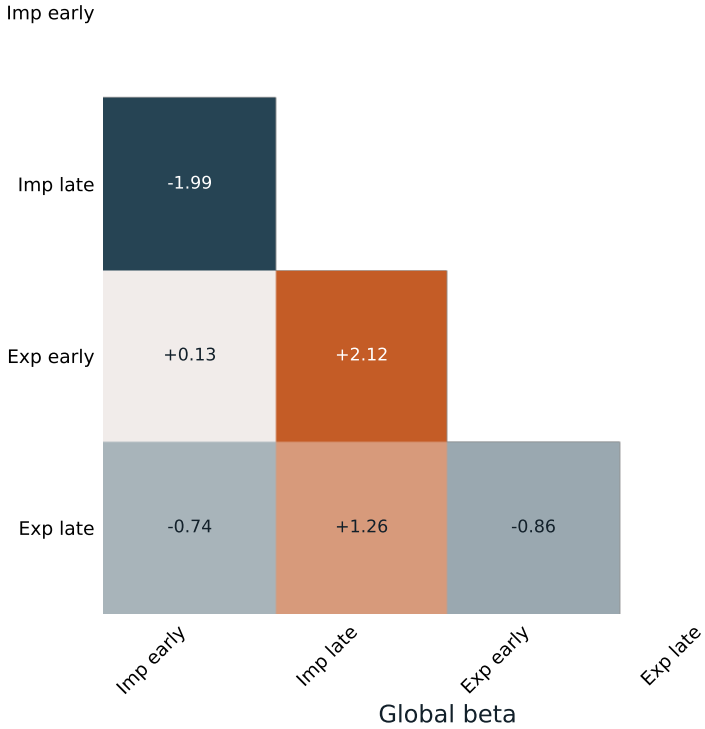
All 10 condition pairs, 16 measures, n=18, 4 s median ICA. Correction runs within measure across the 10 pairs. No cell reaches .05.



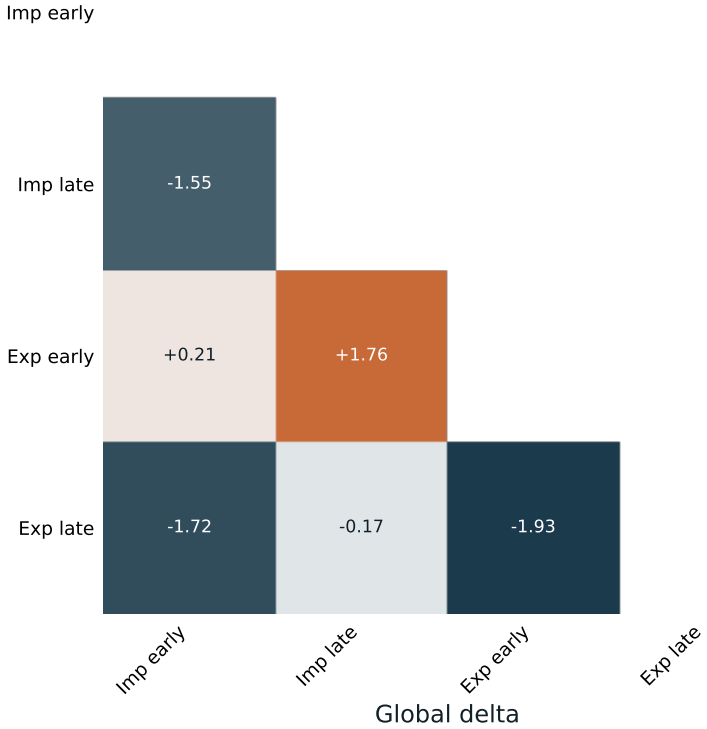
Dataset B, Delta matched control pairwise effects (row minus column)

Each cell is first differenced against its matched no-ad reply. 6 pairs, 16 measures, n=18. Each panel is scaled to its own range. Exploratory.

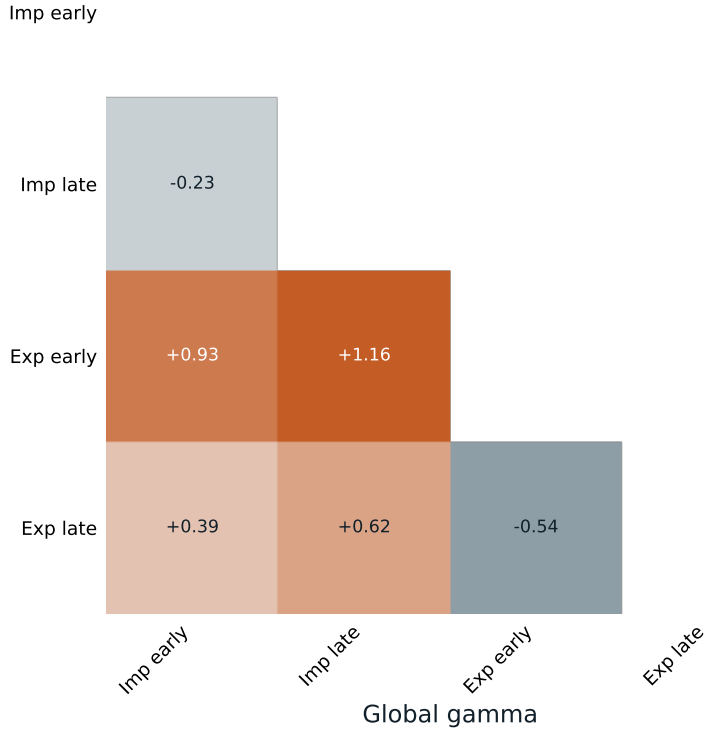
Fz theta *



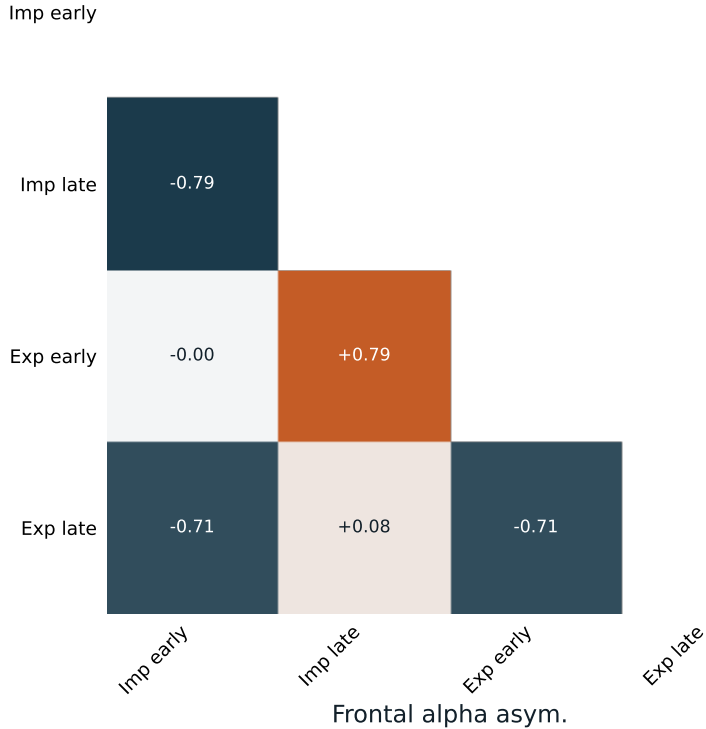
Posterior alpha *



Global theta



Global alpha

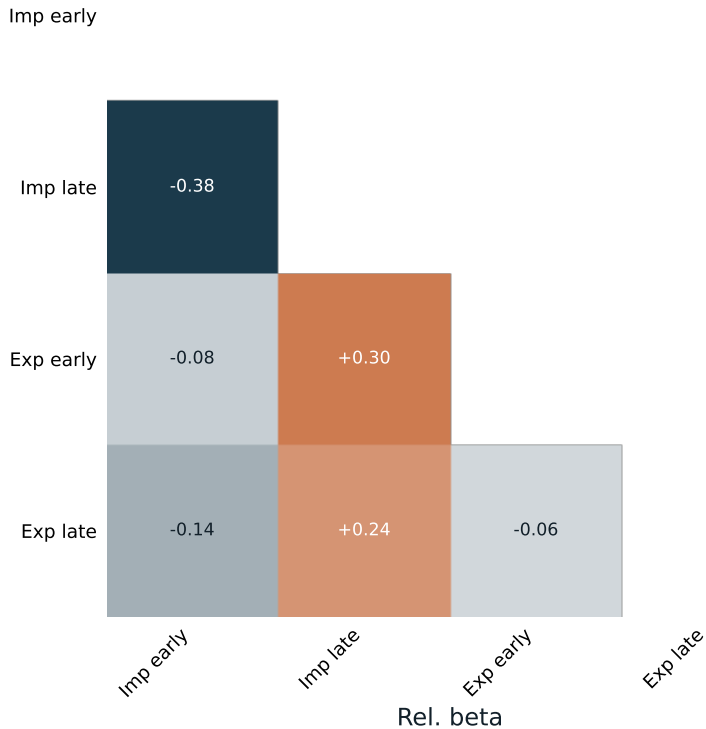
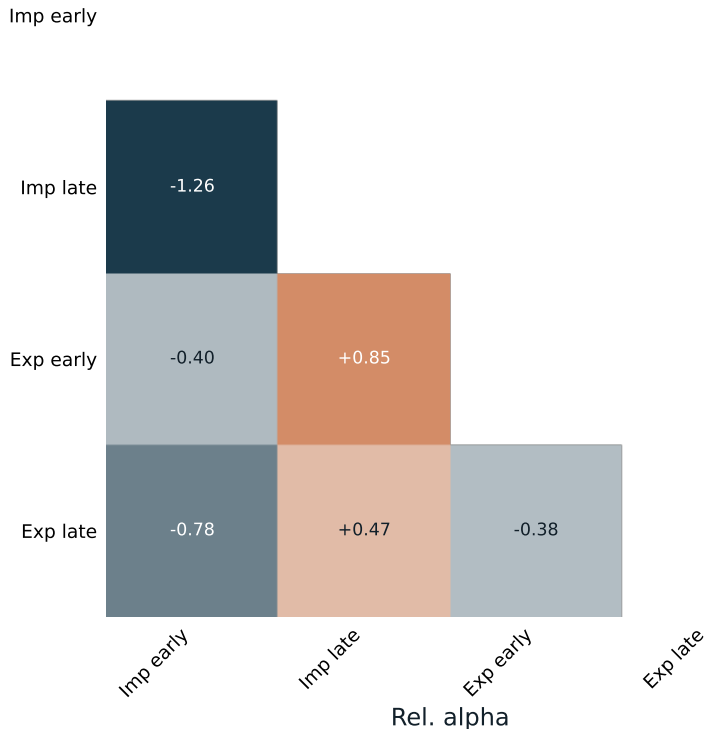
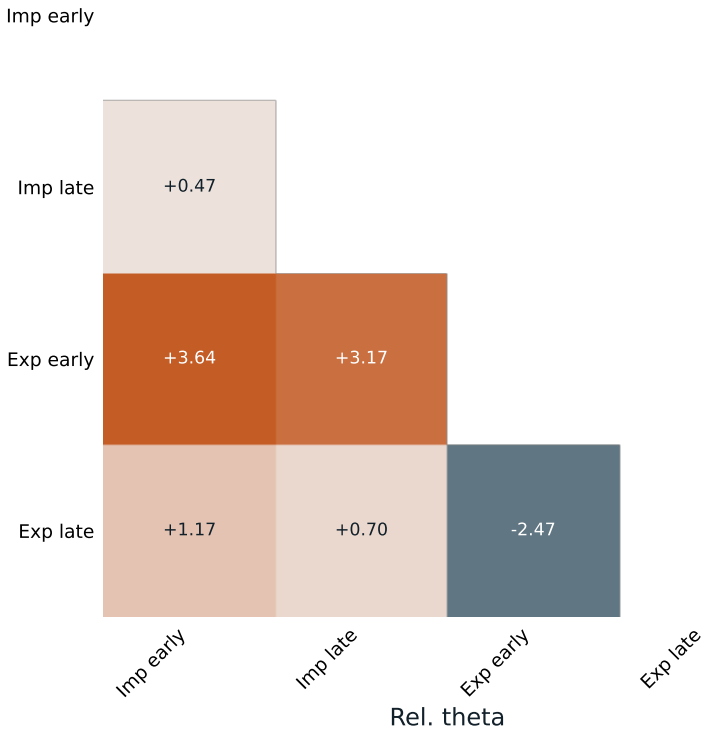
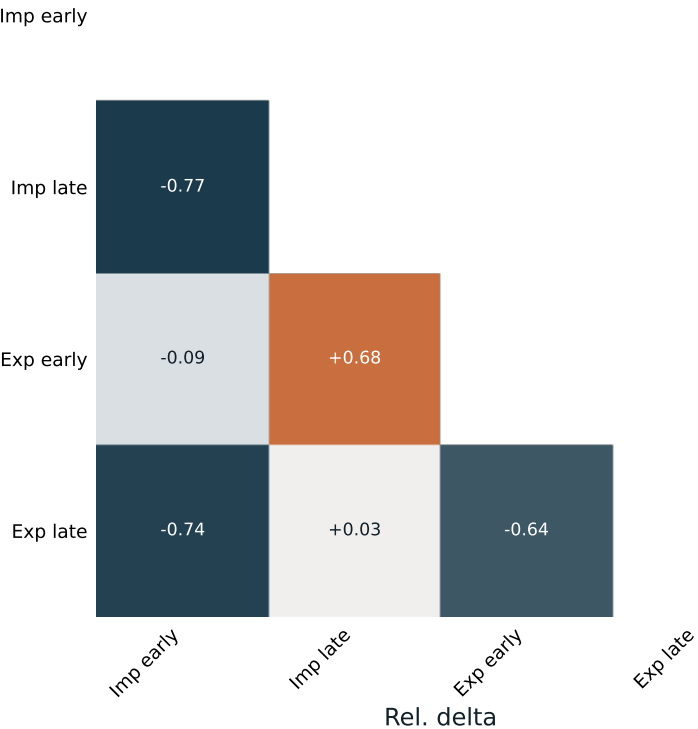


Global beta

Global delta

Global gamma

Frontal alpha asym.

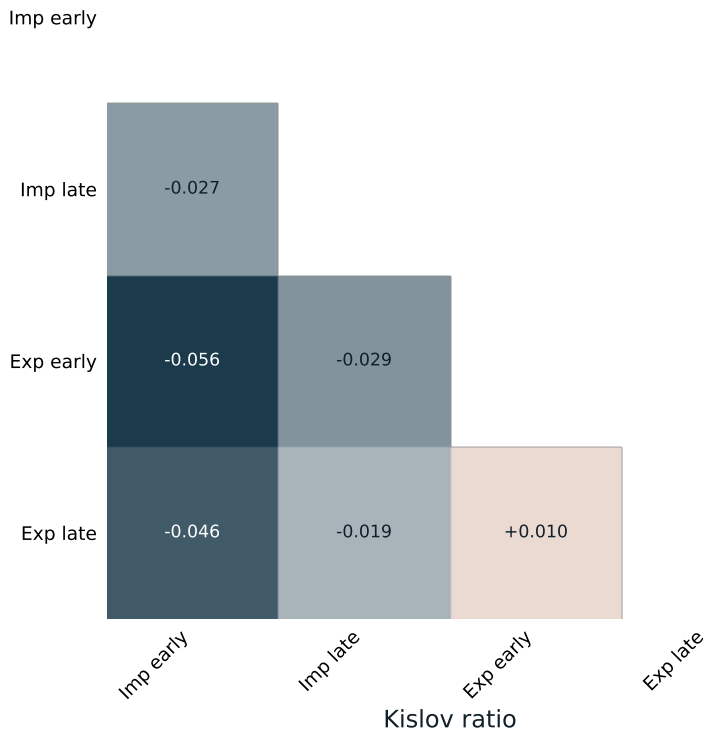
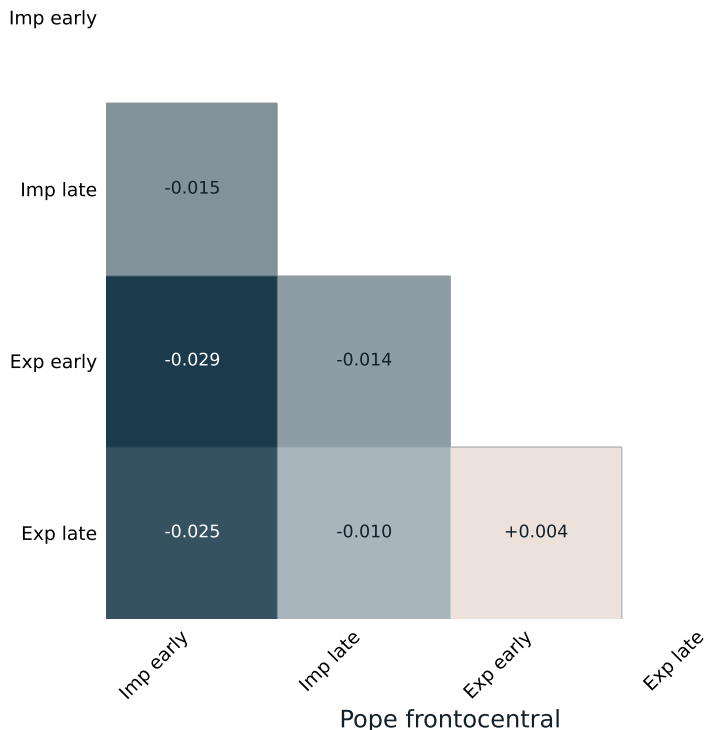
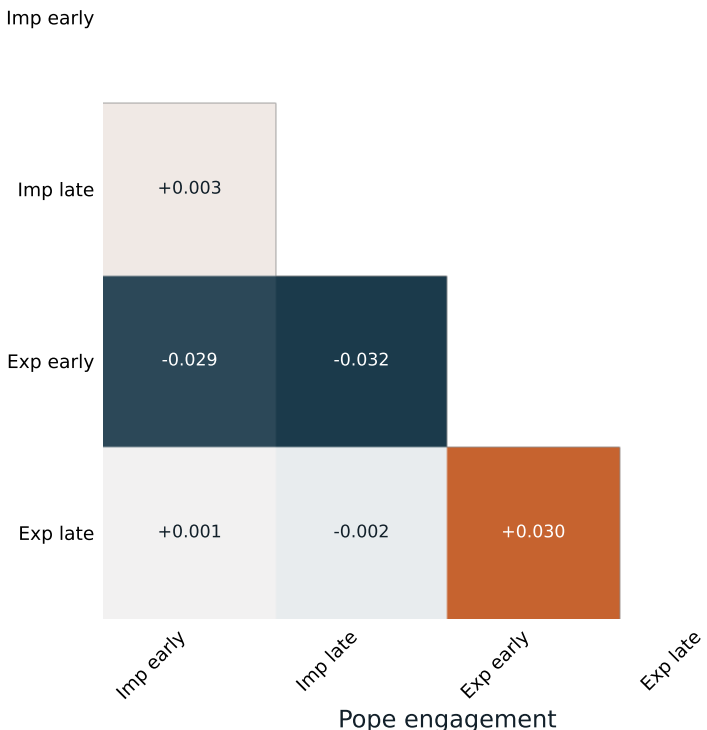
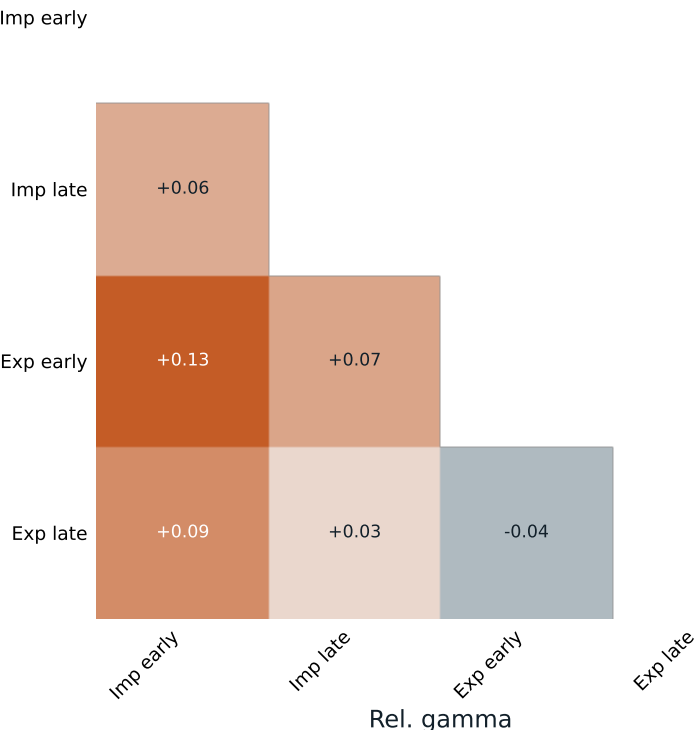


Rel. delta

Rel. theta

Rel. alpha

Rel. beta

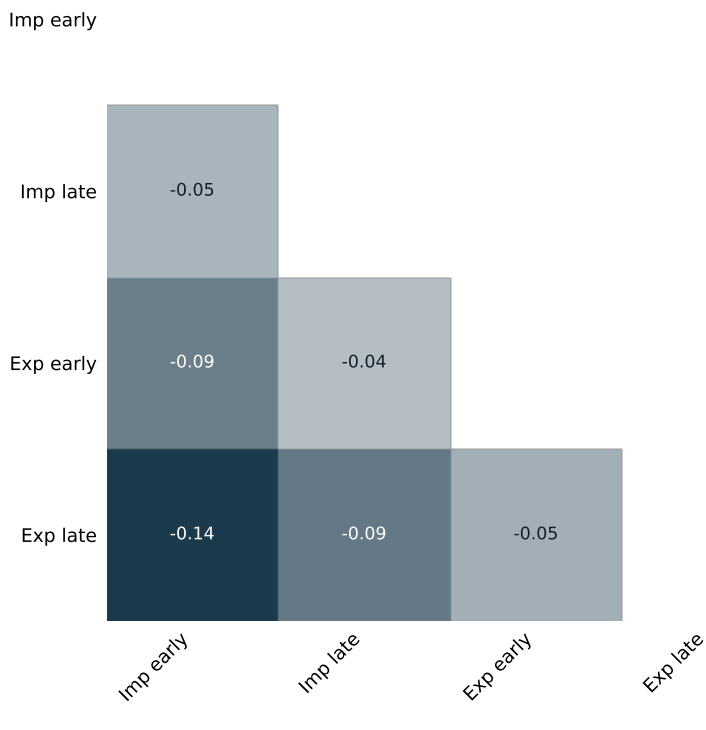
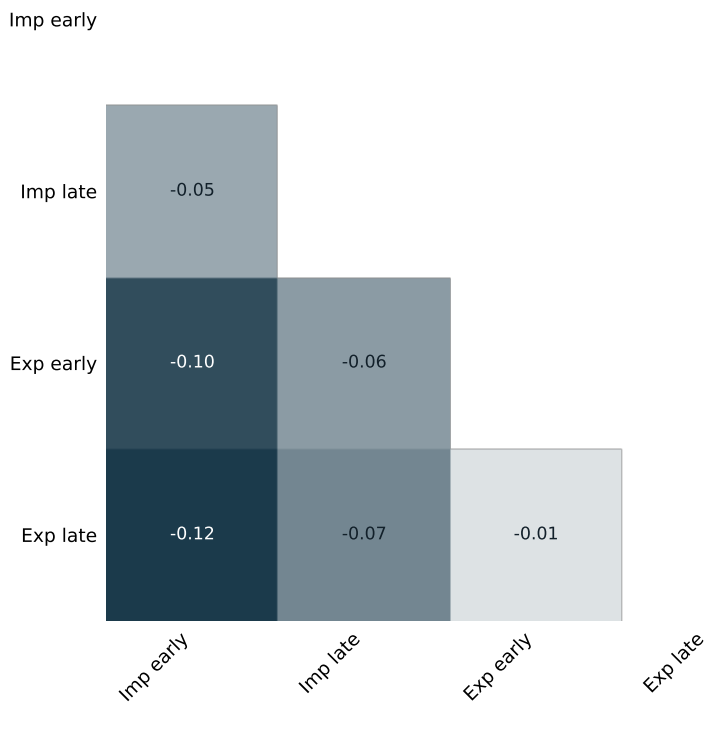
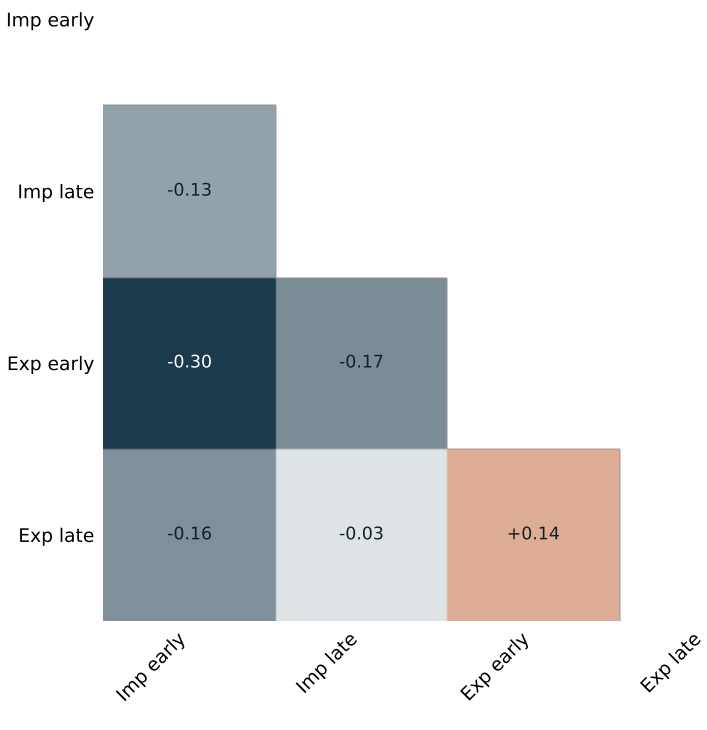
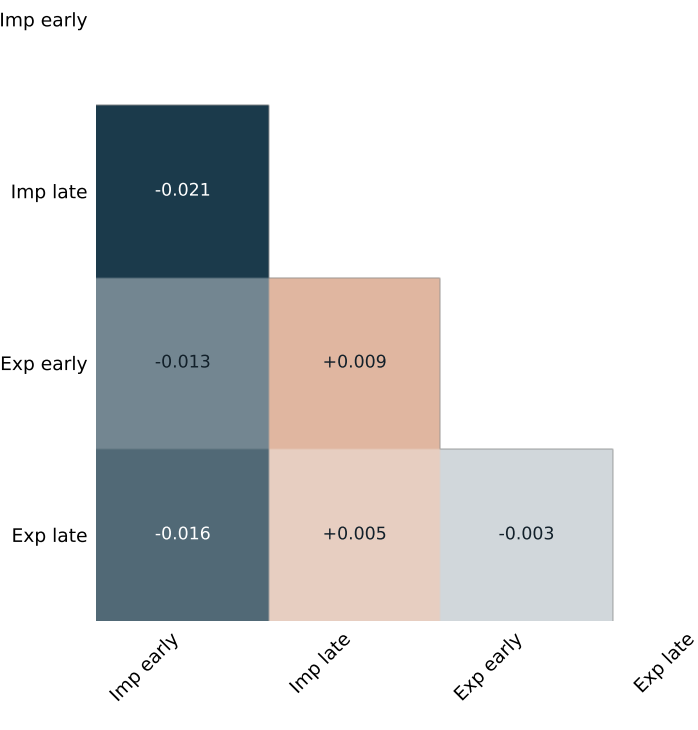


Rel. gamma

Pope engagement

Pope frontocentral

Kislov ratio



Rel. delta

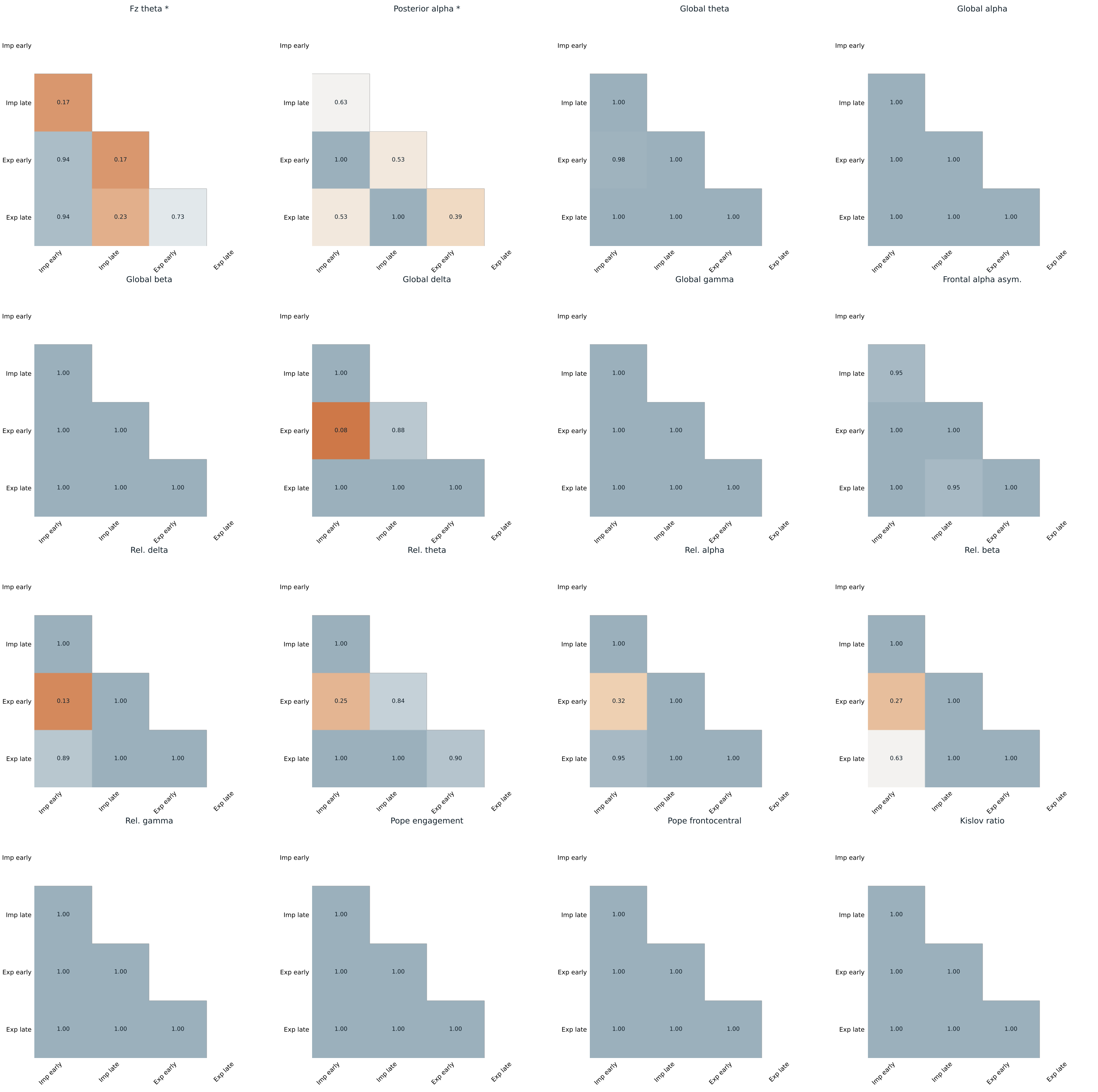
Pope engagement

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Kislov ratio

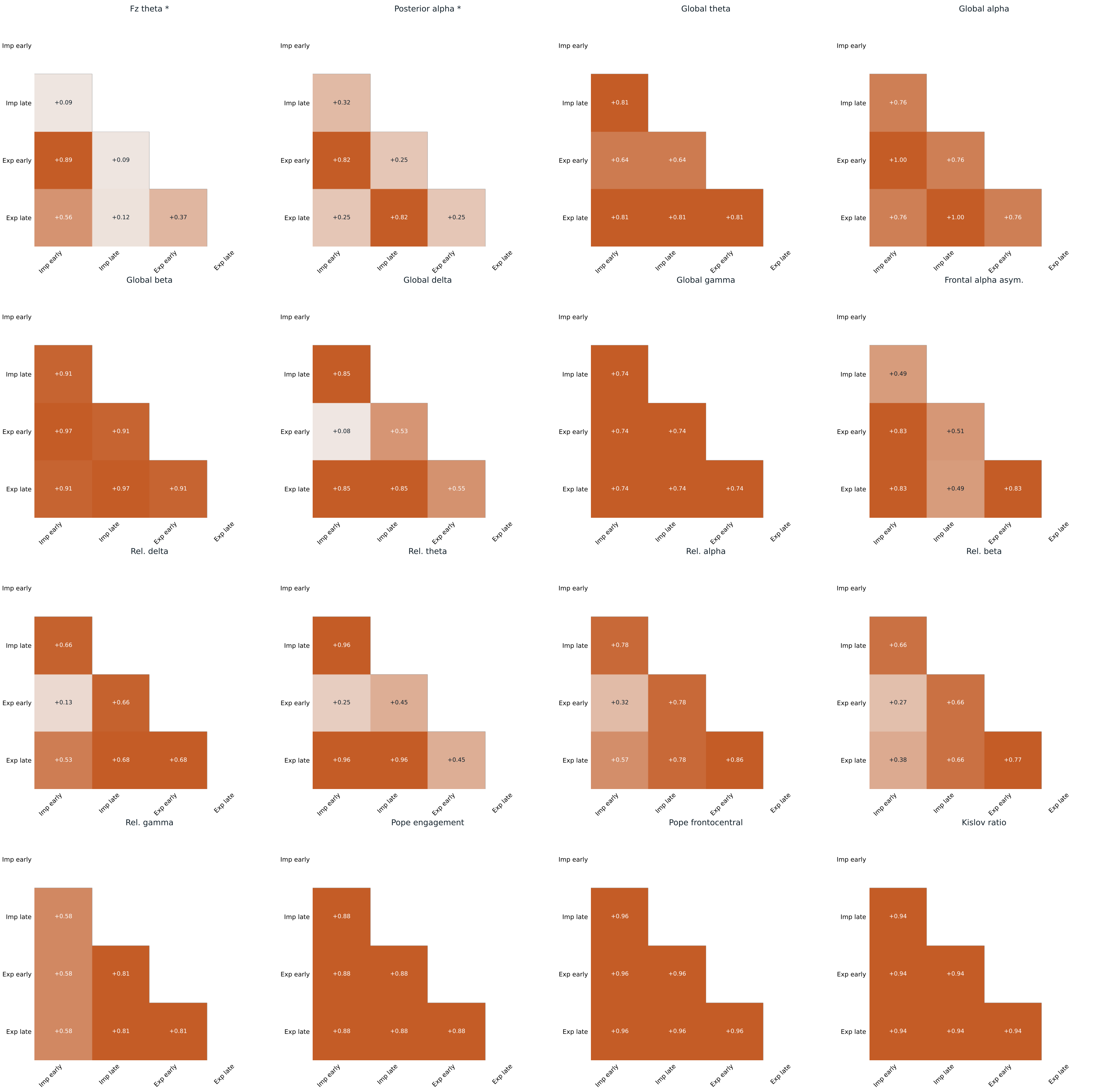
Dataset B, Delta matched control pairwise Holm p

Each cell is first differenced against its matched no-ad reply. 6 pairs, 16 measures, n=18. No cell reaches .05.



Dataset B, Delta matched control pairwise Benjamini-Hochberg q

Each cell is first differenced against its matched no-ad reply. 6 pairs, 16 measures, n=18. No cell reaches .05.



Dataset B, Delta matched control pairwise Benjamini-Yekutieli q

Each cell is first differenced against its matched no-ad reply. 6 pairs, 16 measures, n=18. No cell reaches .05.



Dataset B, raw post-pre pairwise effects (row minus column)

No control subtraction. Same-timing pairs match the Delta board exactly; cross-timing pairs do not. Each panel is scaled to its own range. Exploratory.



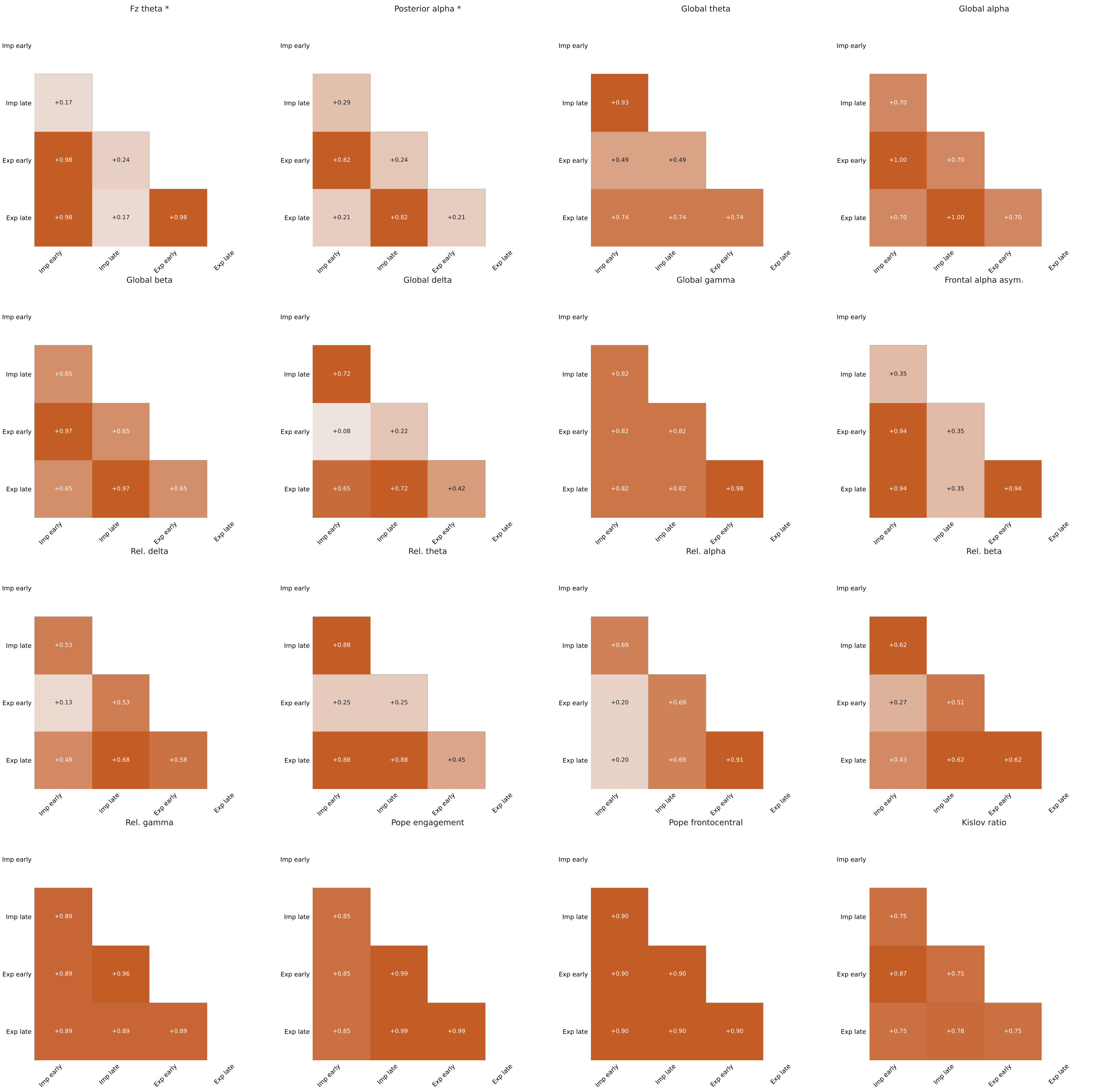
Dataset B, raw post-pre pairwise Holm p

No control subtraction. Same-timing pairs match the Delta board exactly; cross-timing pairs do not. No cell reaches .05.



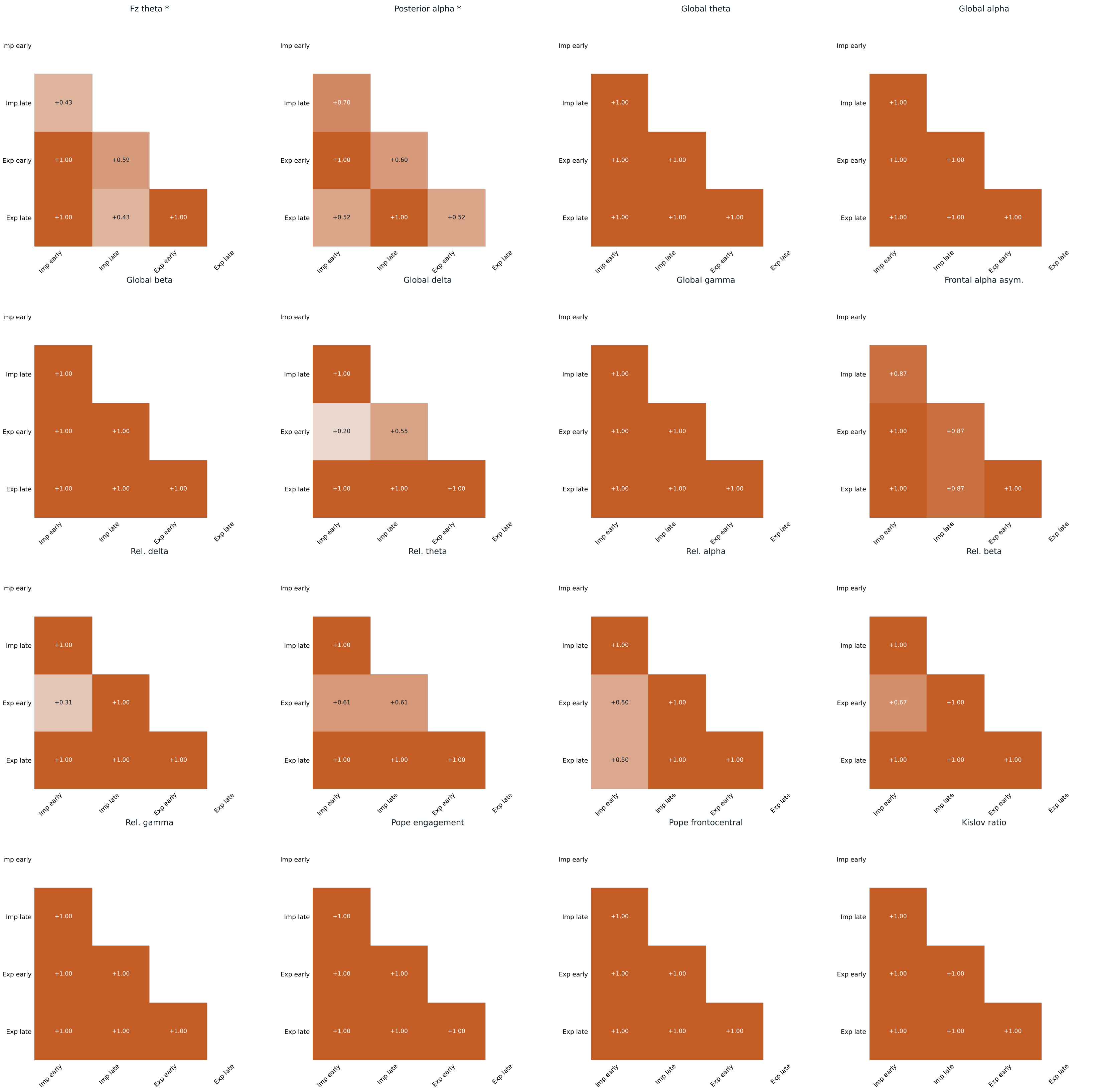
Dataset B, raw post-pre pairwise Benjamini-Hochberg q

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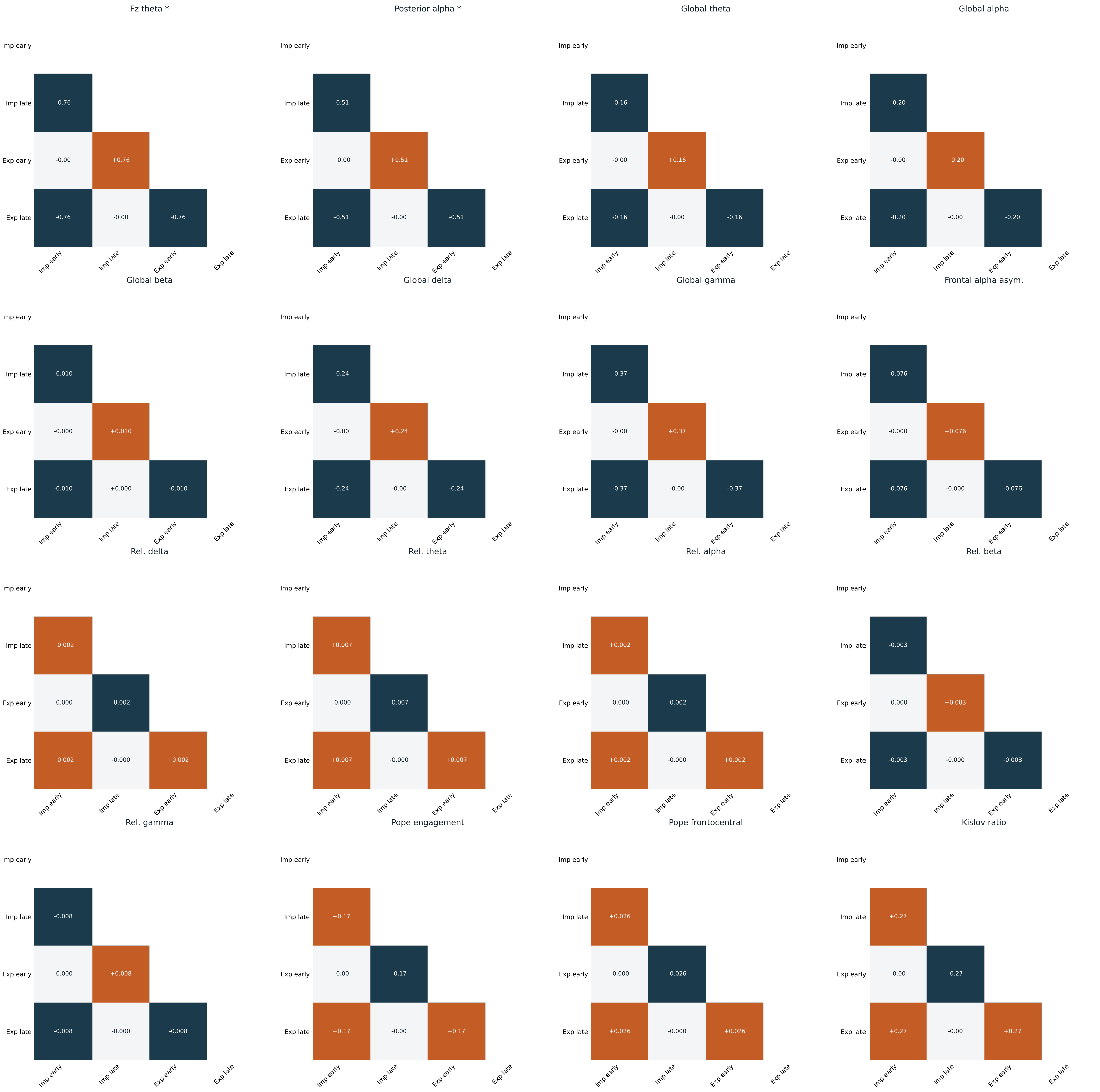
Dataset B, raw post-pre pairwise Benjamini-Yekutieli q

No control subtraction. Same-timing pairs match the Delta board exactly; cross-timing pairs do not. No cell reaches .05.



Dataset B: how much the matched control moves each pair

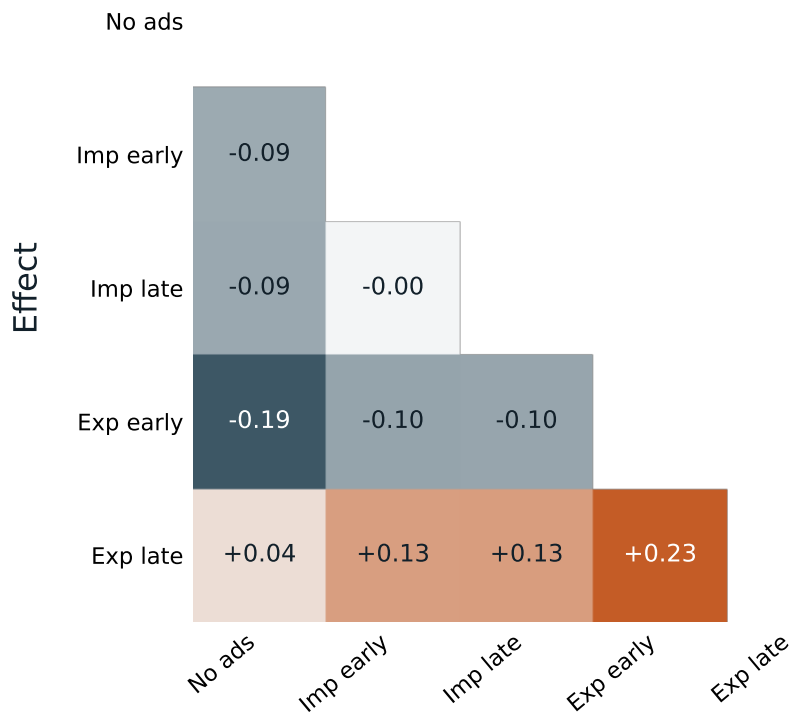
Delta space minus raw space, dB. Same-timing pairs are exactly zero because the shared control cancels; every cross-timing pair carries the control drift.



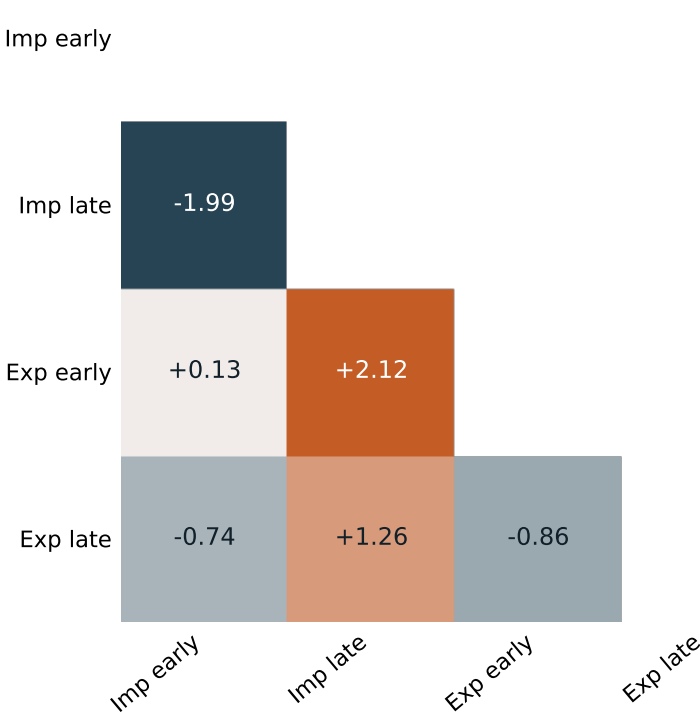
Fz theta *

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

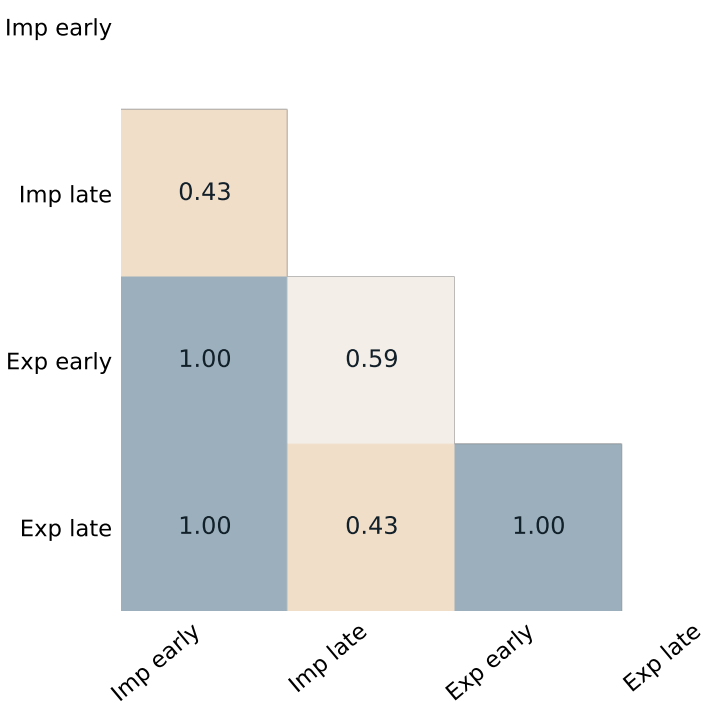
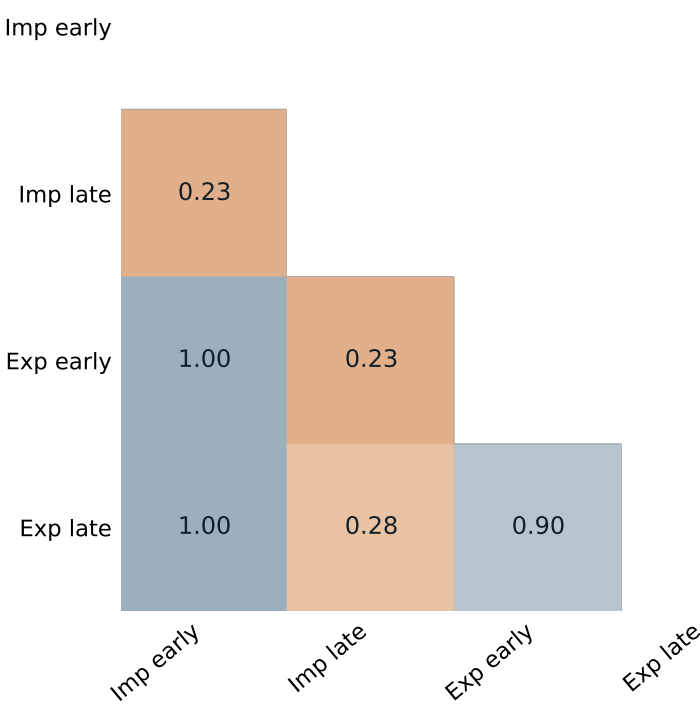
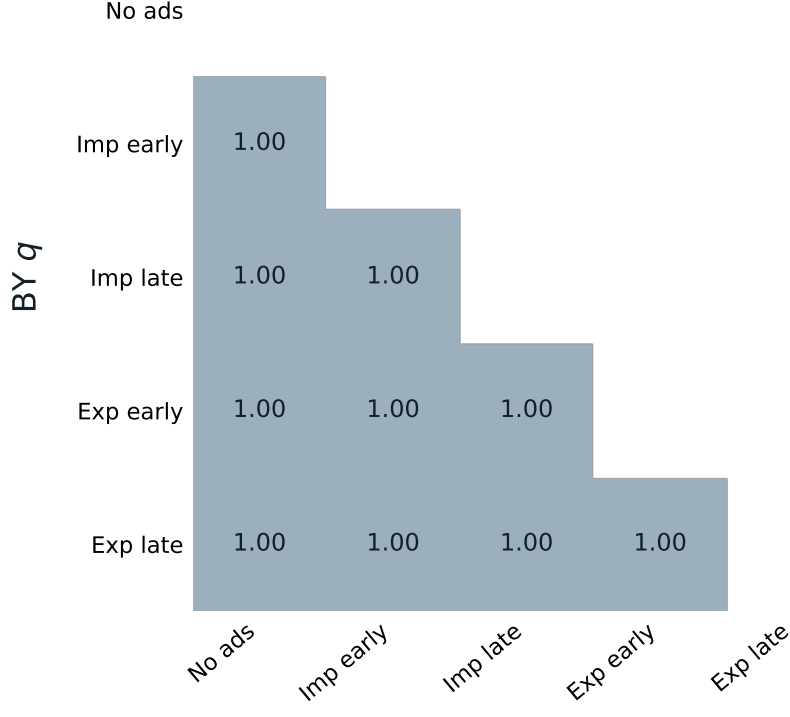
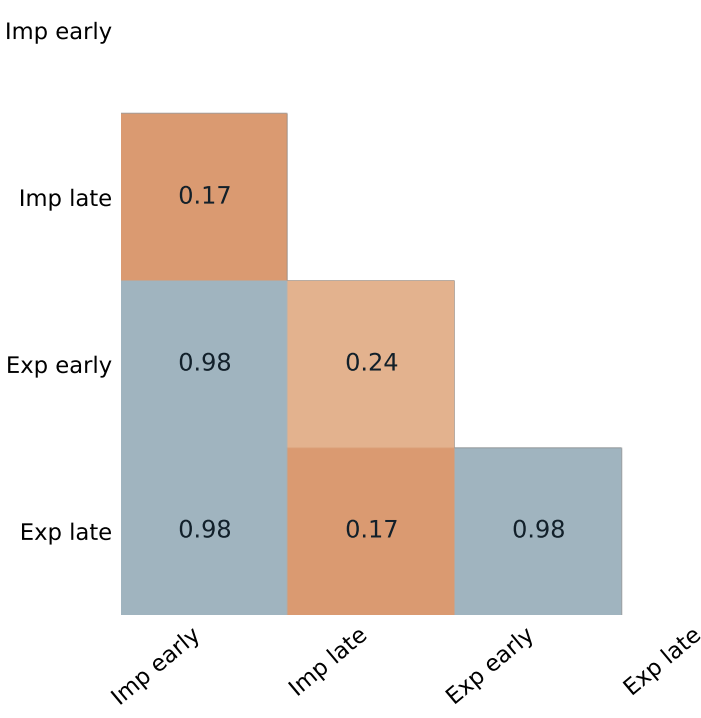
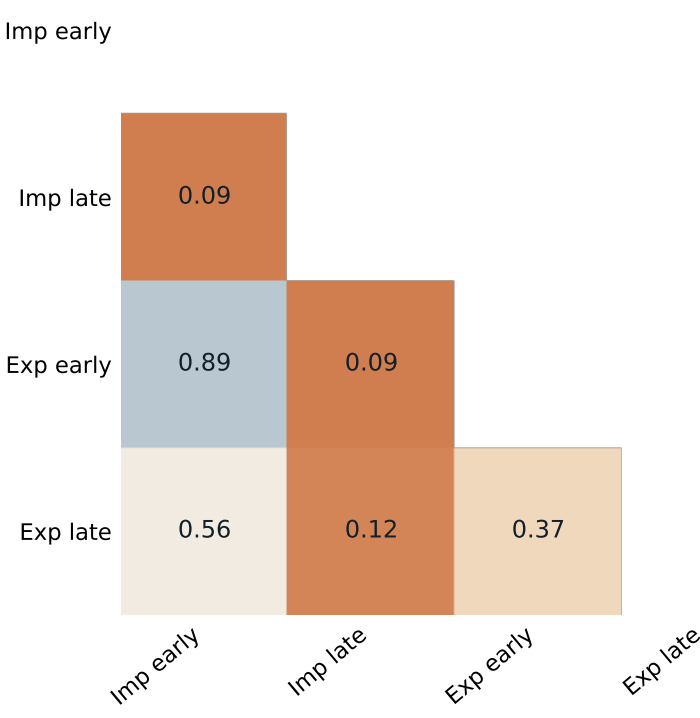
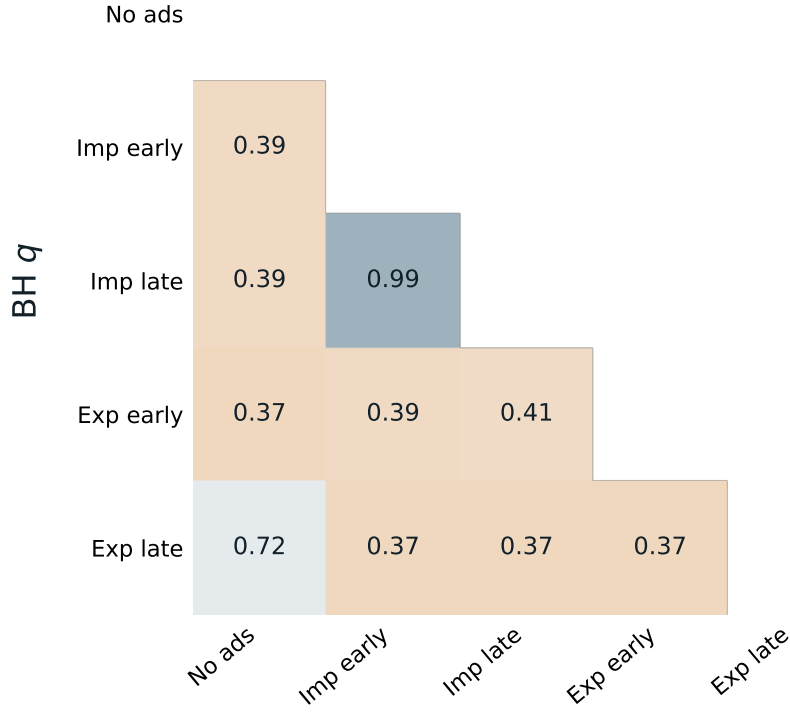
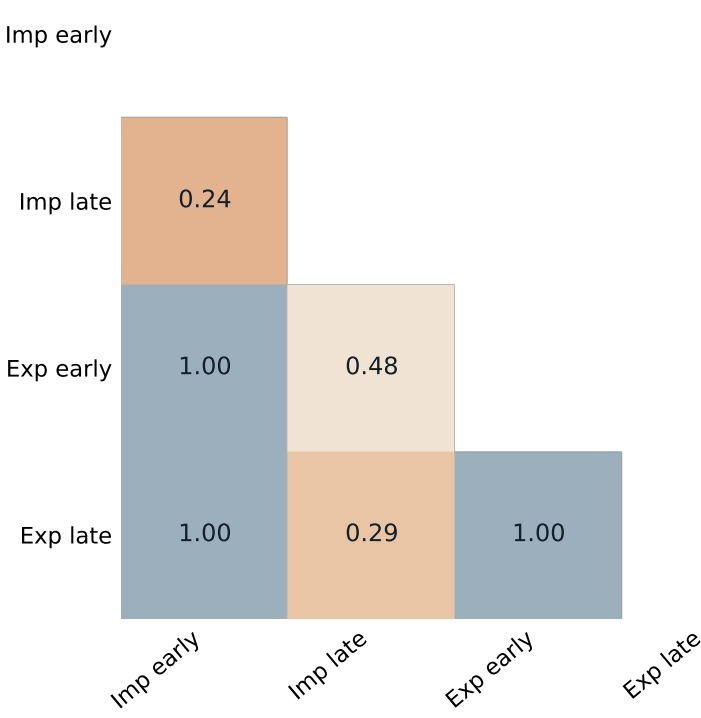
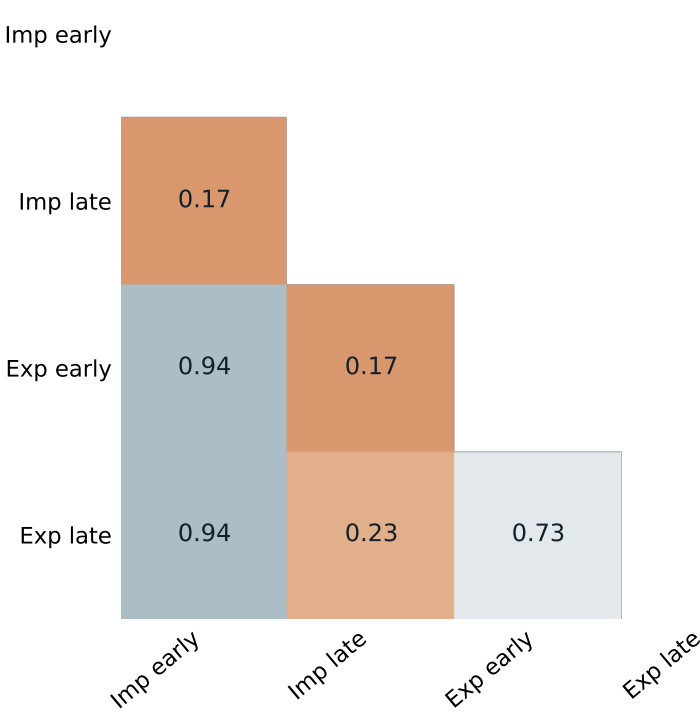
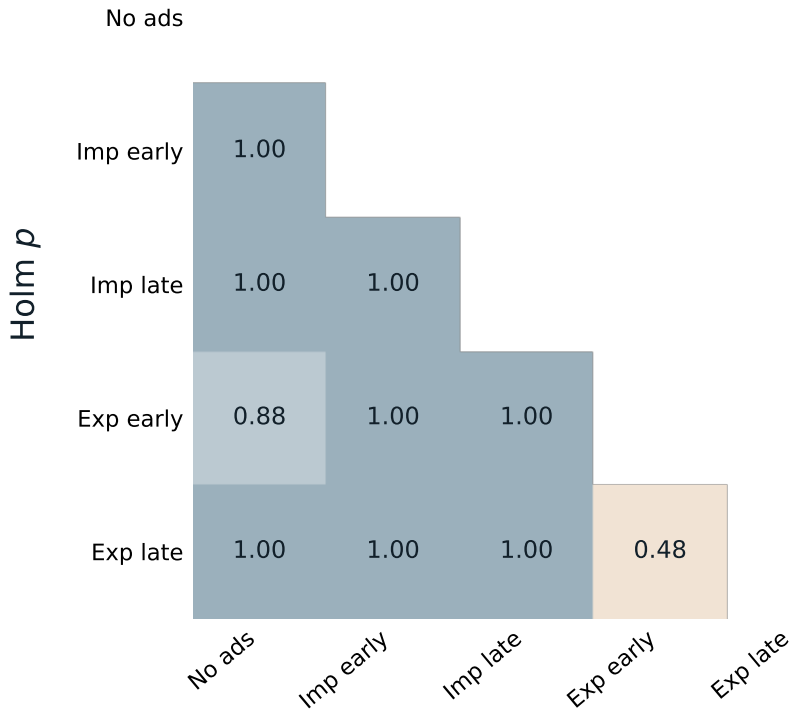
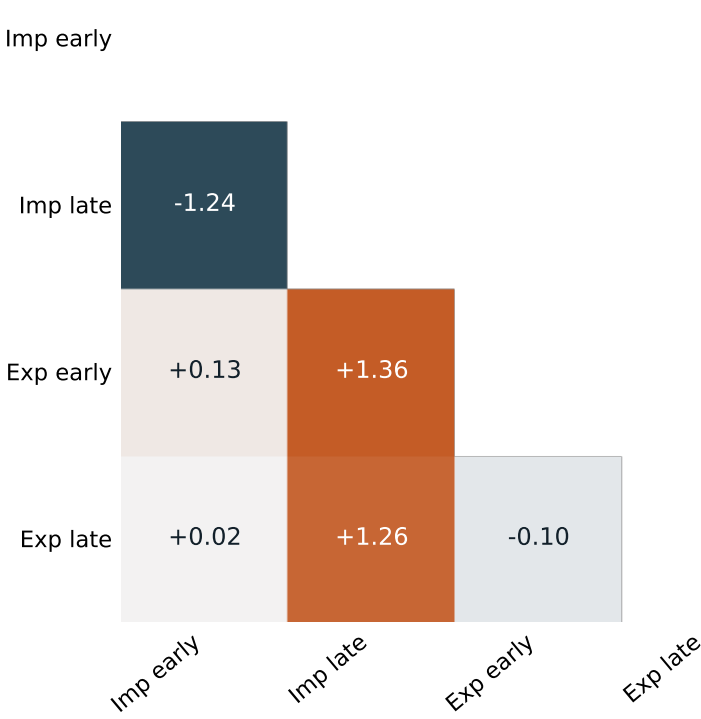
Dataset A (condition medians)



Dataset B, Δ matched control



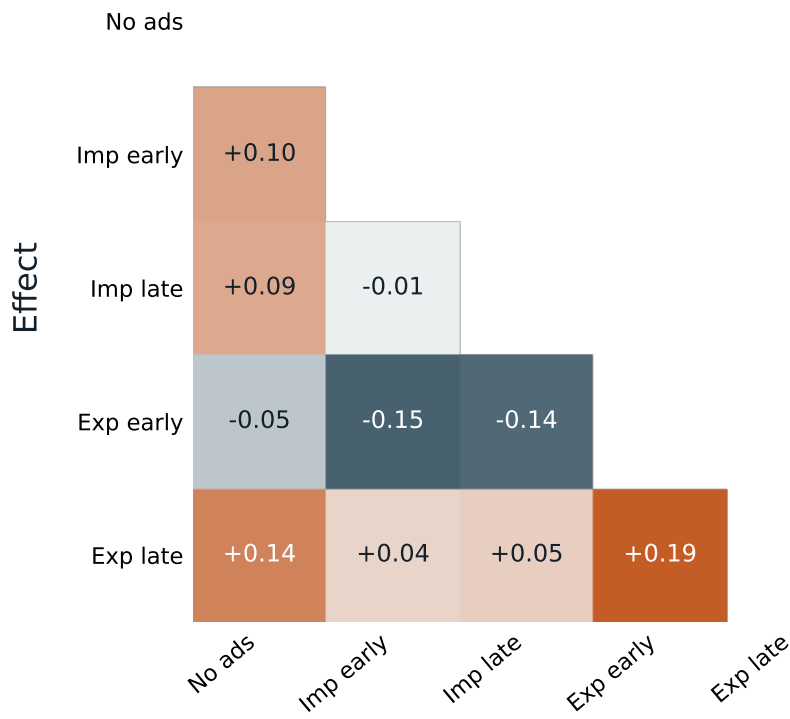
Dataset B, raw post–pre



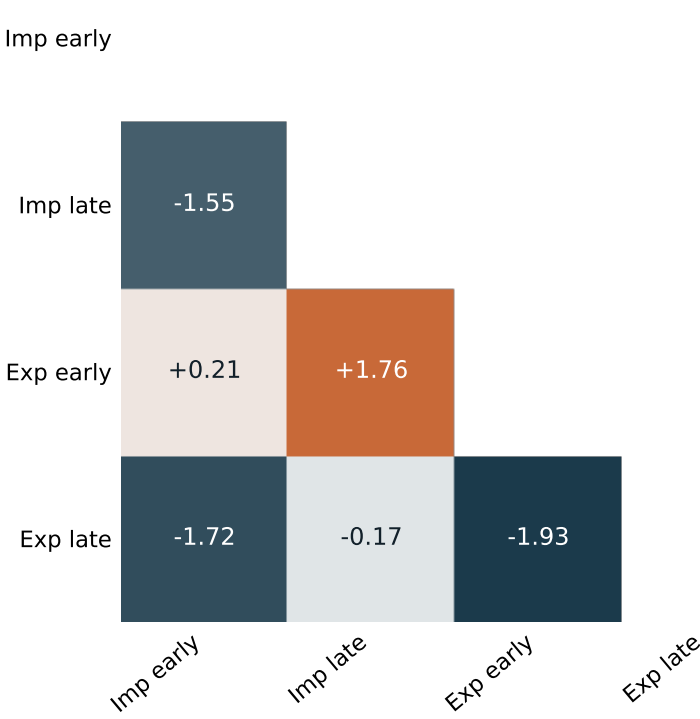
Posterior alpha *

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

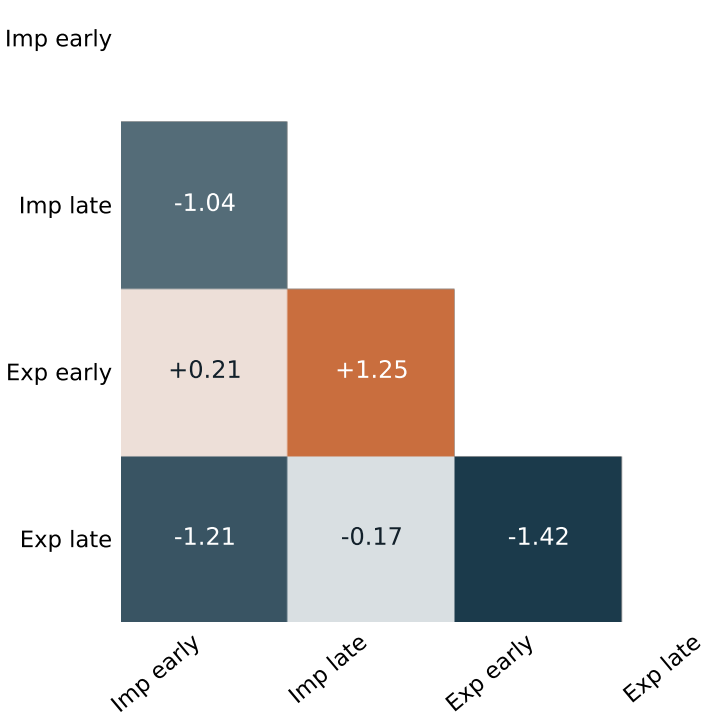
Dataset A (condition medians)



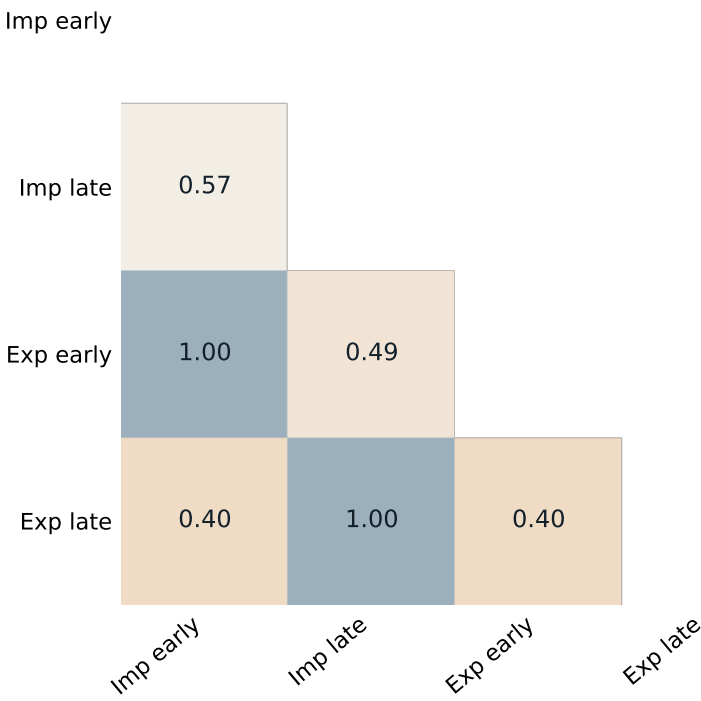
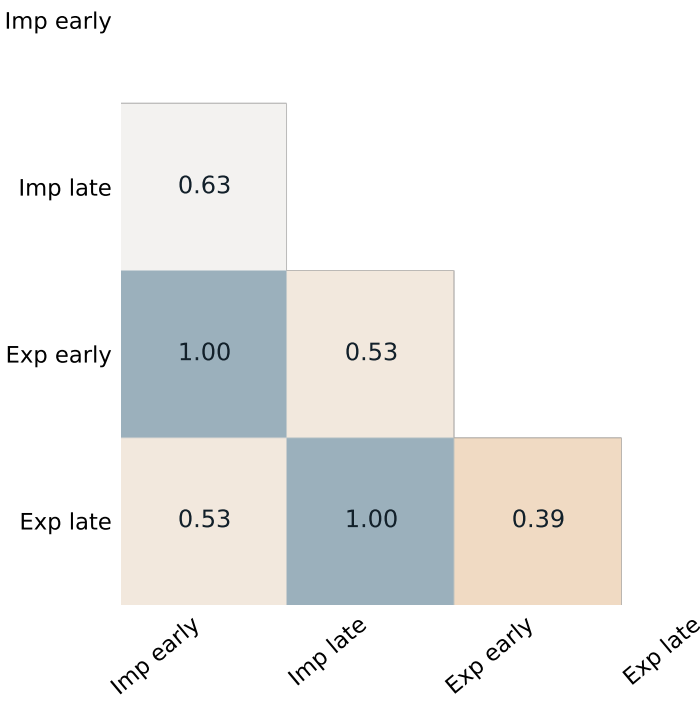
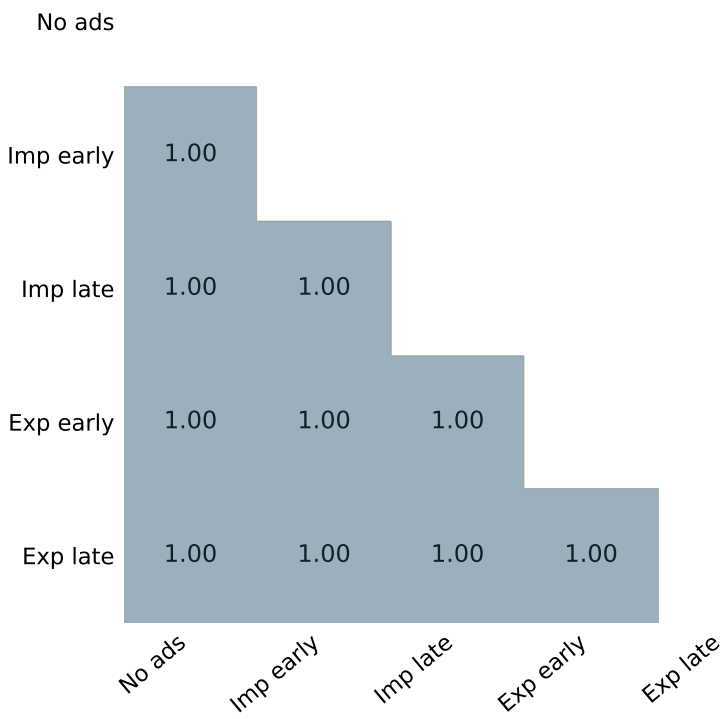
Dataset B, Δ matched control



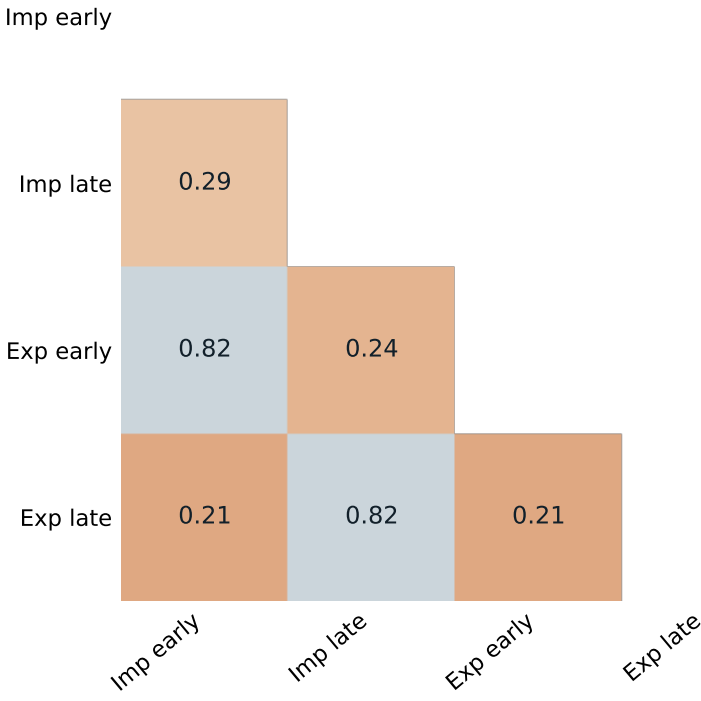
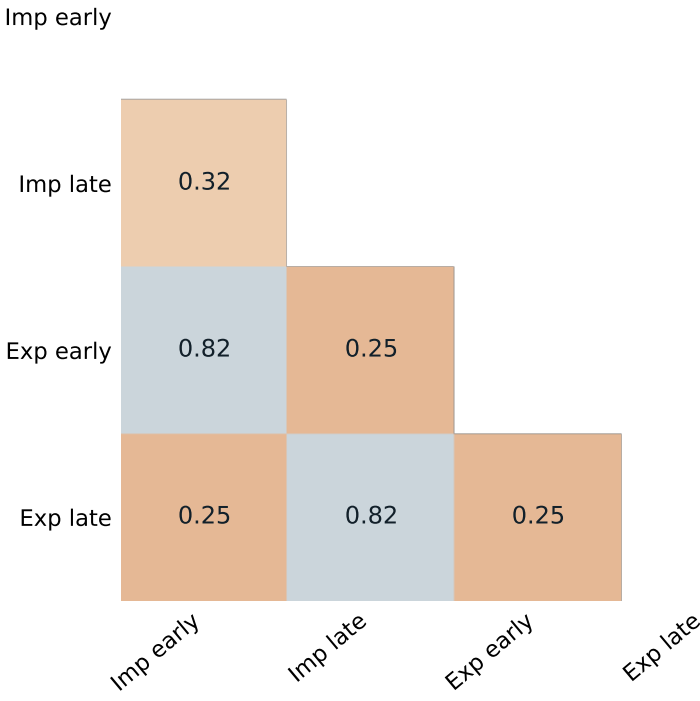
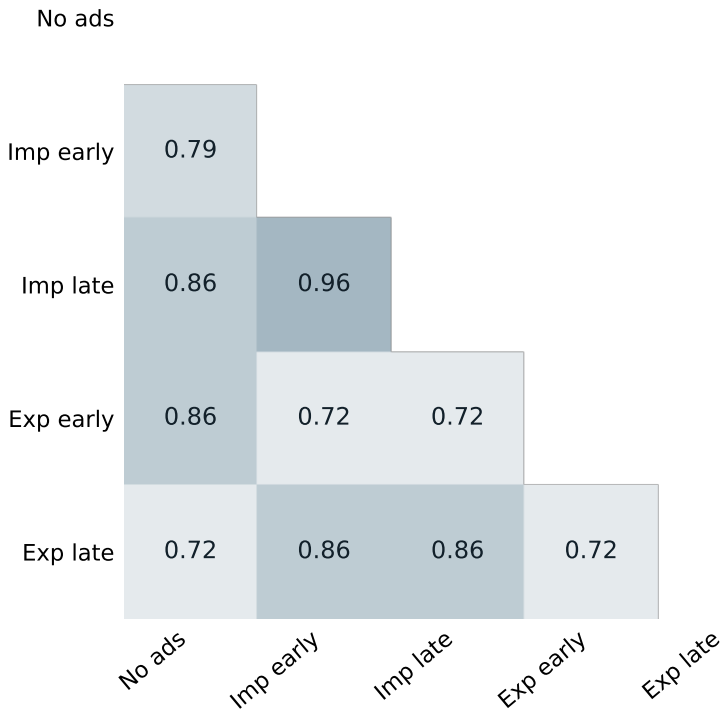
Dataset B, raw post-pre



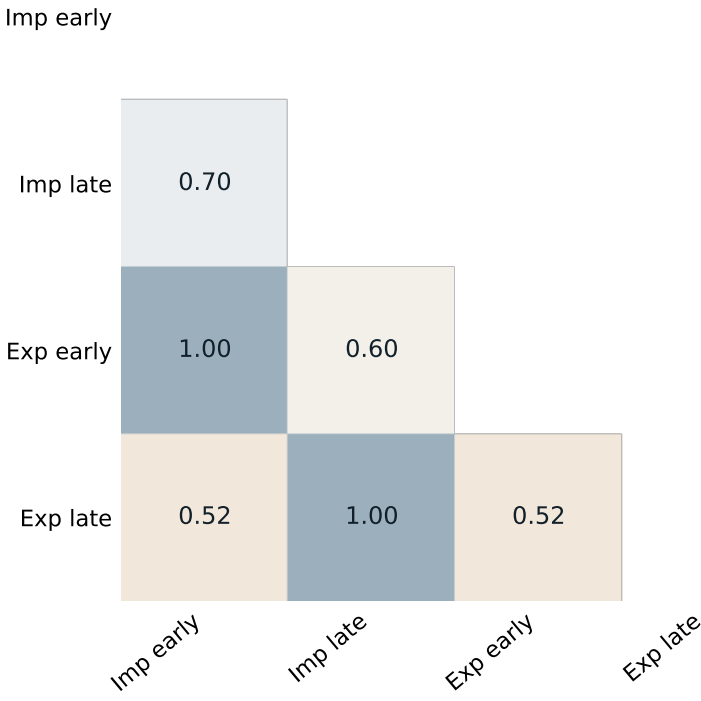
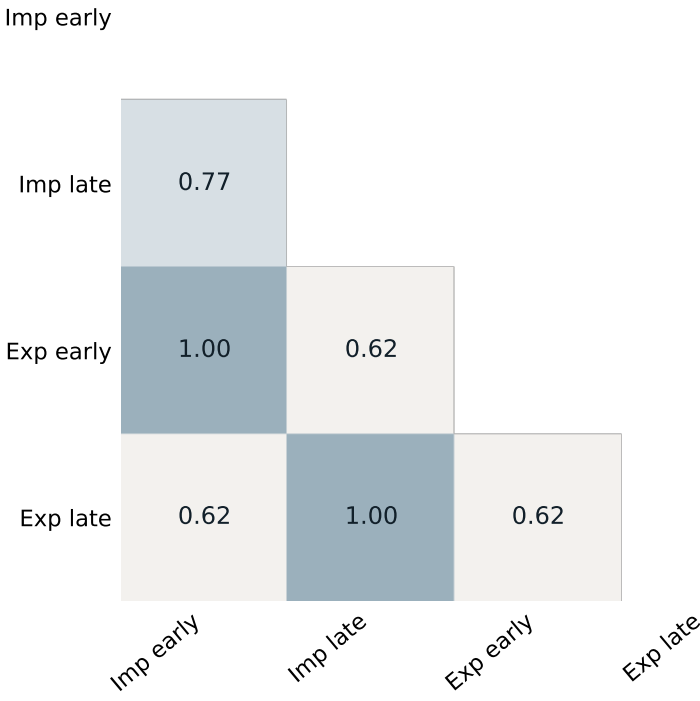
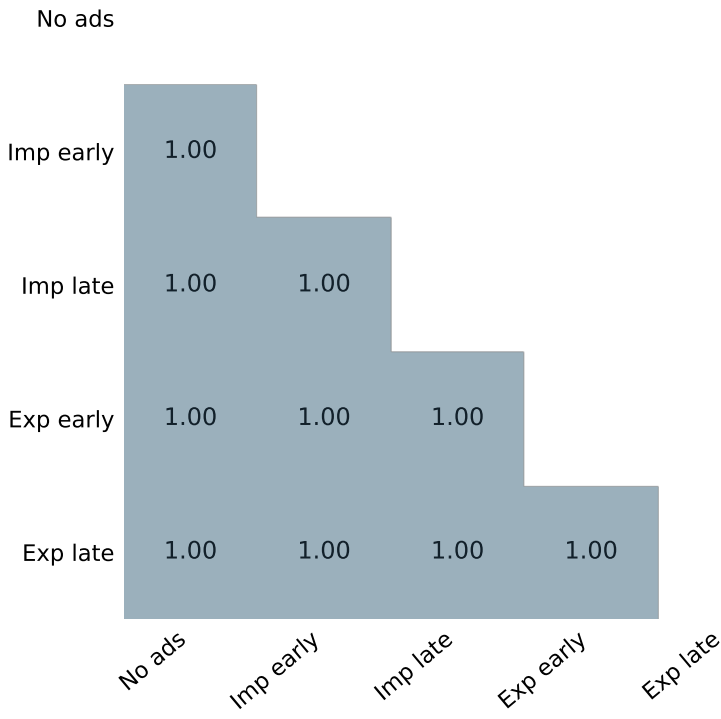
Holm p



BH q



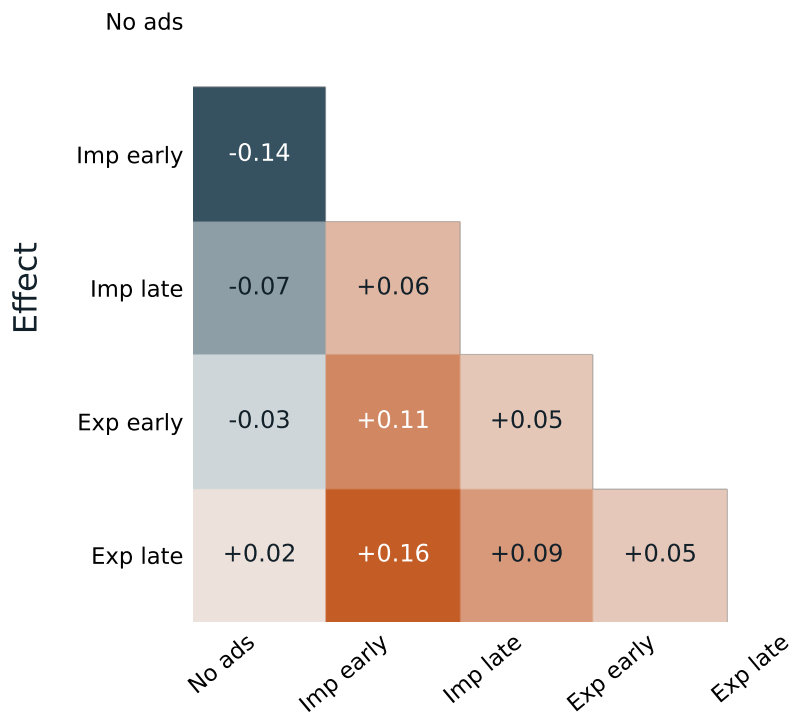
BY q



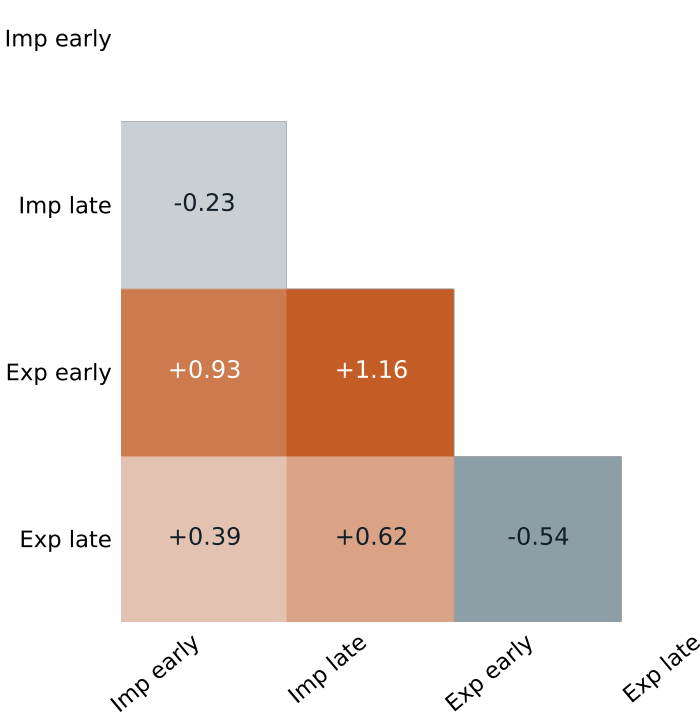
Global theta

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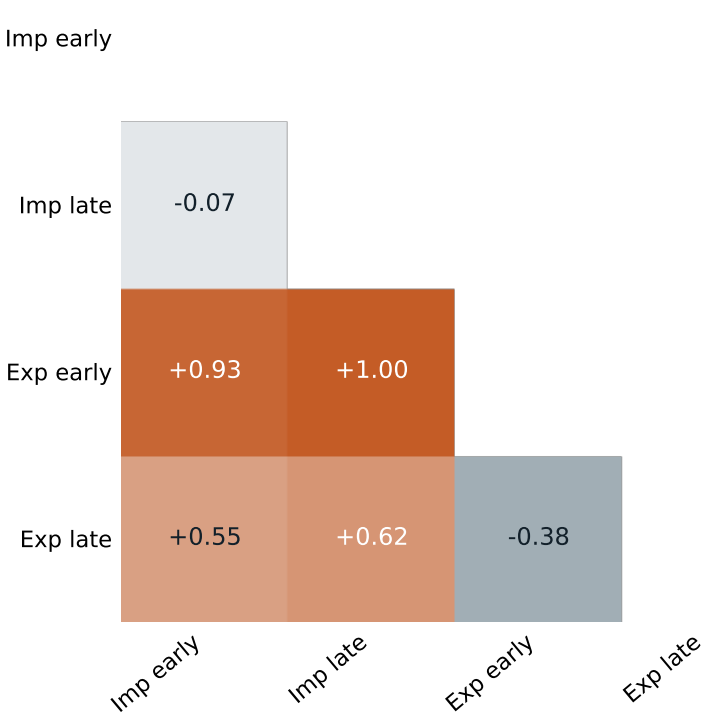
Dataset A (condition medians)



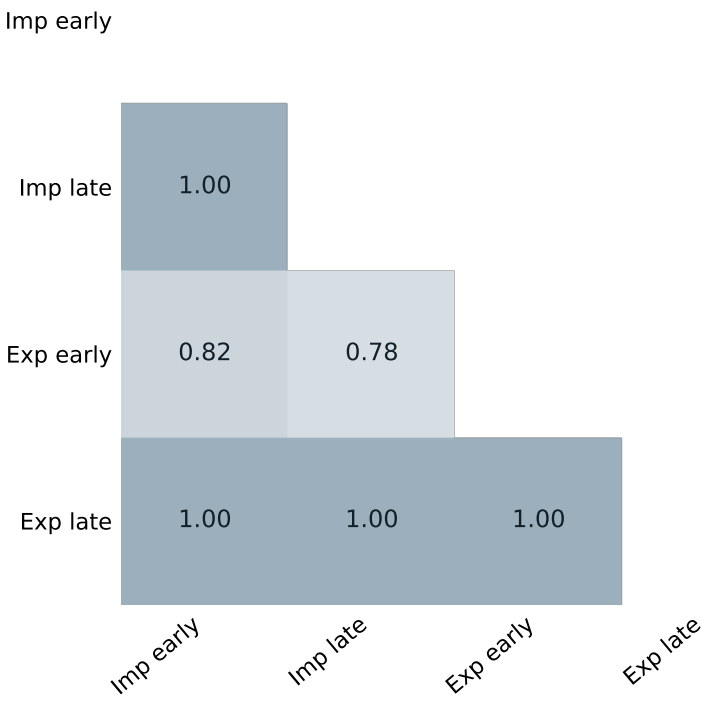
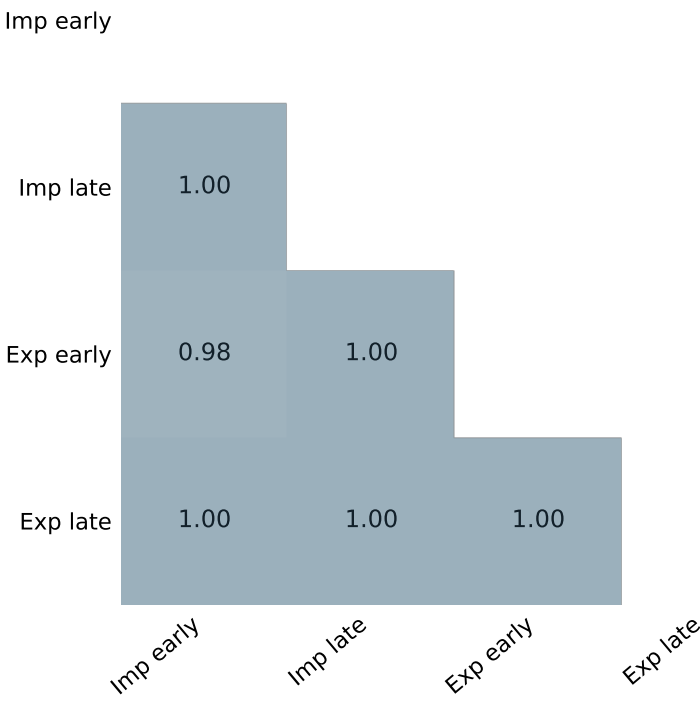
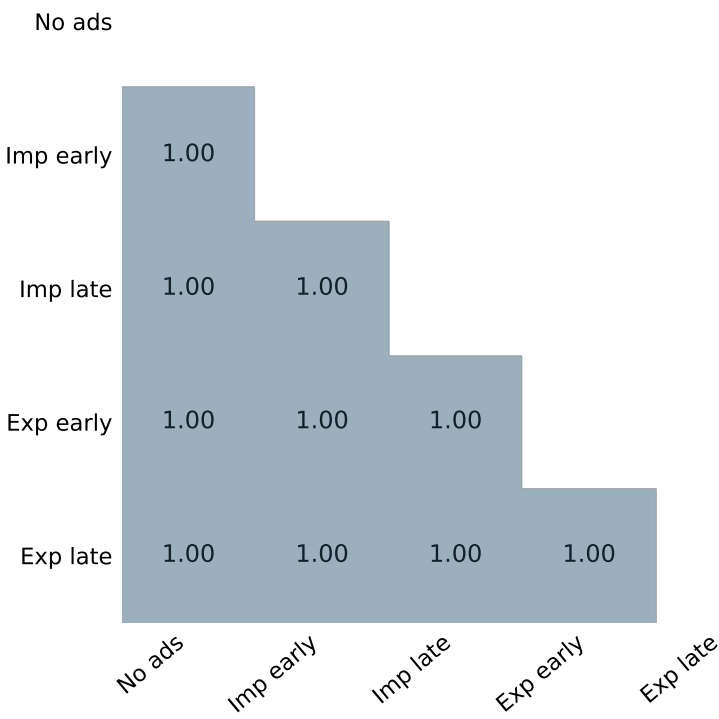
Dataset B, Δ matched control



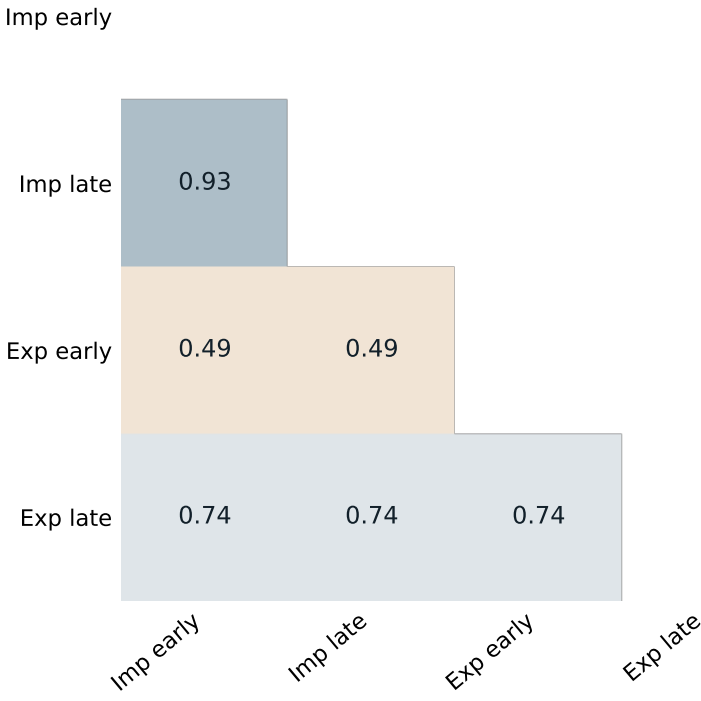
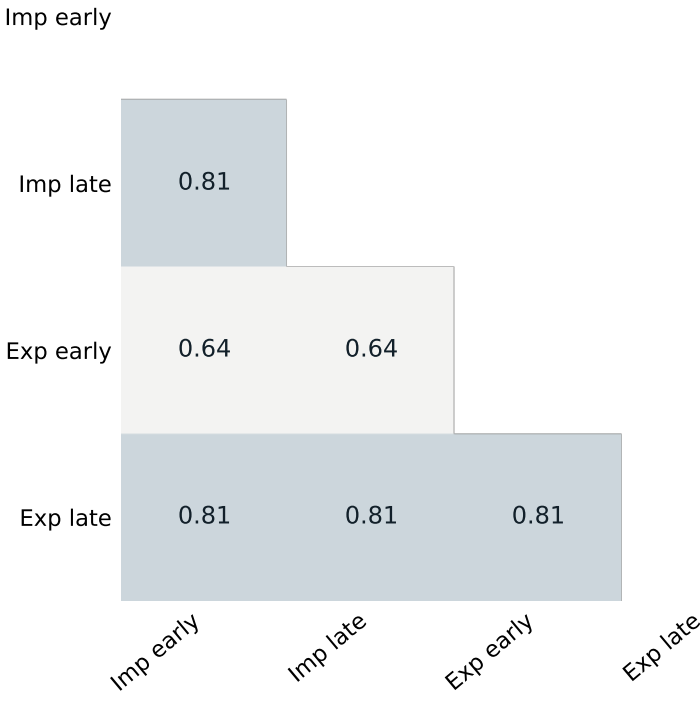
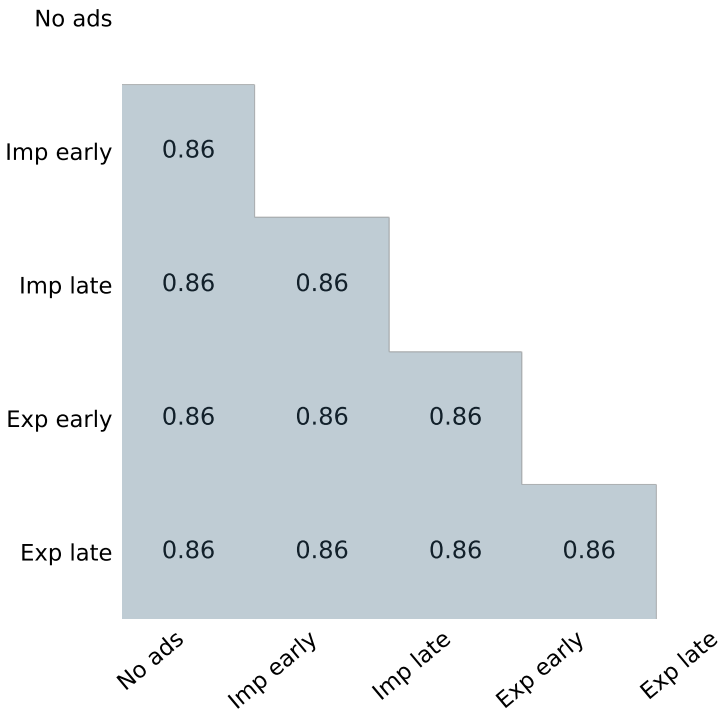
Dataset B, raw post-pre



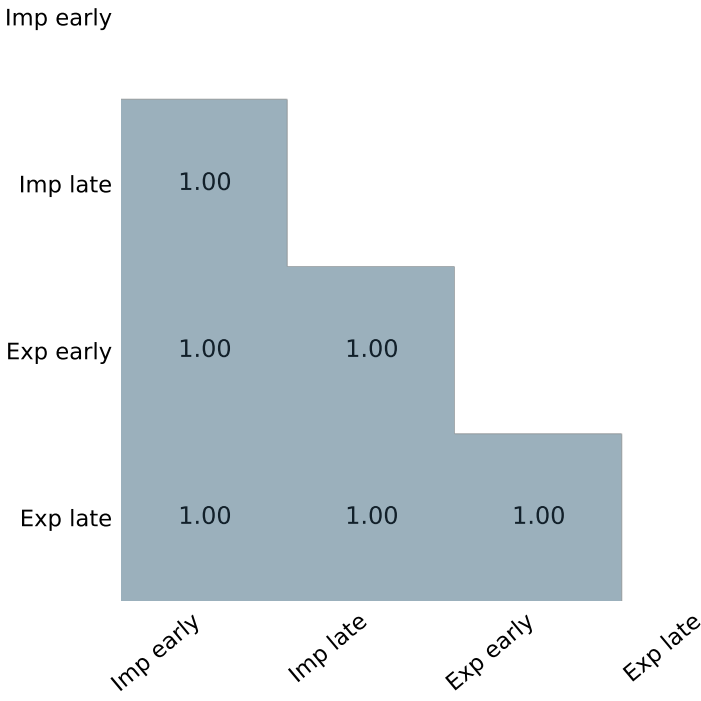
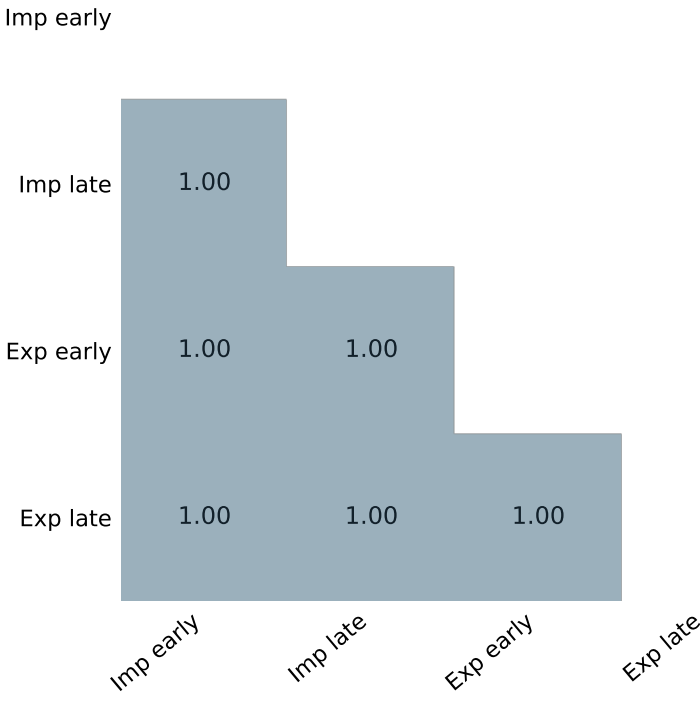
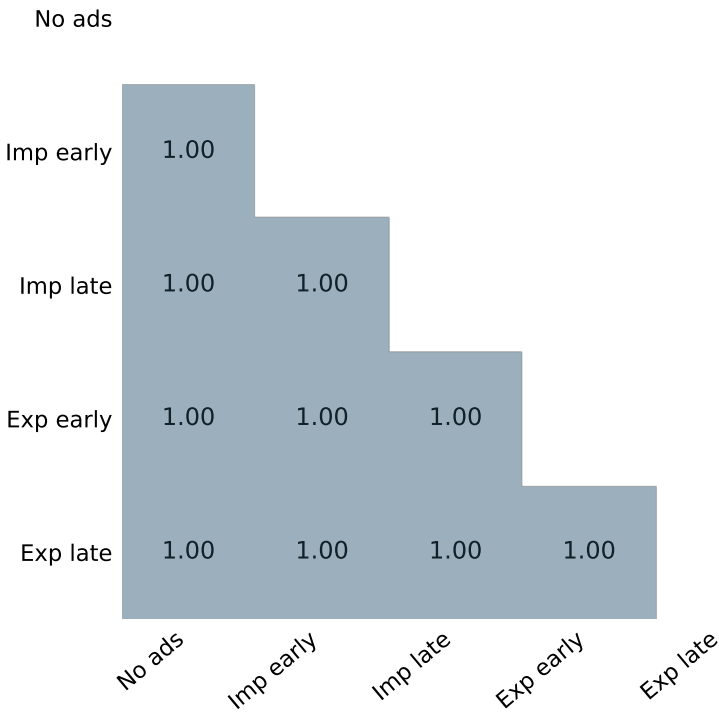
Holm p



BH q



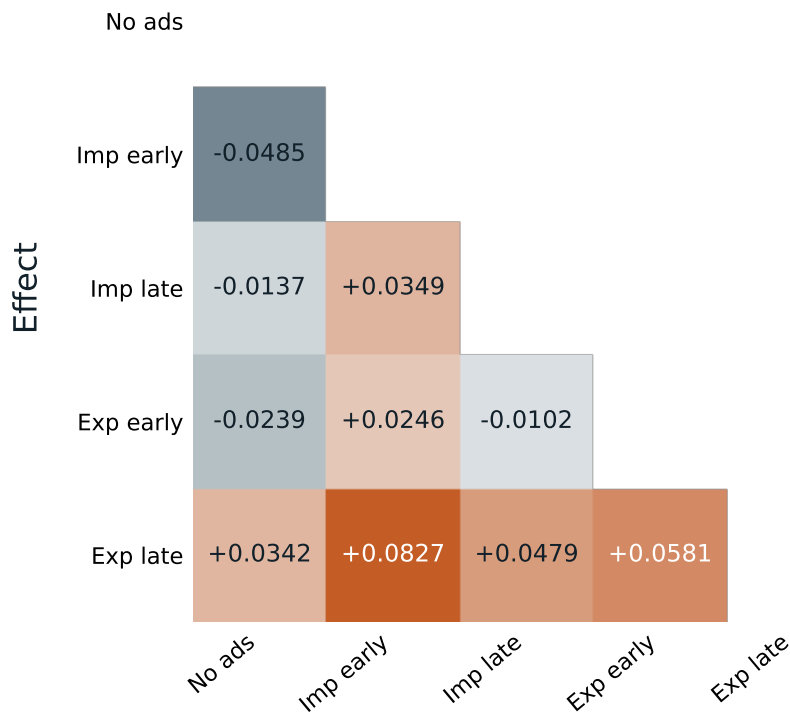
BY q



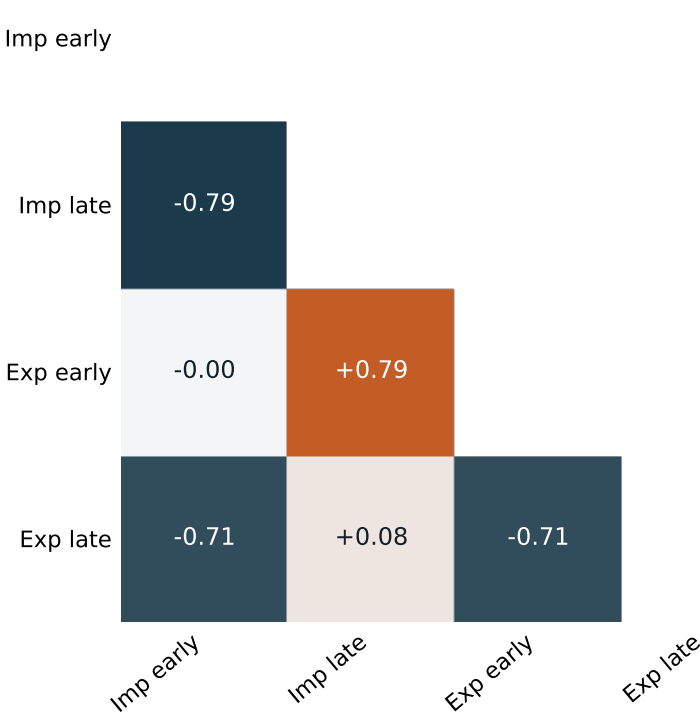
Global alpha

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

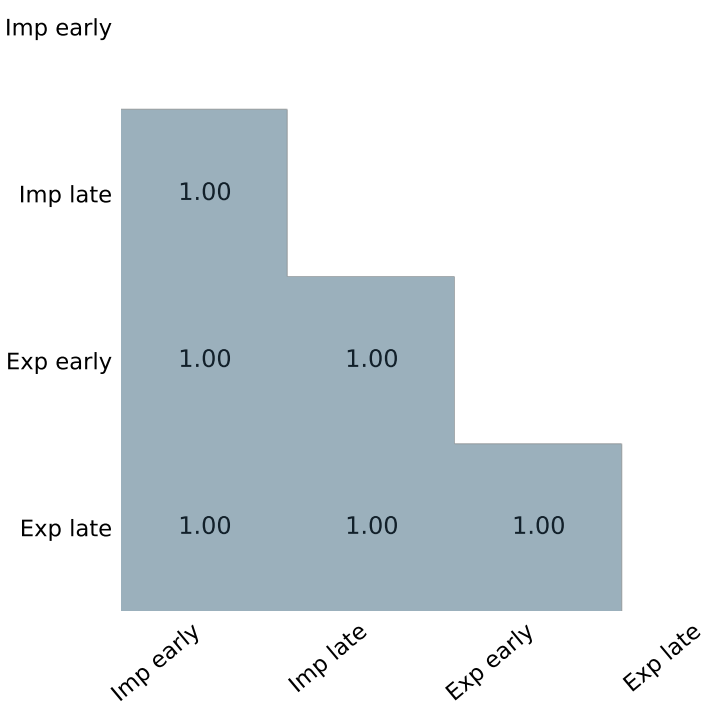
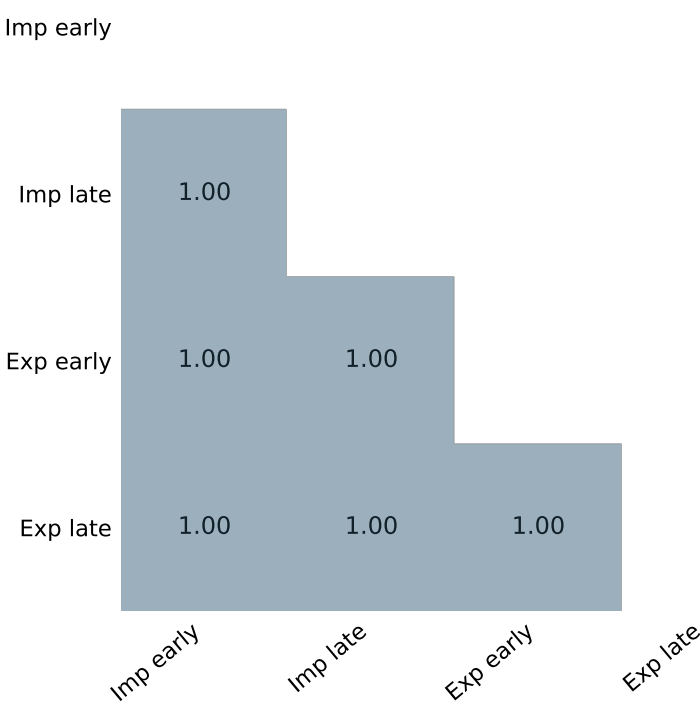
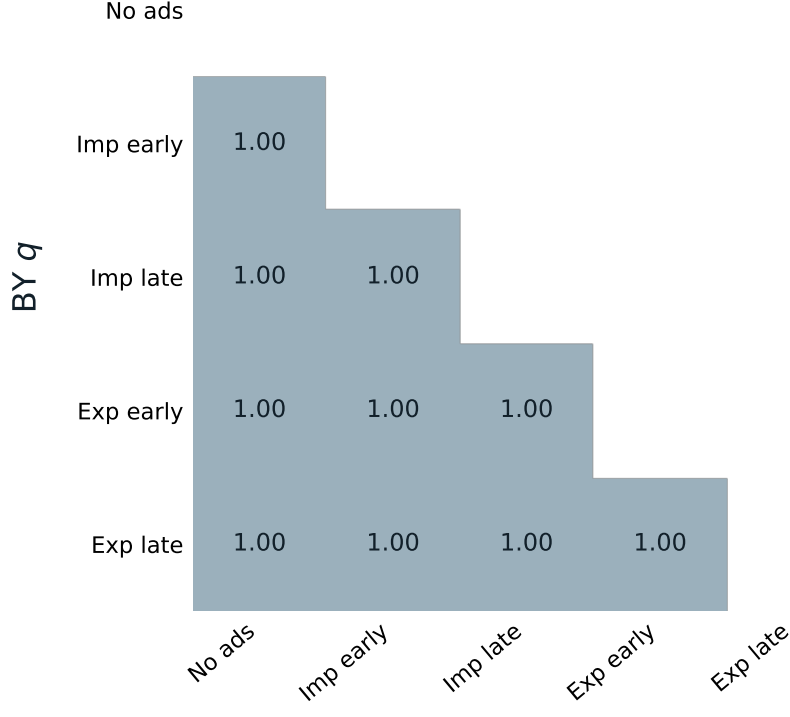
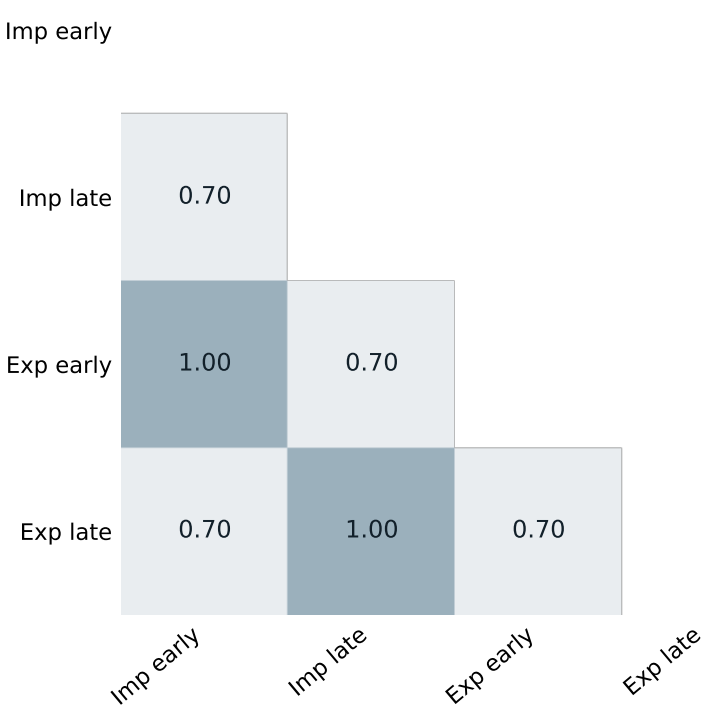
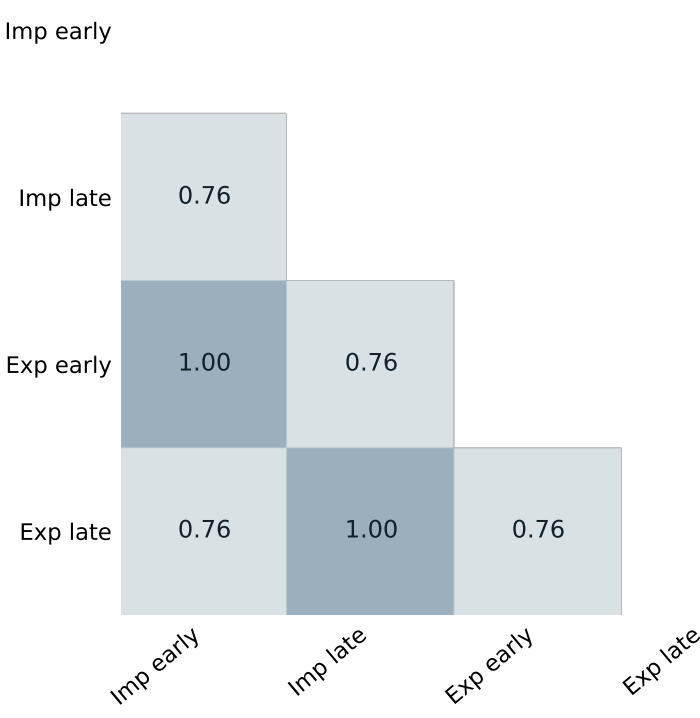
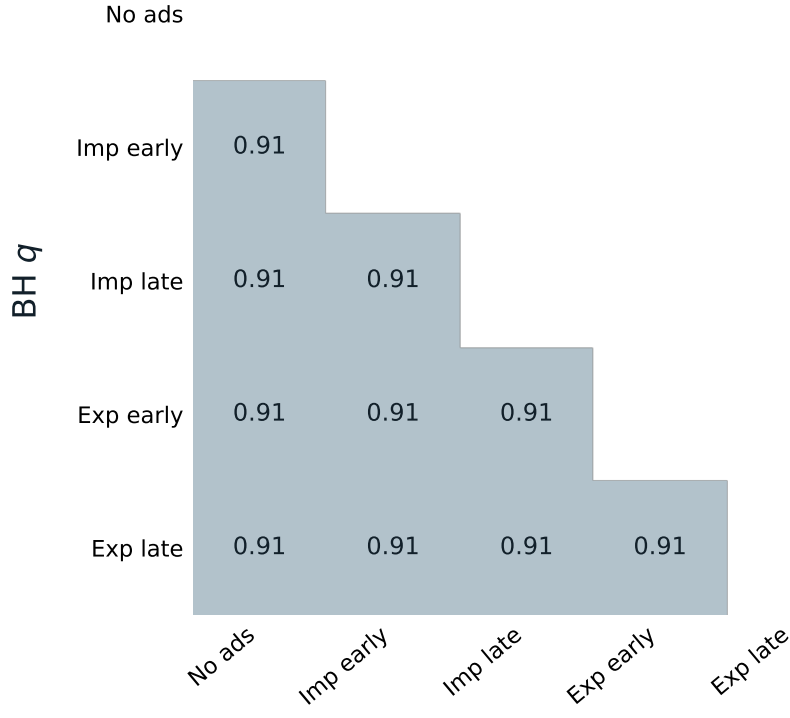
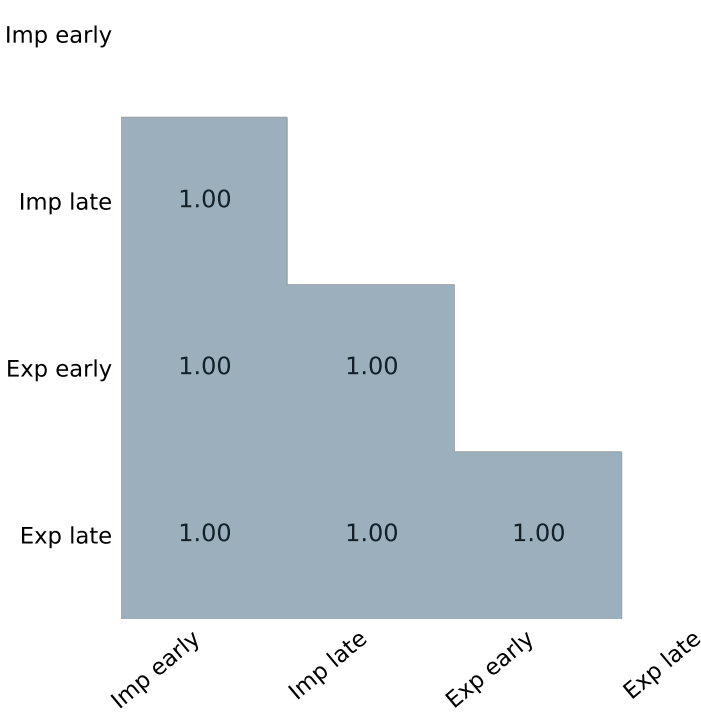
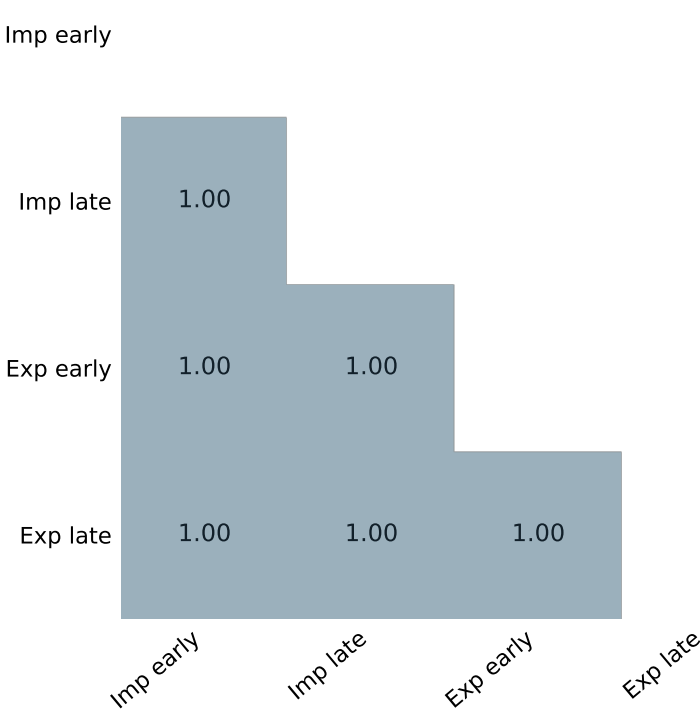
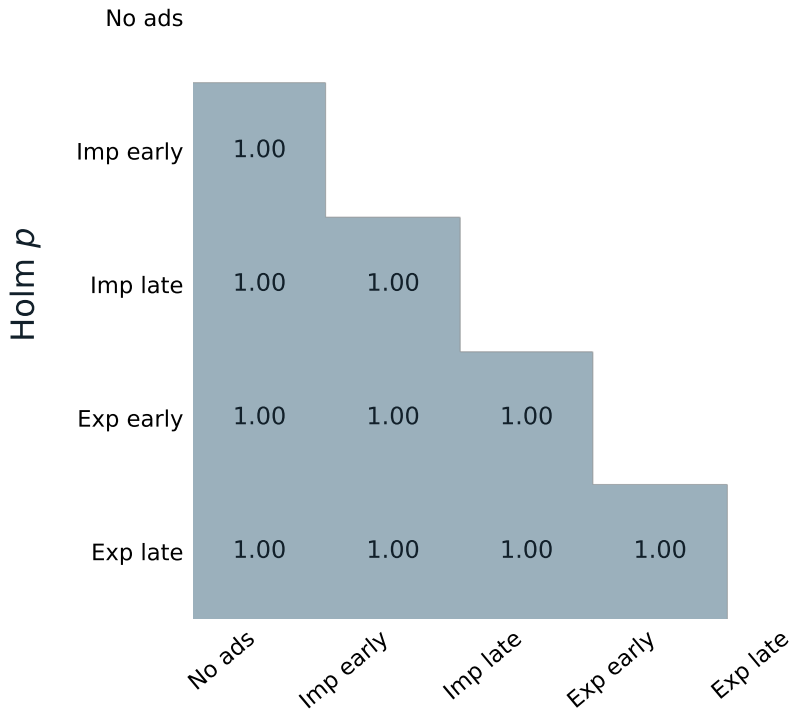
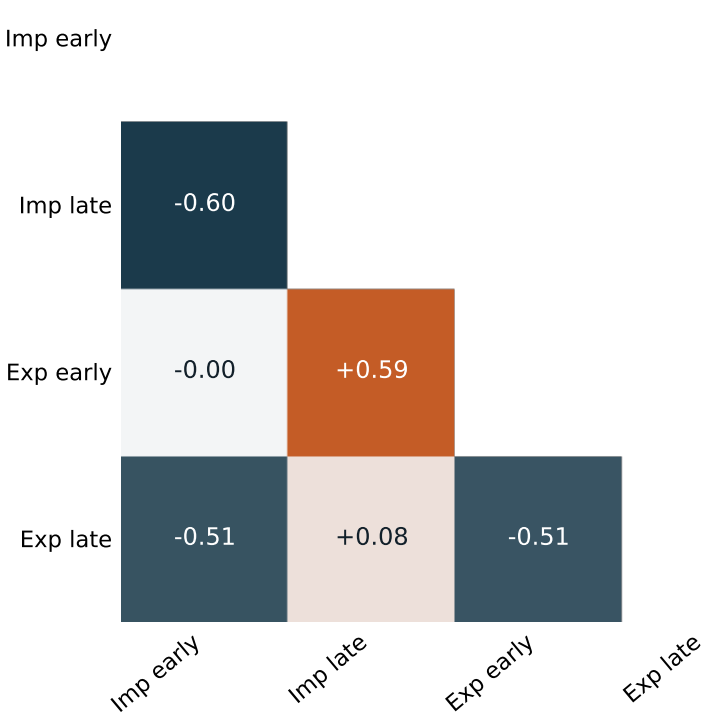
Dataset A (condition medians)



Dataset B, Δ matched control



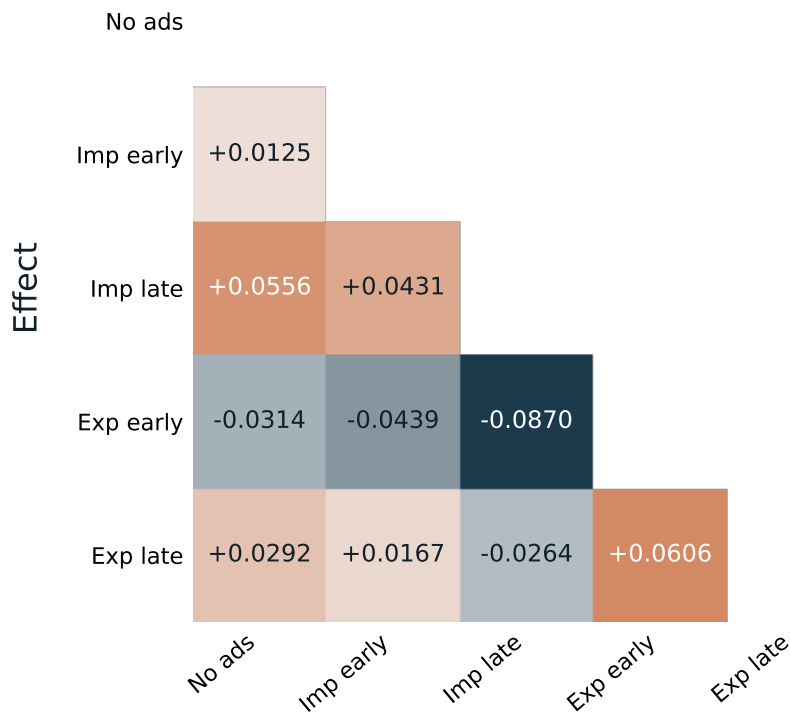
Dataset B, raw post–pre



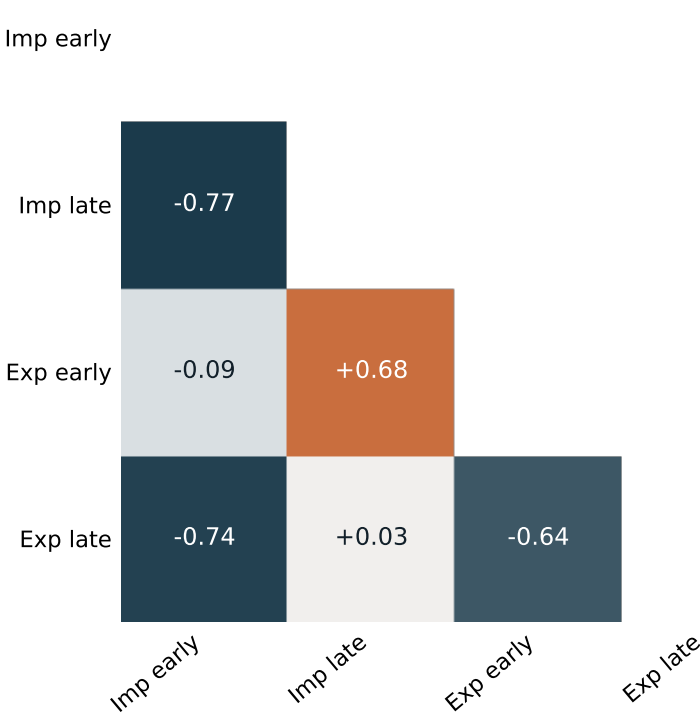
Global beta

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

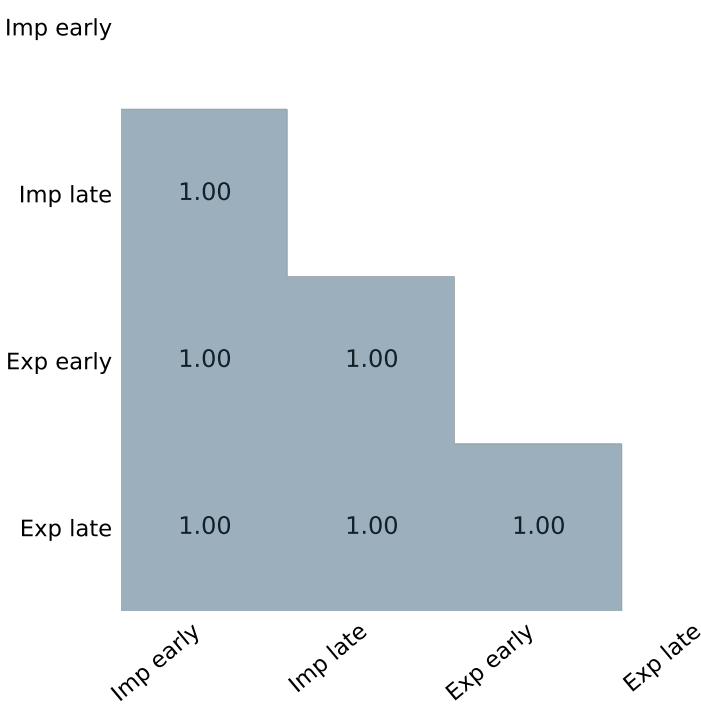
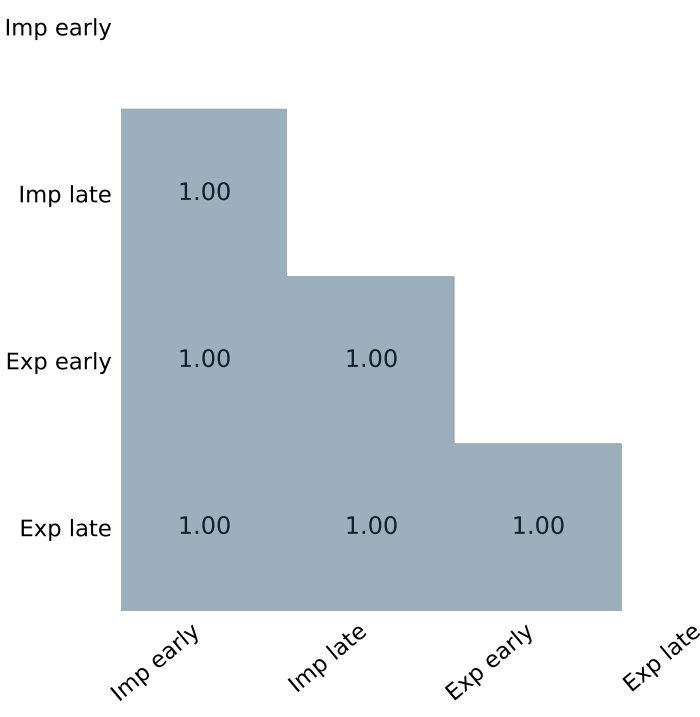
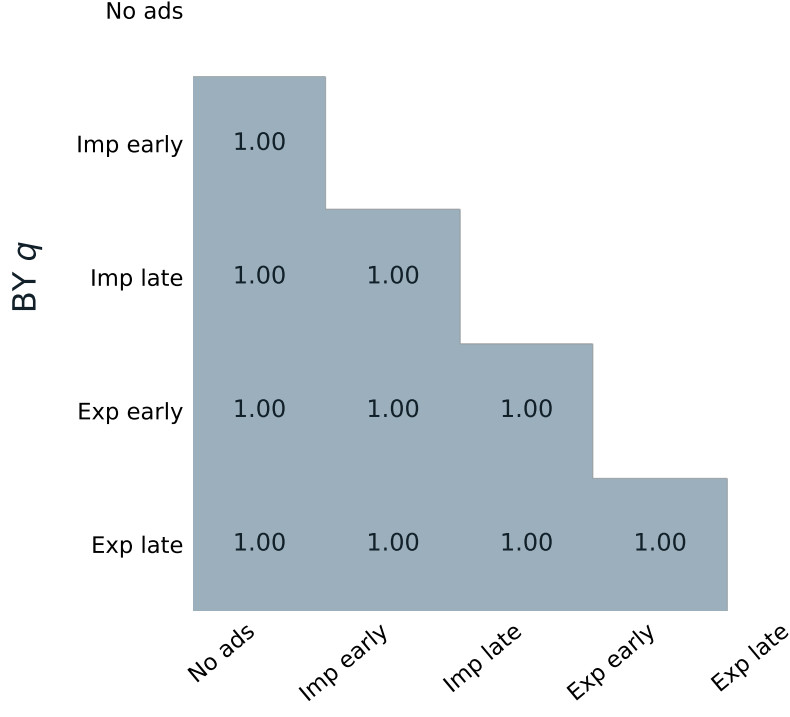
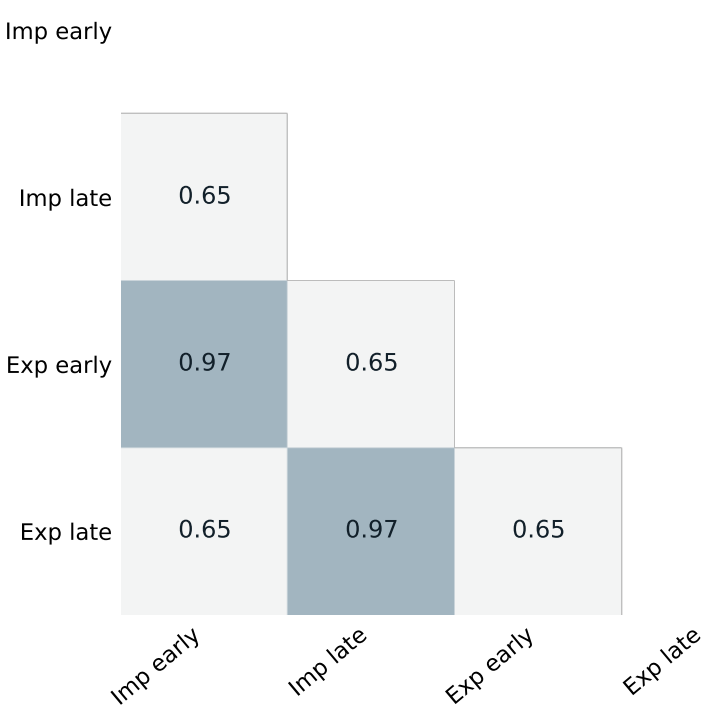
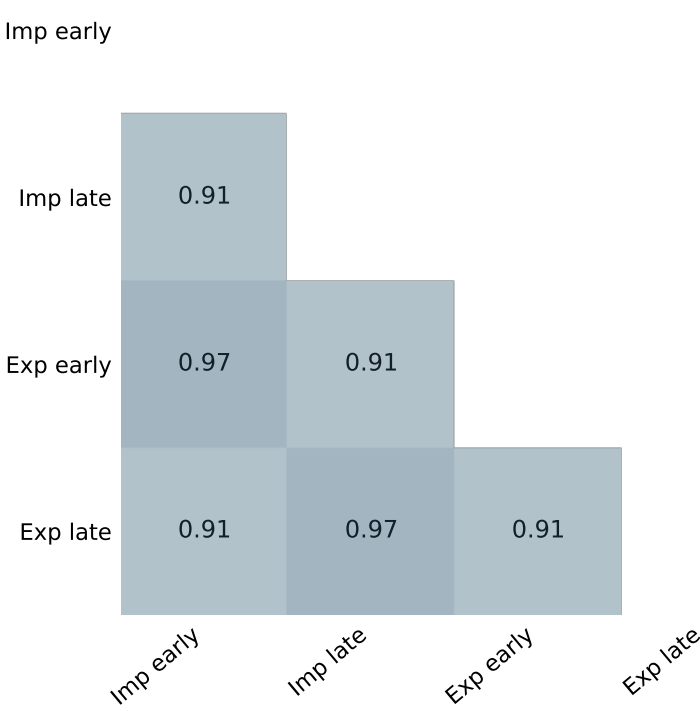
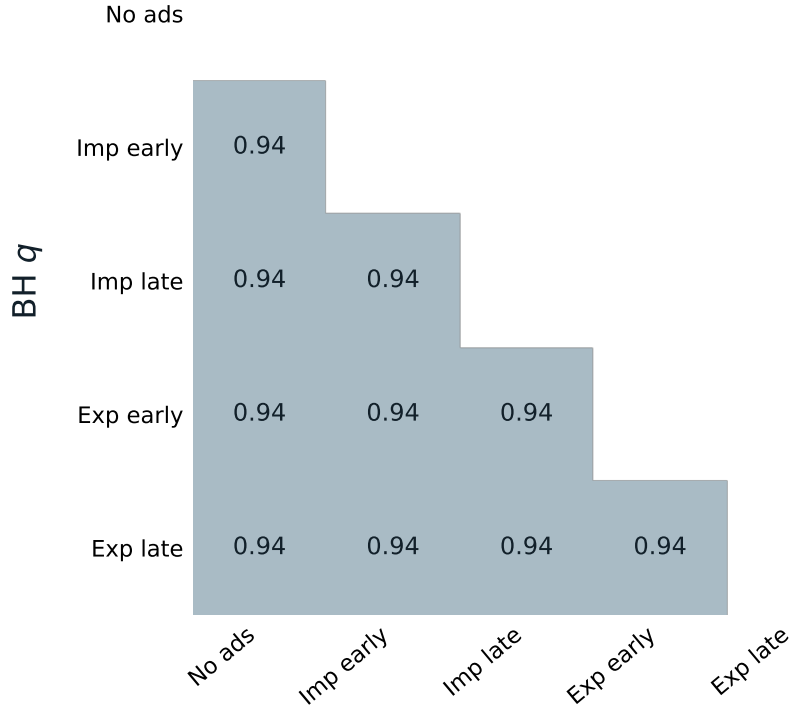
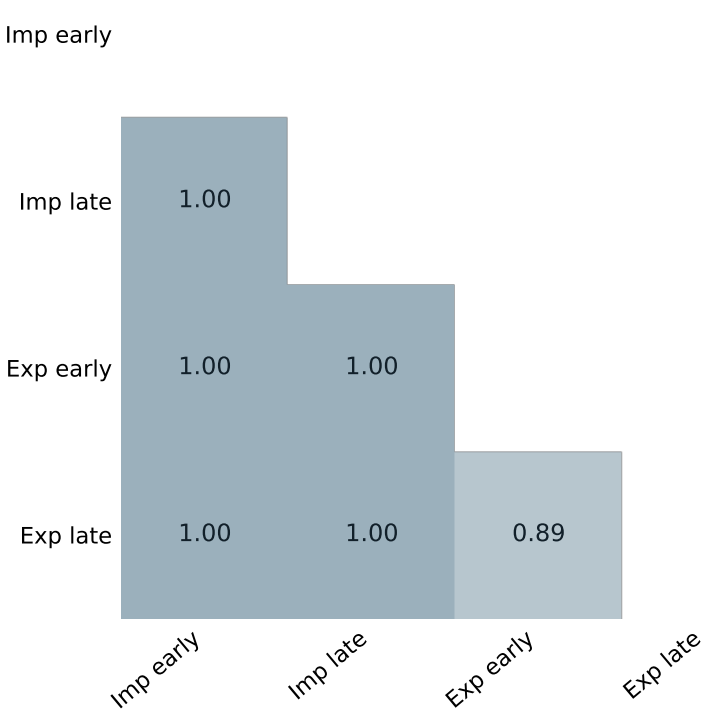
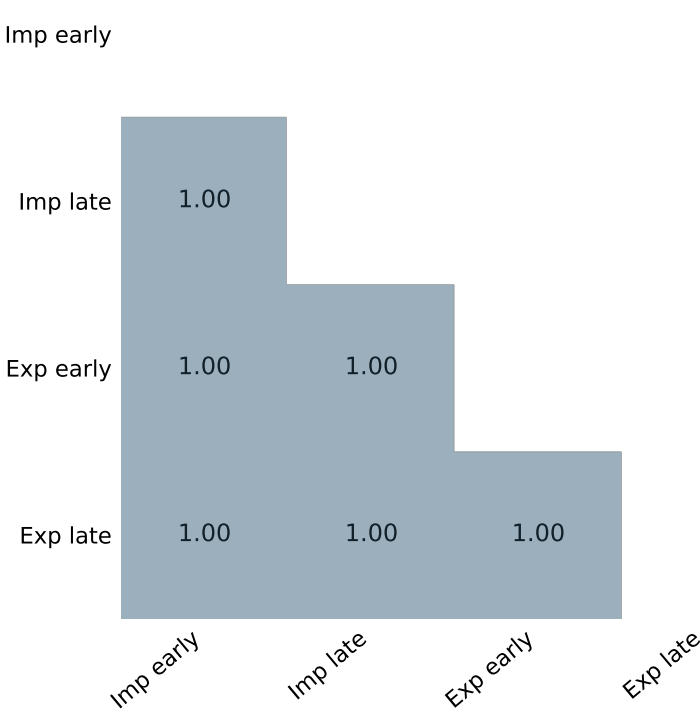
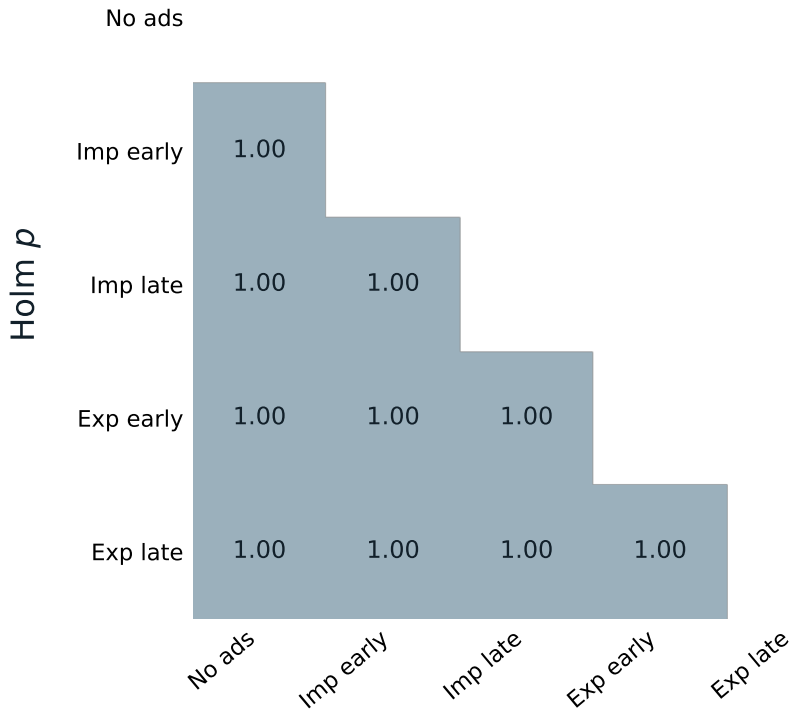
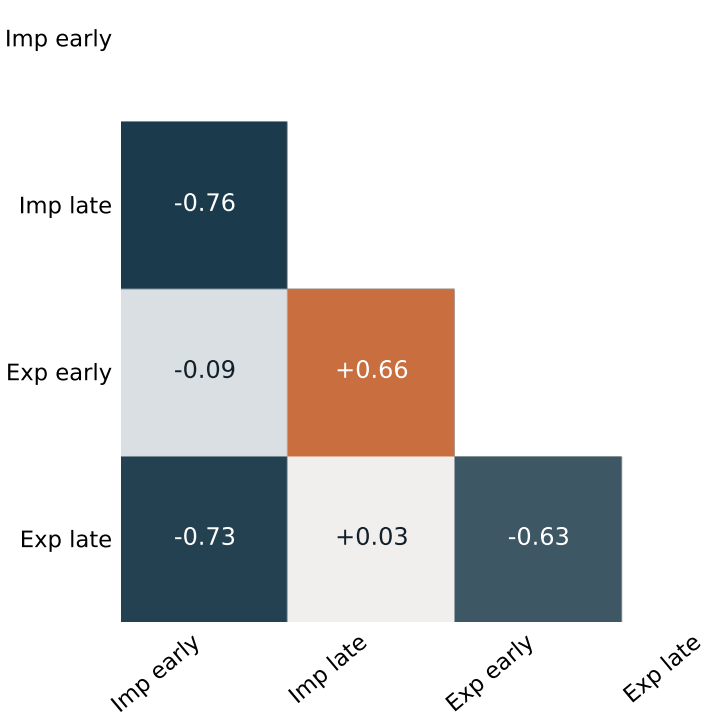
Dataset A (condition medians)



Dataset B, Δ matched control



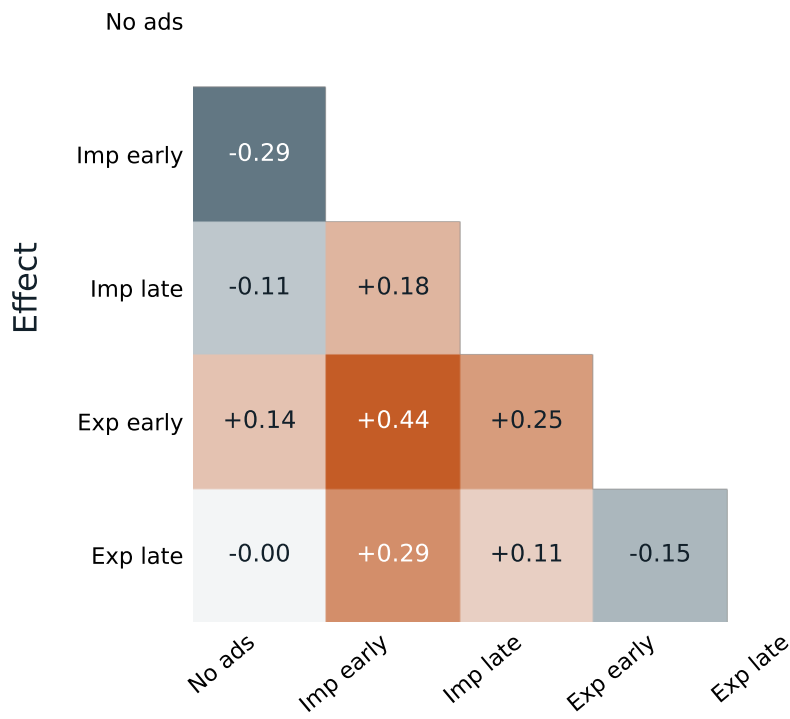
Dataset B, raw post–pre



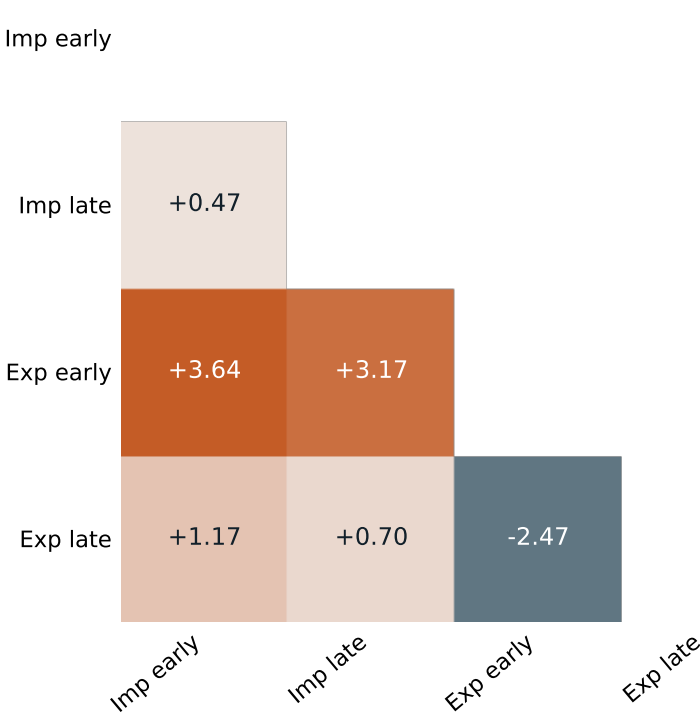
Global delta

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

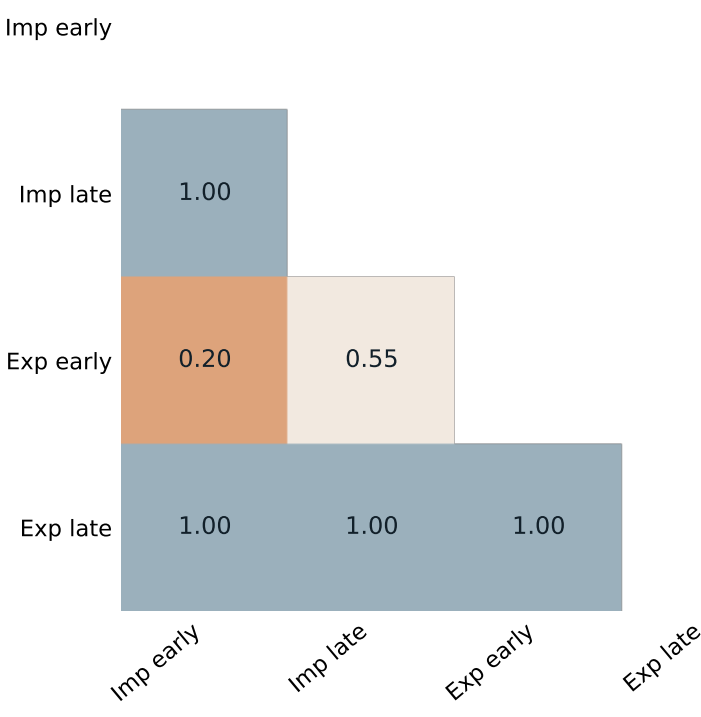
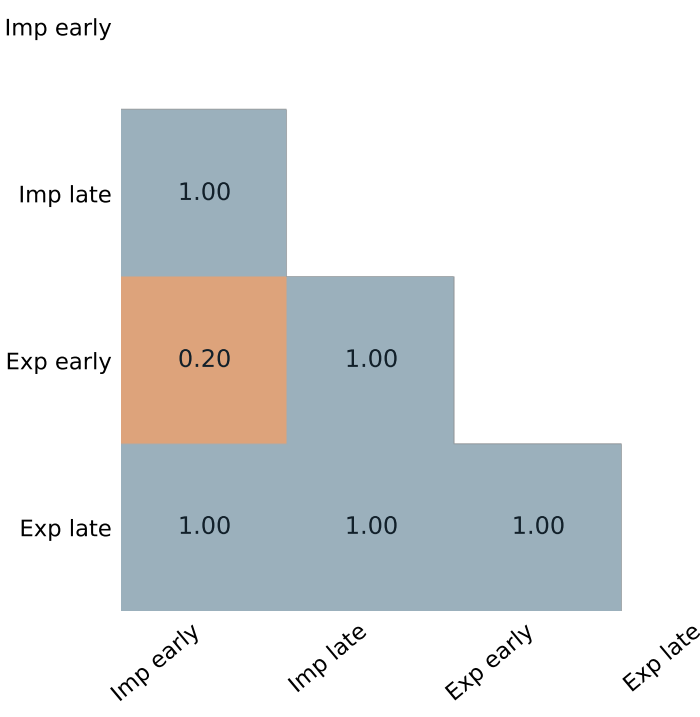
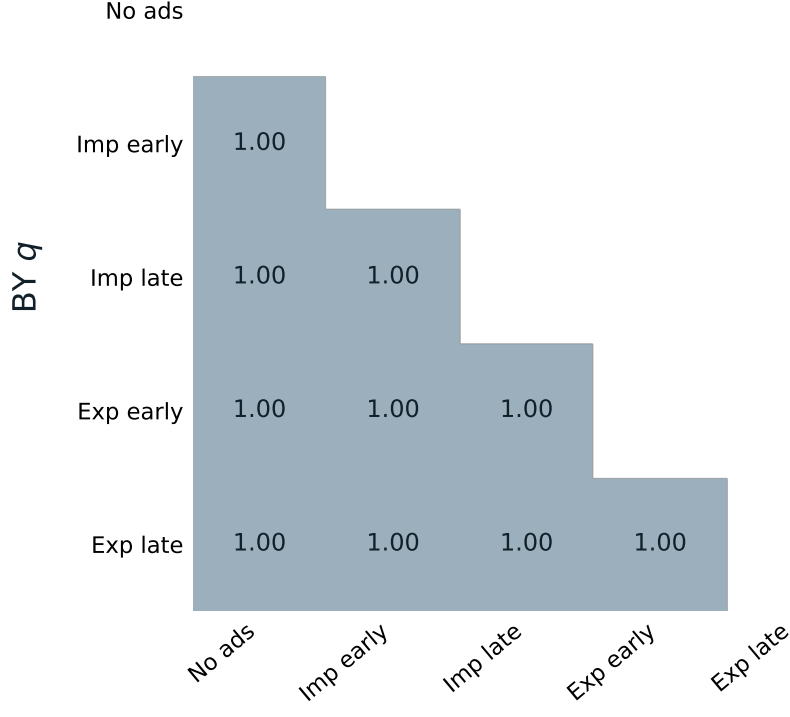
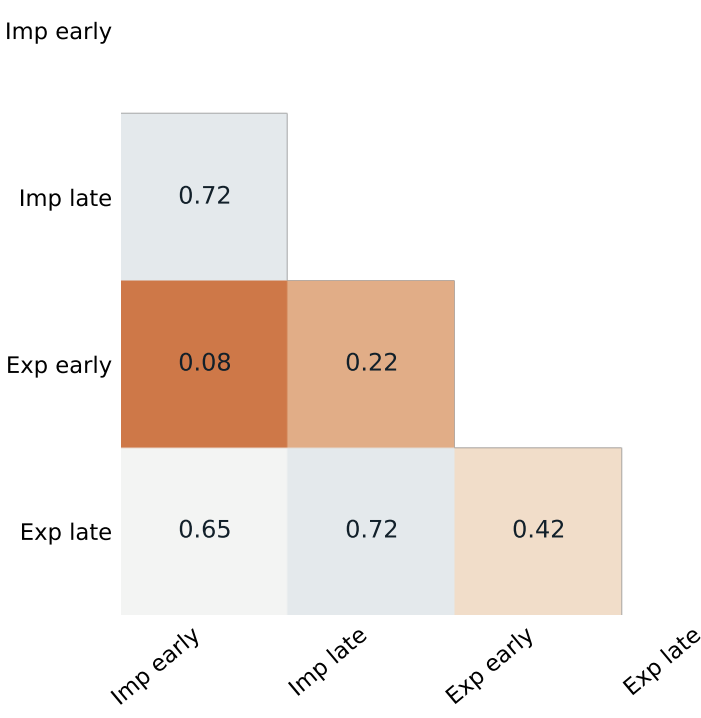
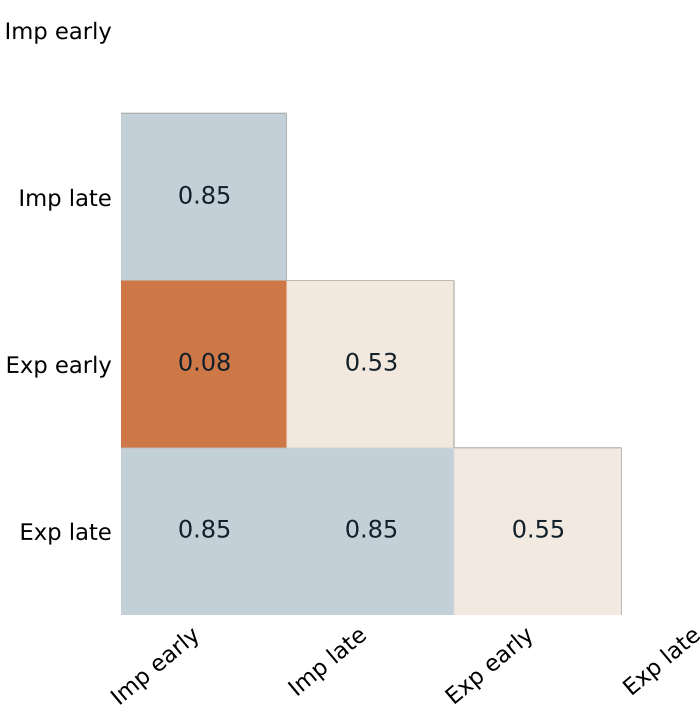
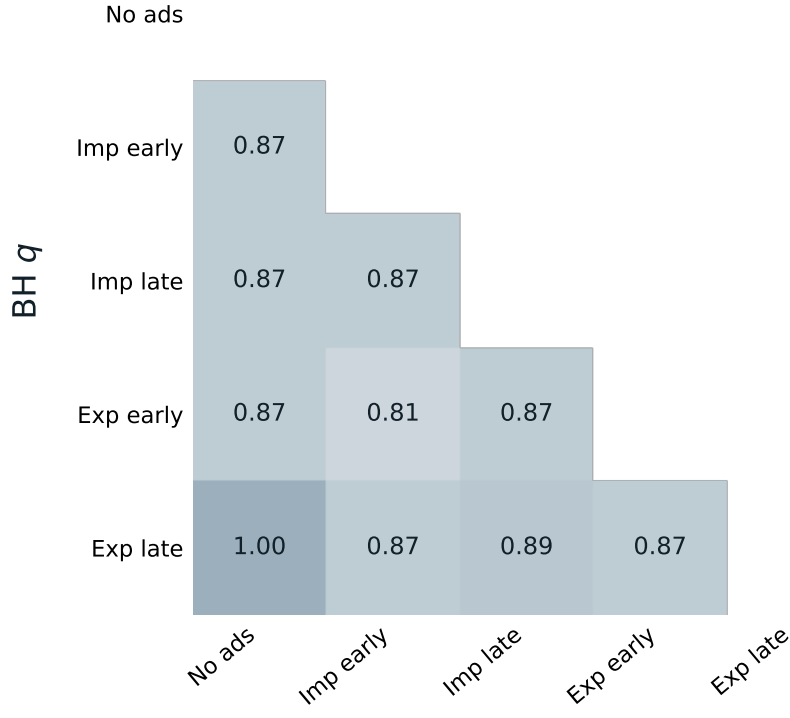
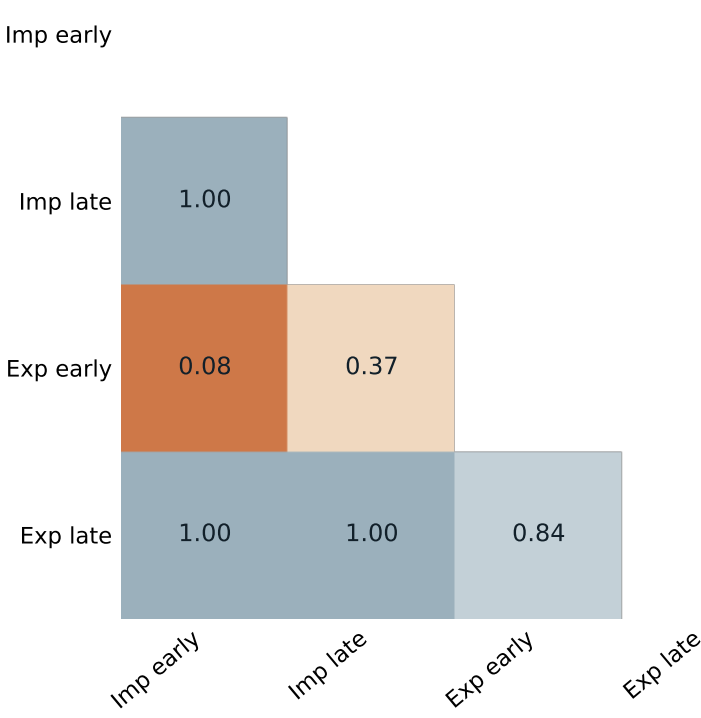
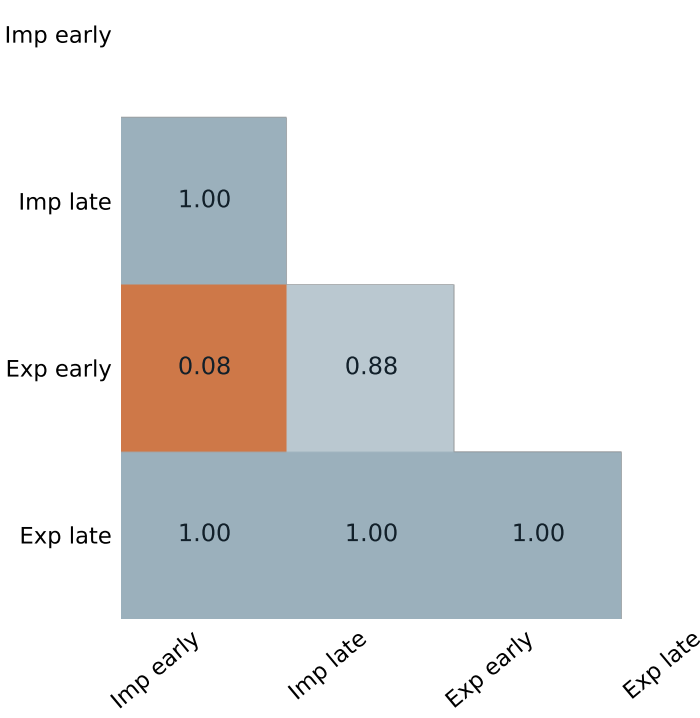
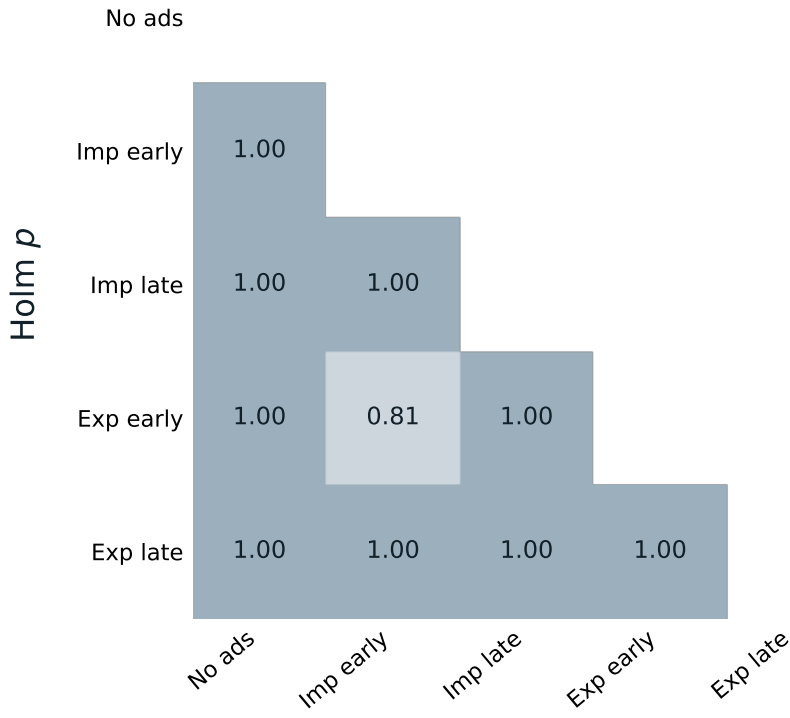
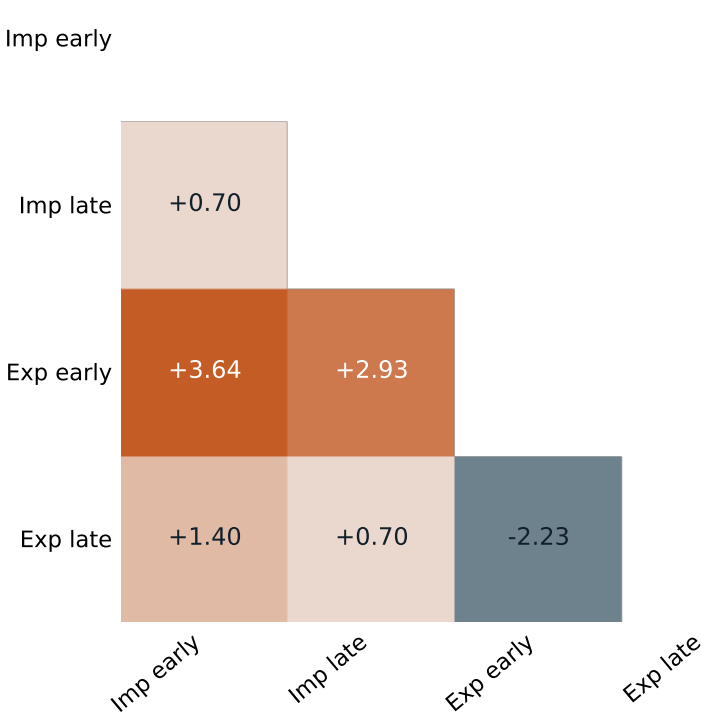
Dataset A (condition medians)



Dataset B, Δ matched control



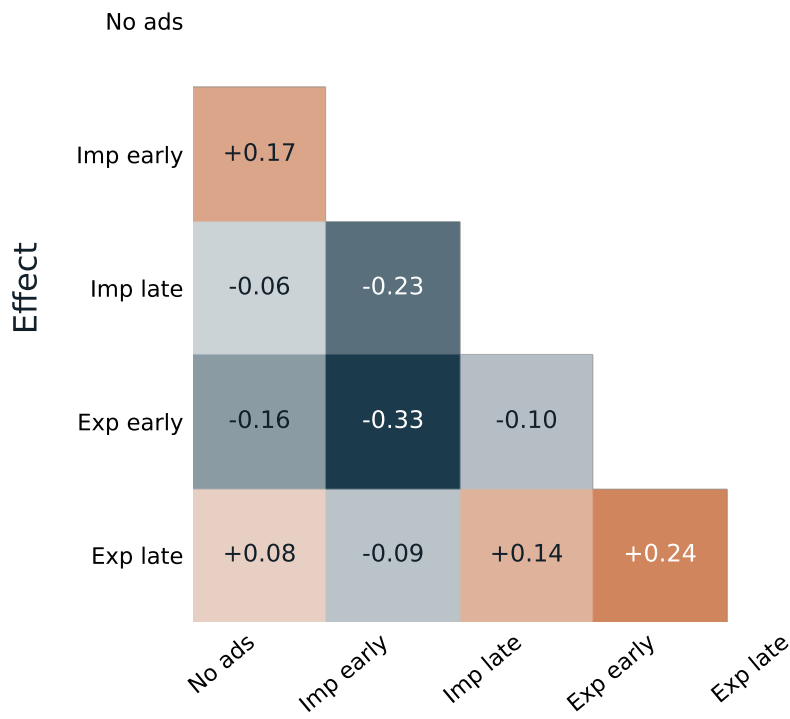
Dataset B, raw post–pre



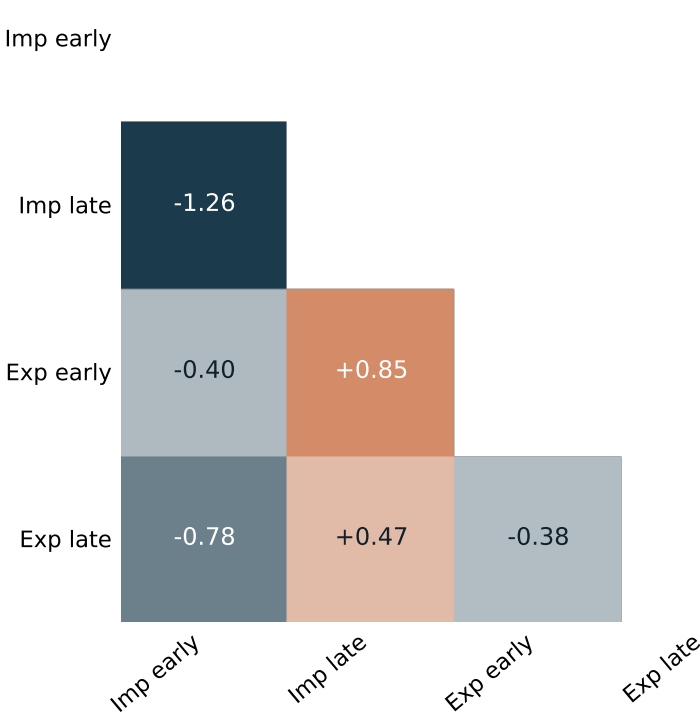
Global gamma

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

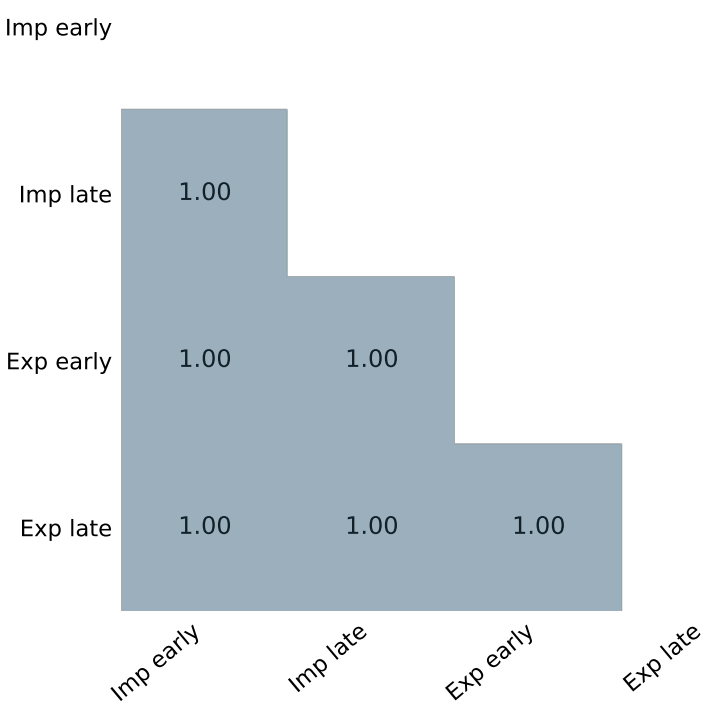
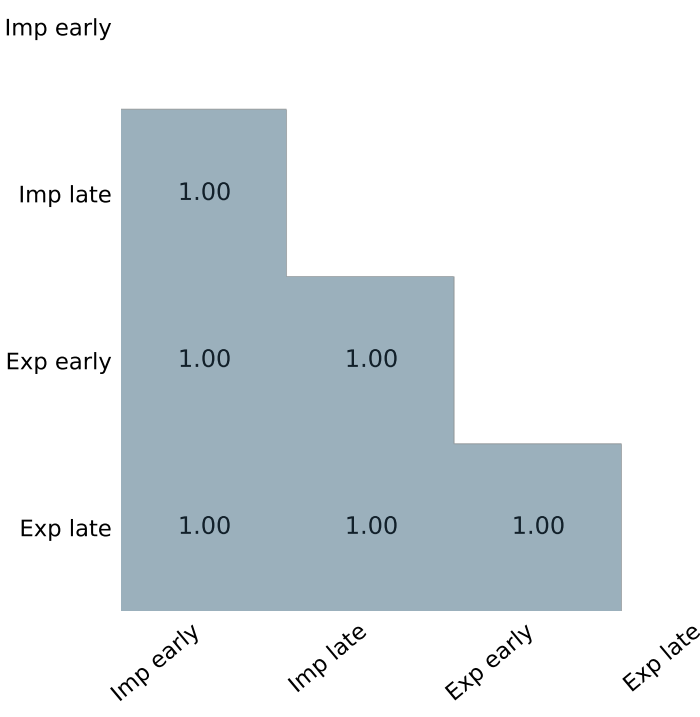
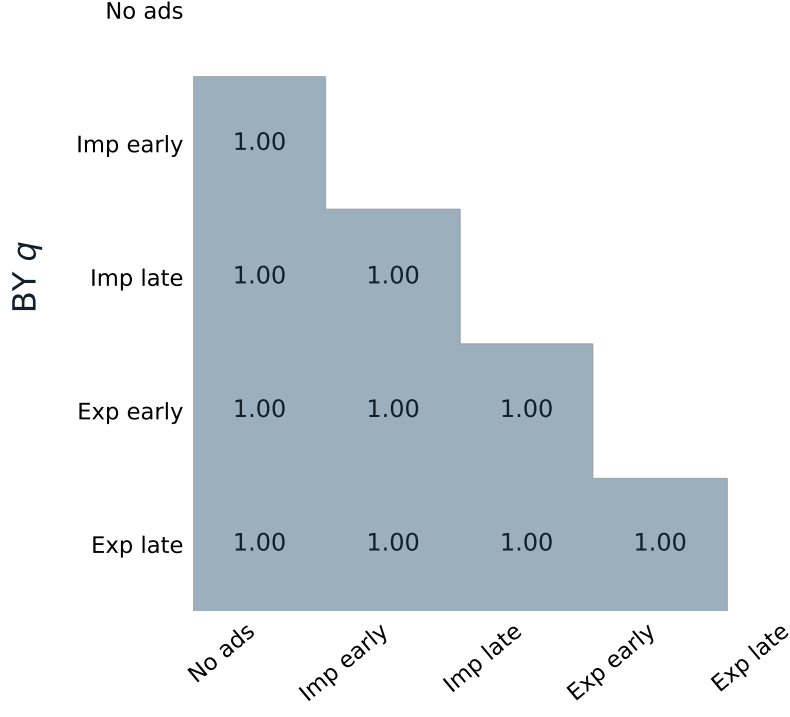
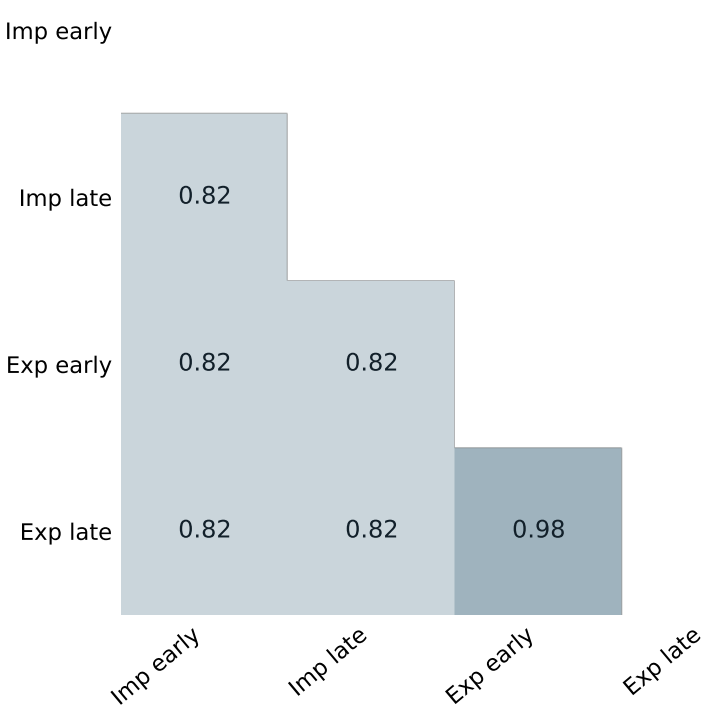
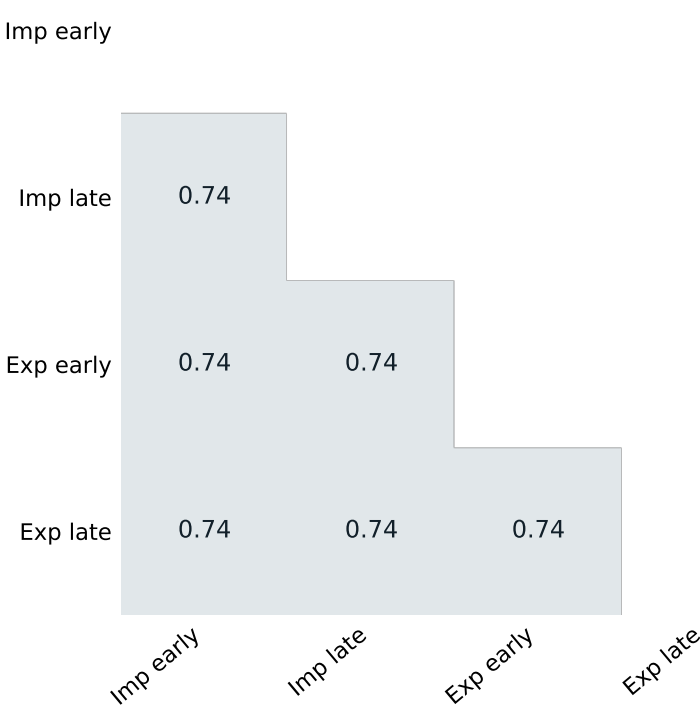
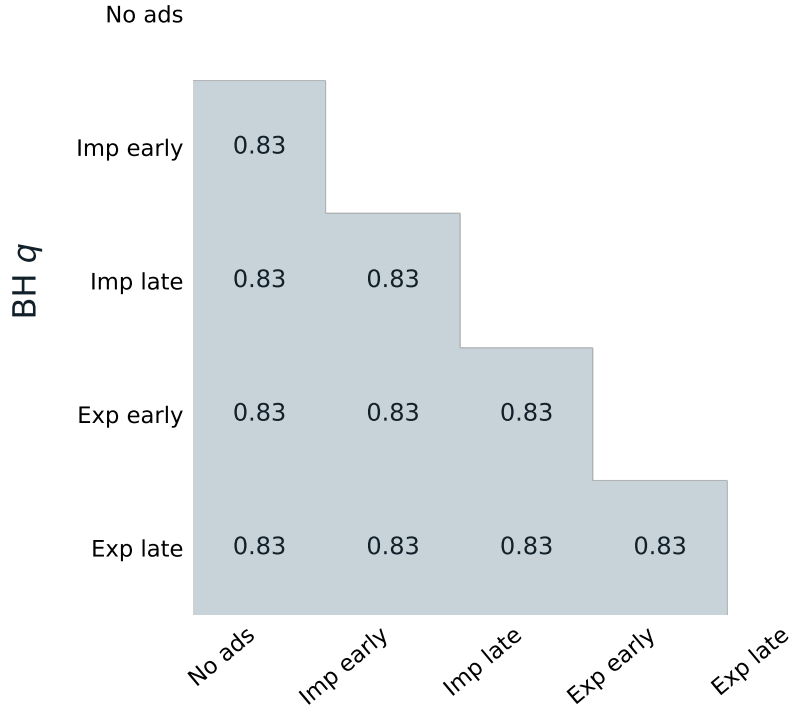
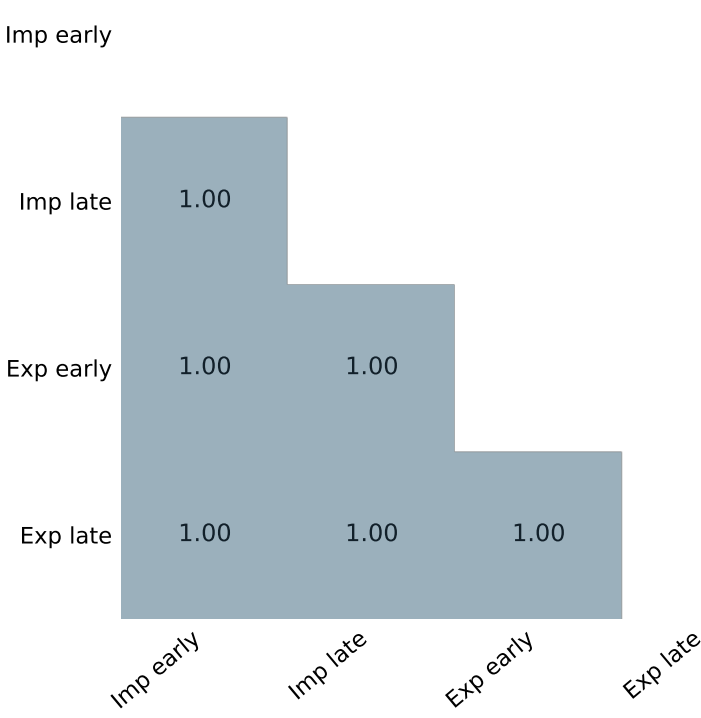
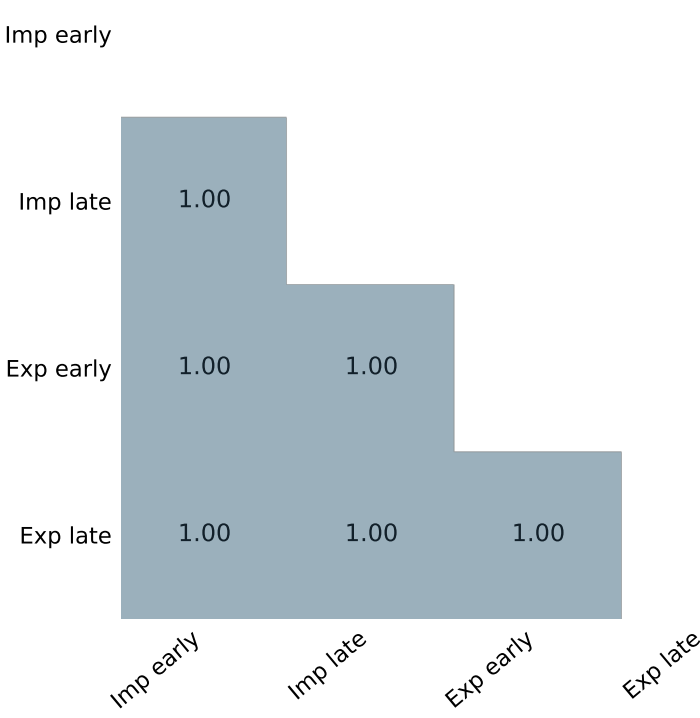
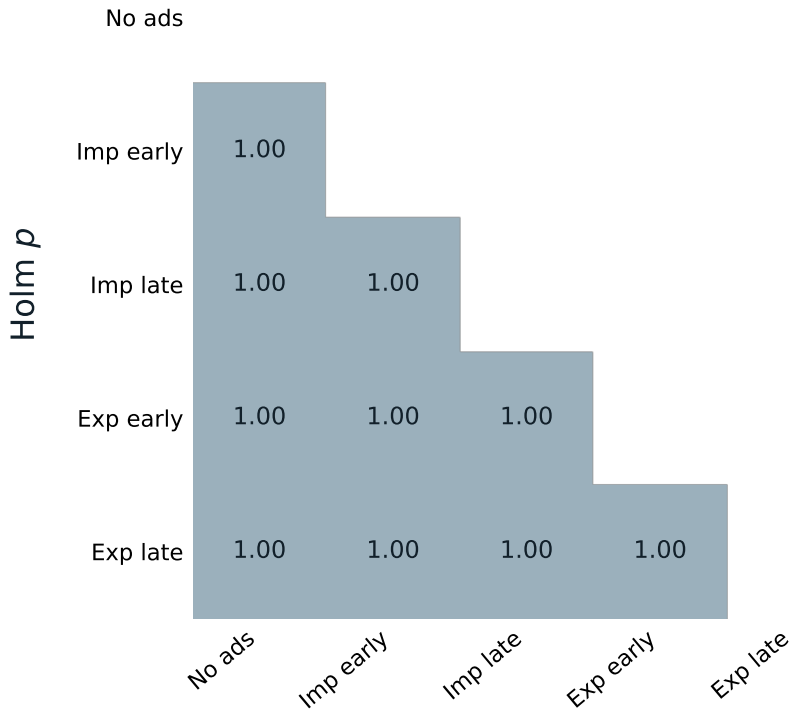
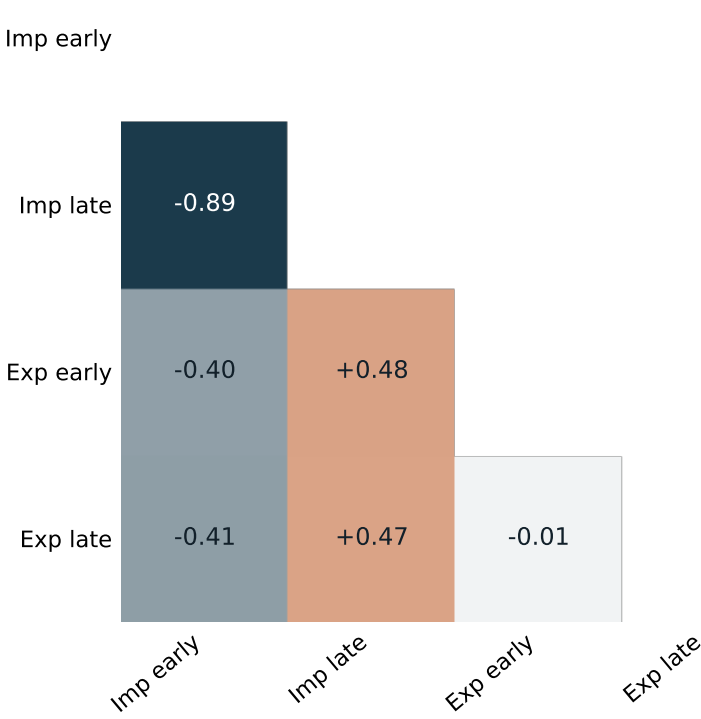
Dataset A (condition medians)



Dataset B, Δ matched control



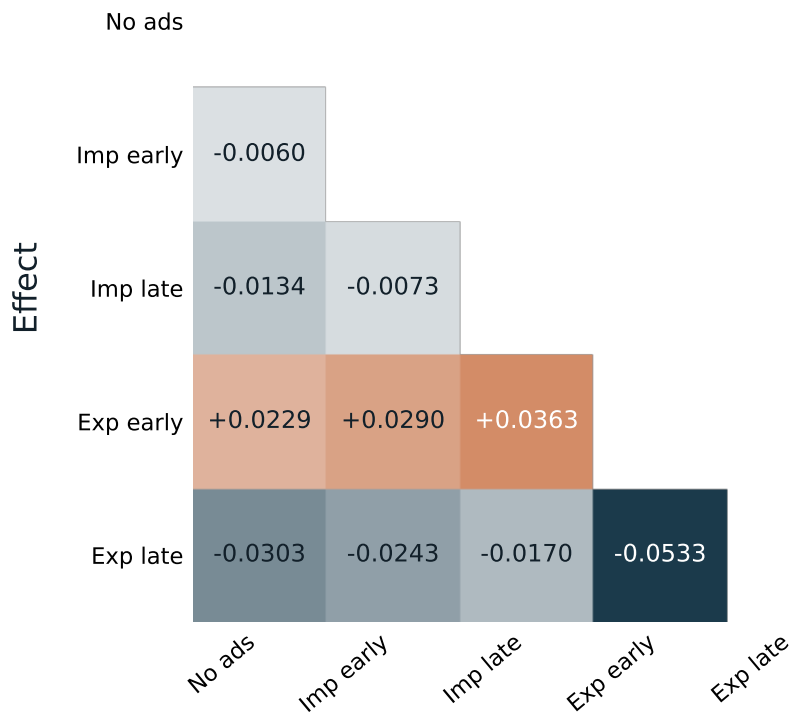
Dataset B, raw post–pre



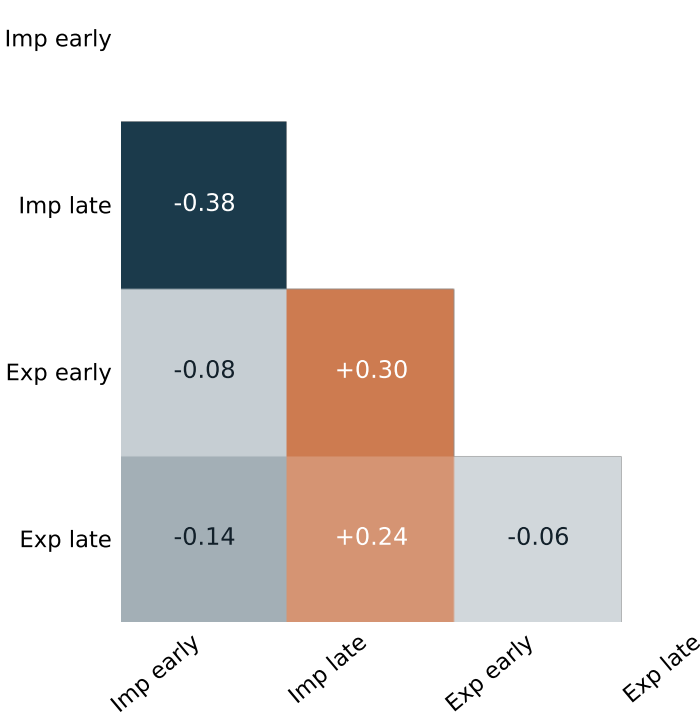
Frontal alpha asym.

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

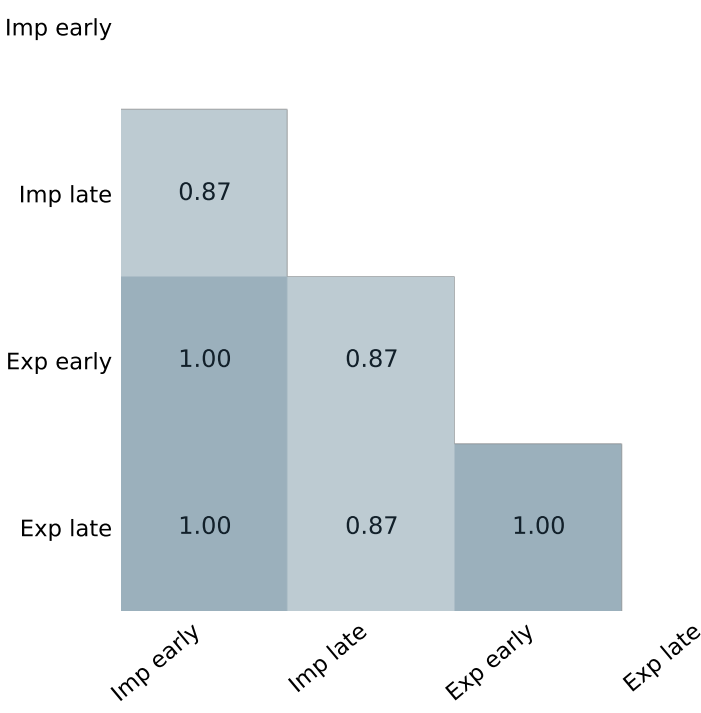
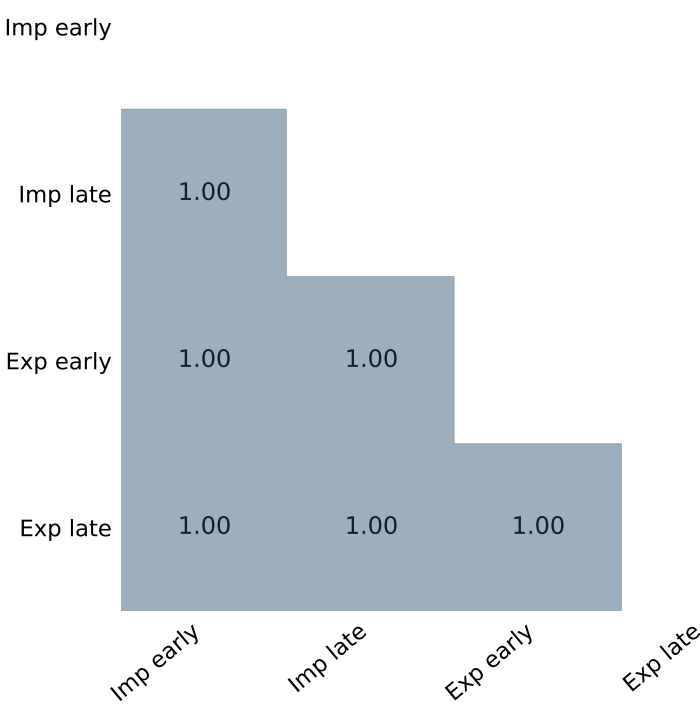
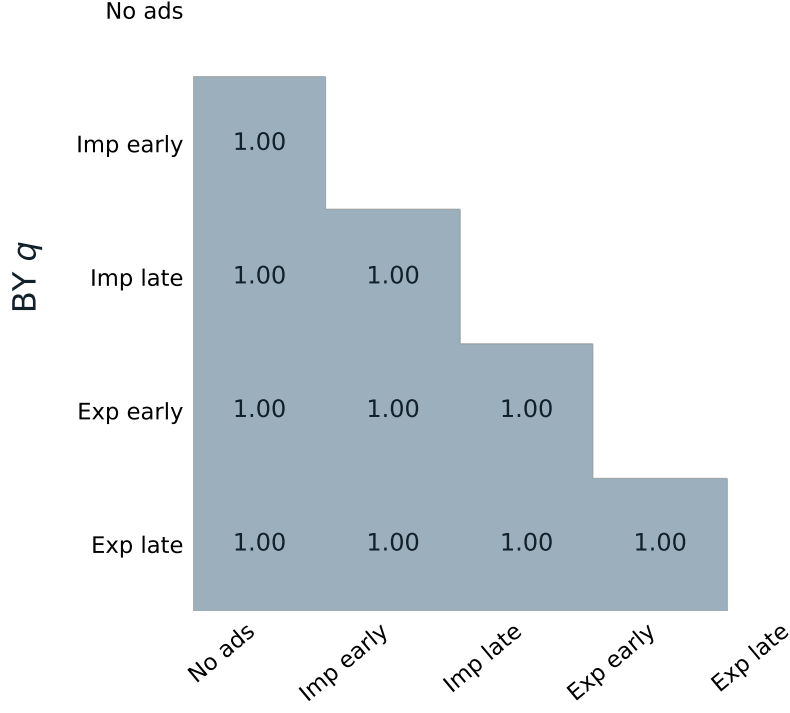
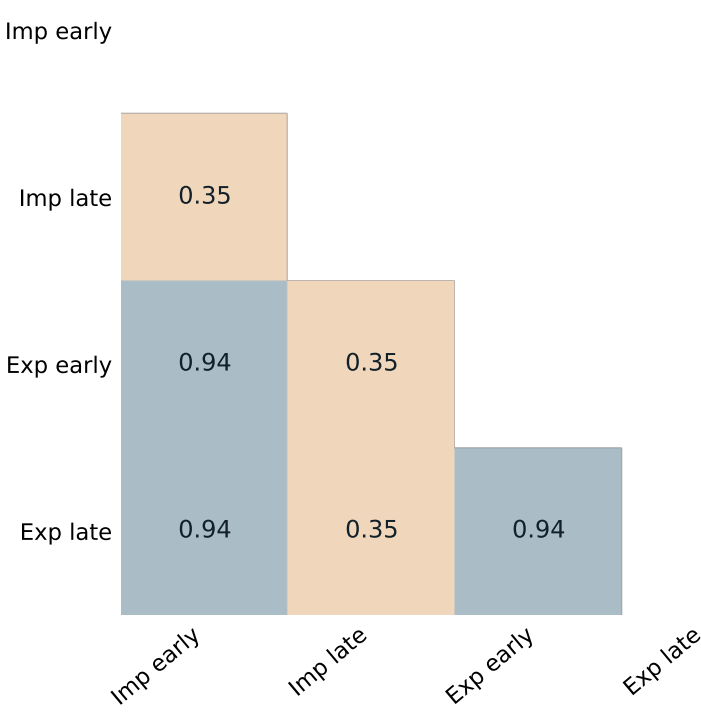
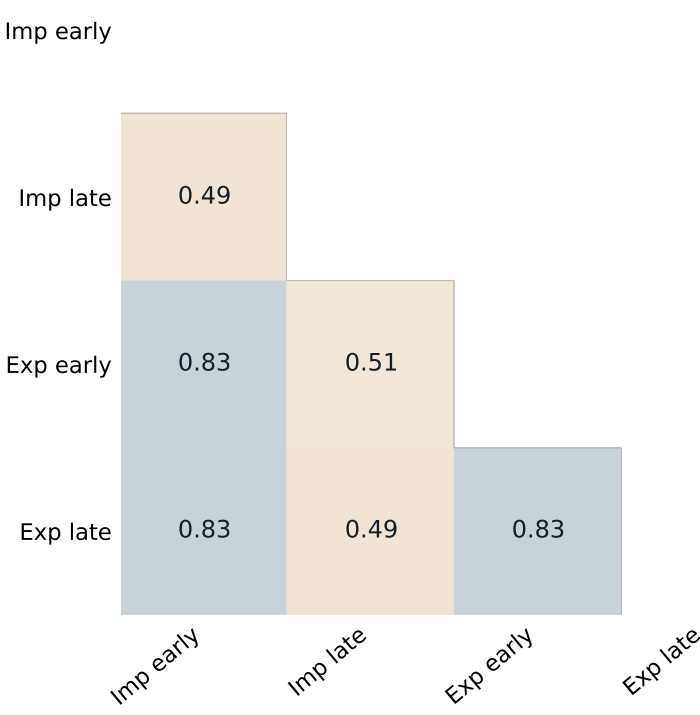
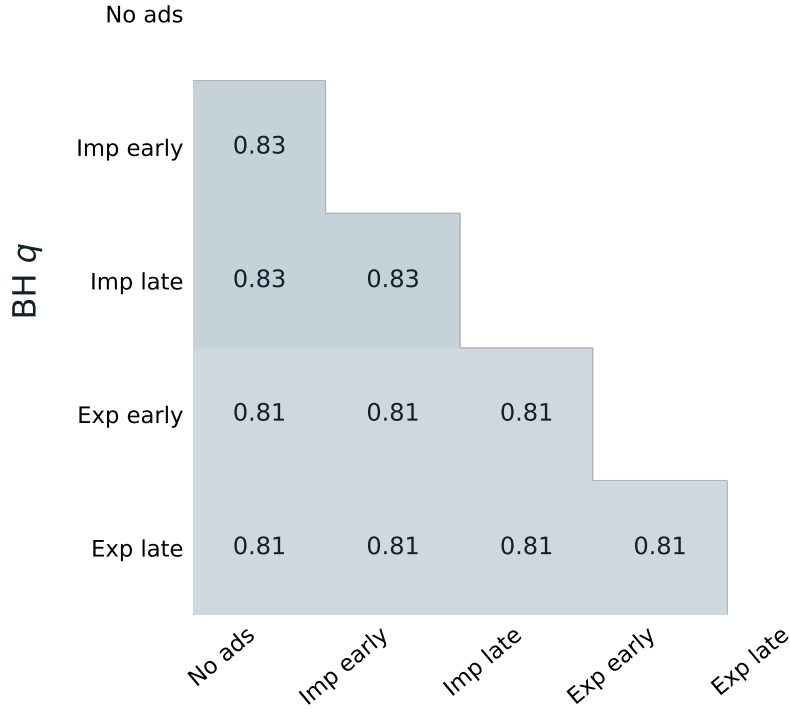
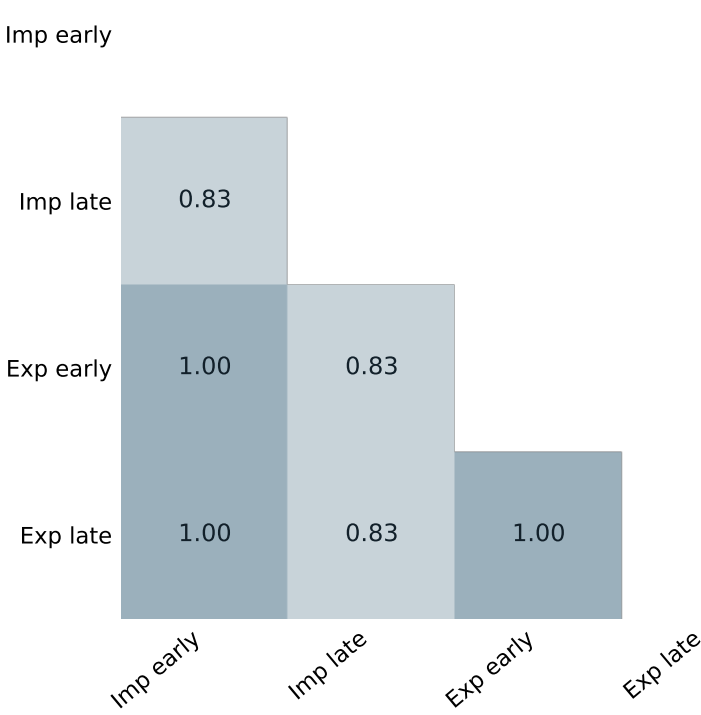
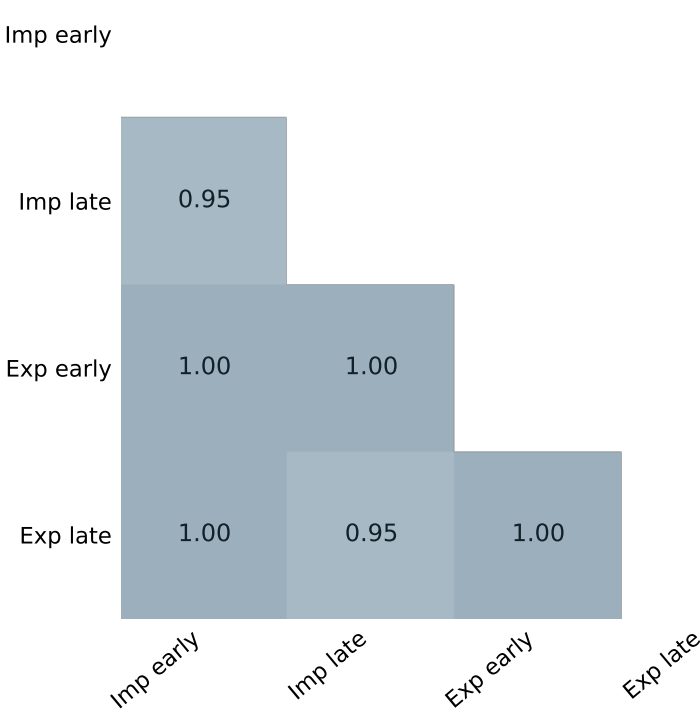
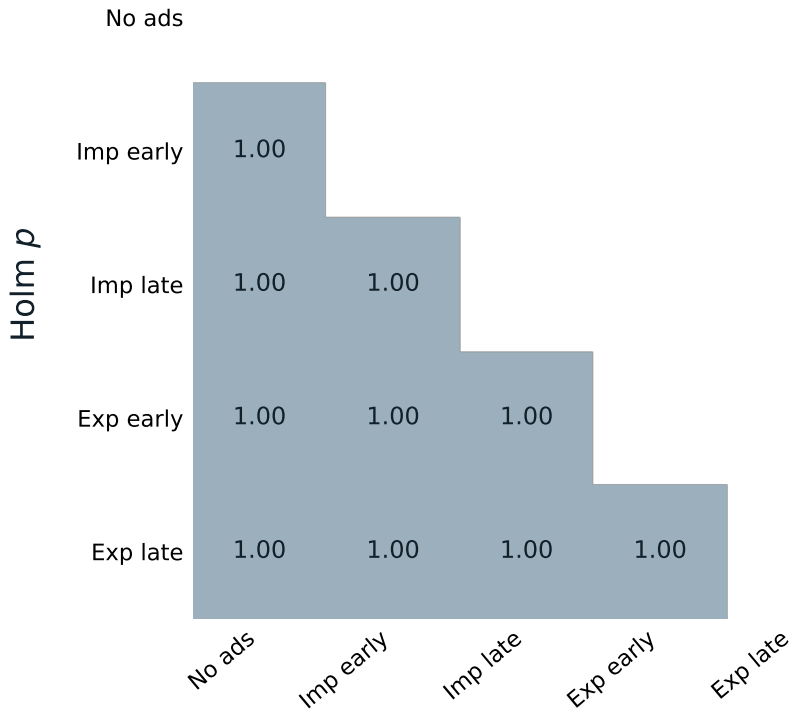
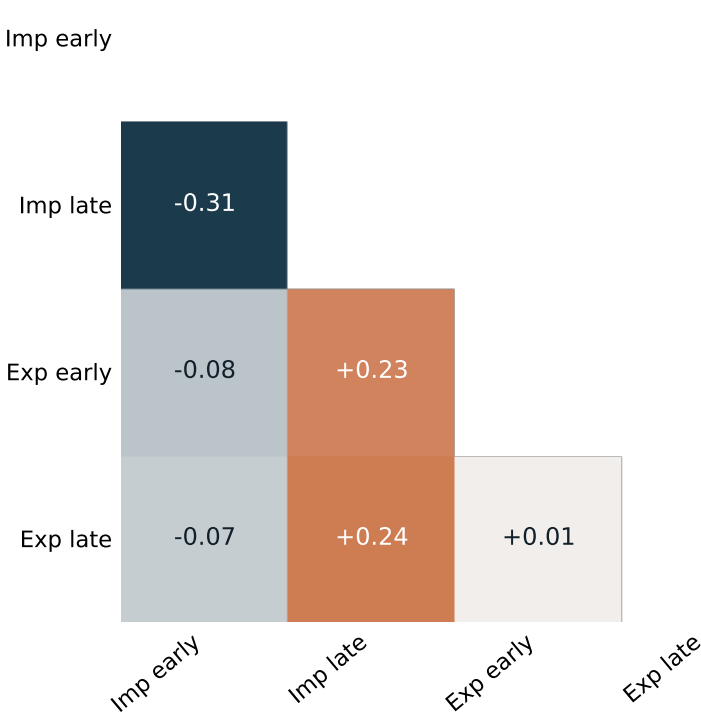
Dataset A (condition medians)



Dataset B, Δ matched control



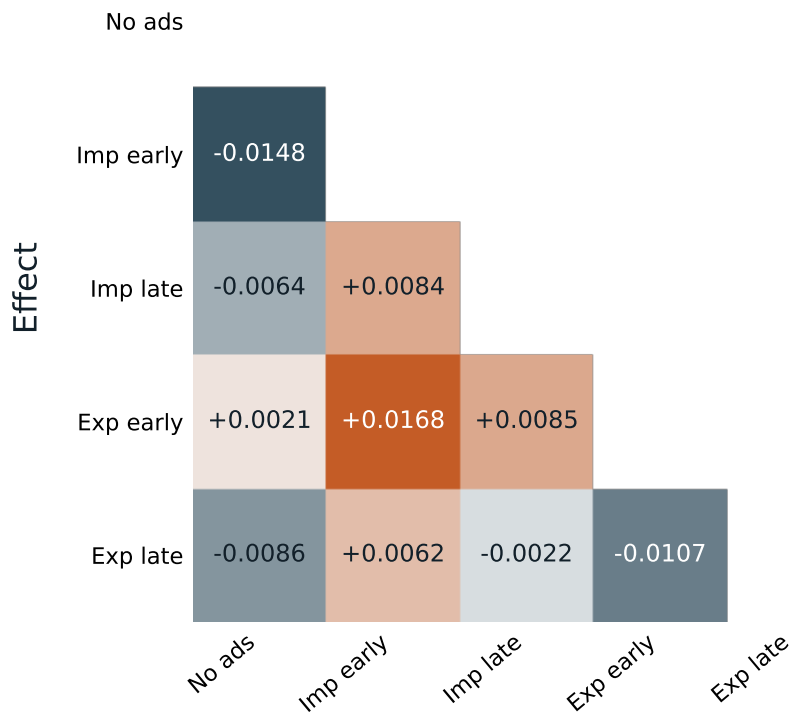
Dataset B, raw post–pre



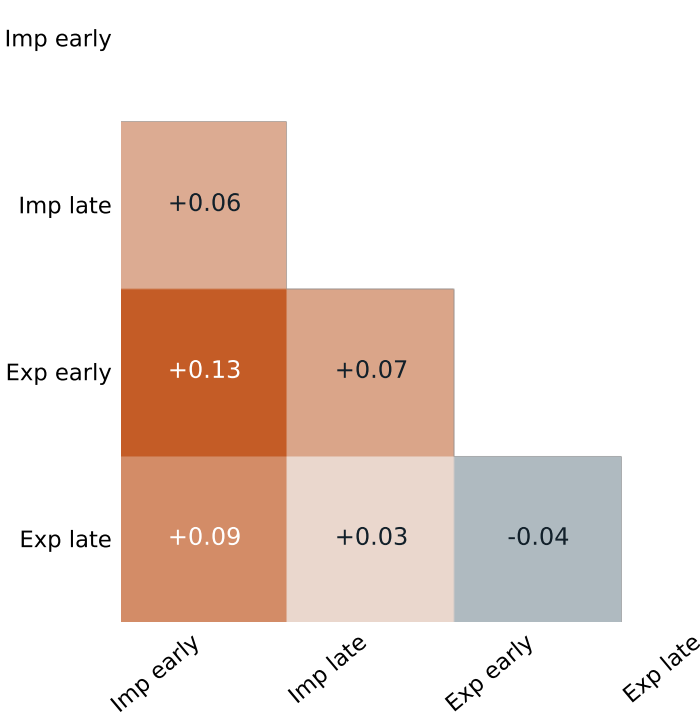
Rel. delta

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

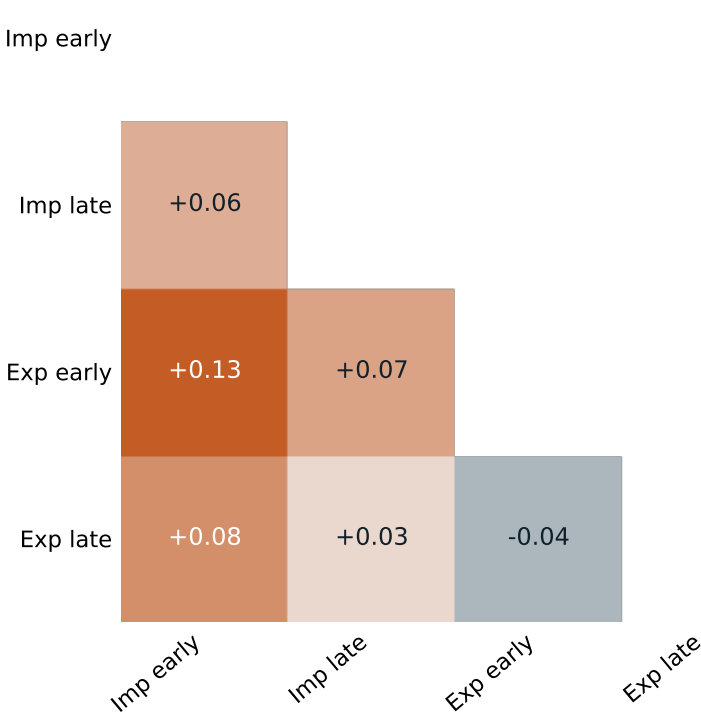
Dataset A (condition medians)



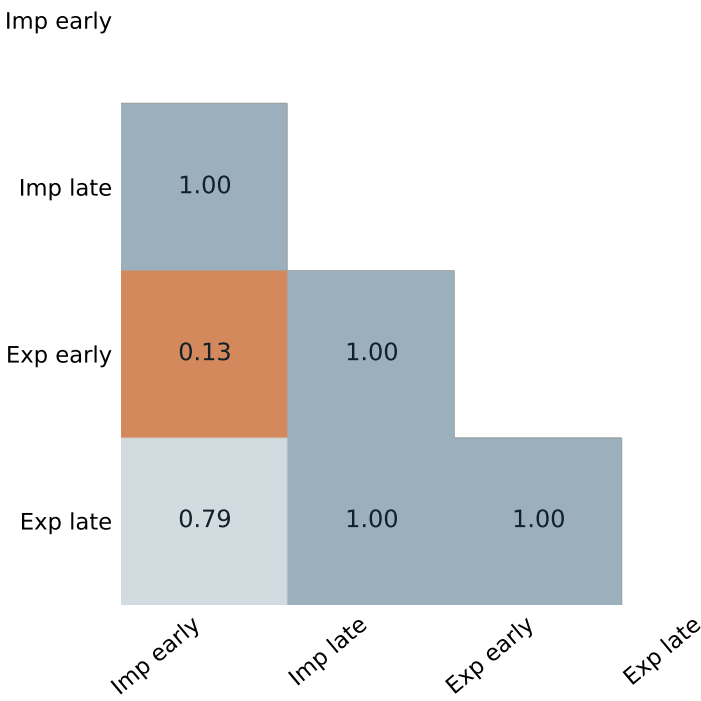
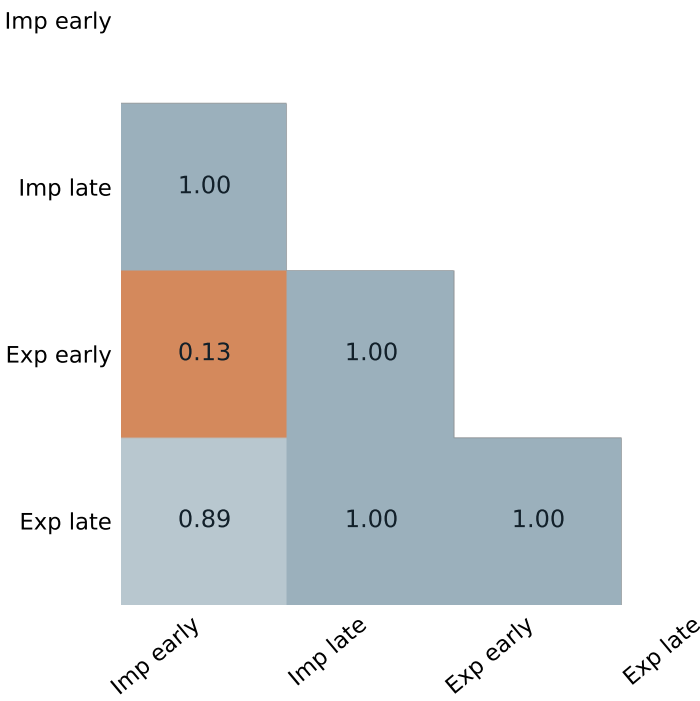
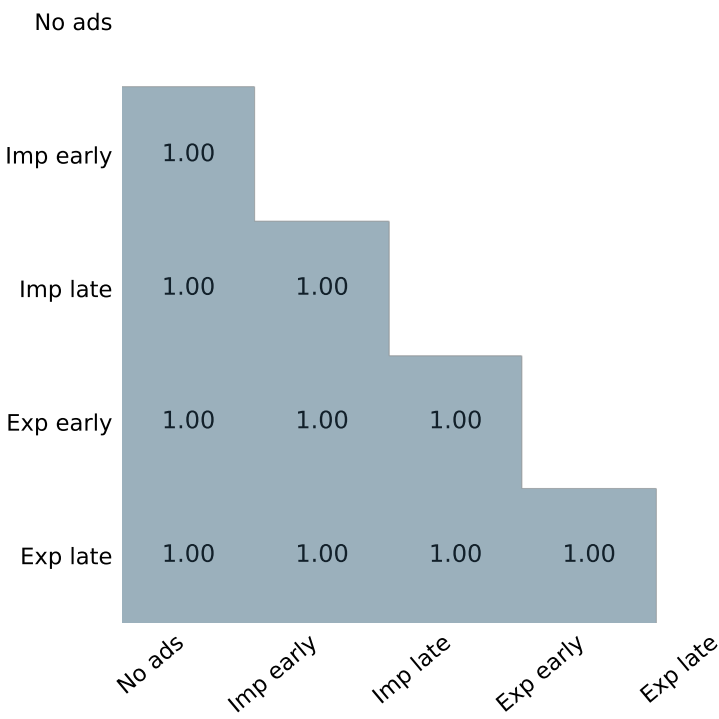
Dataset B, Δ matched control



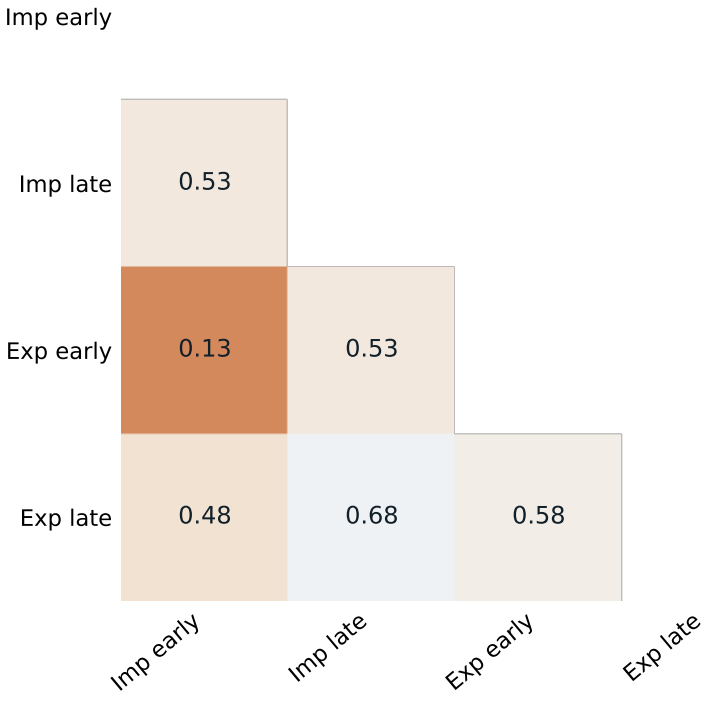
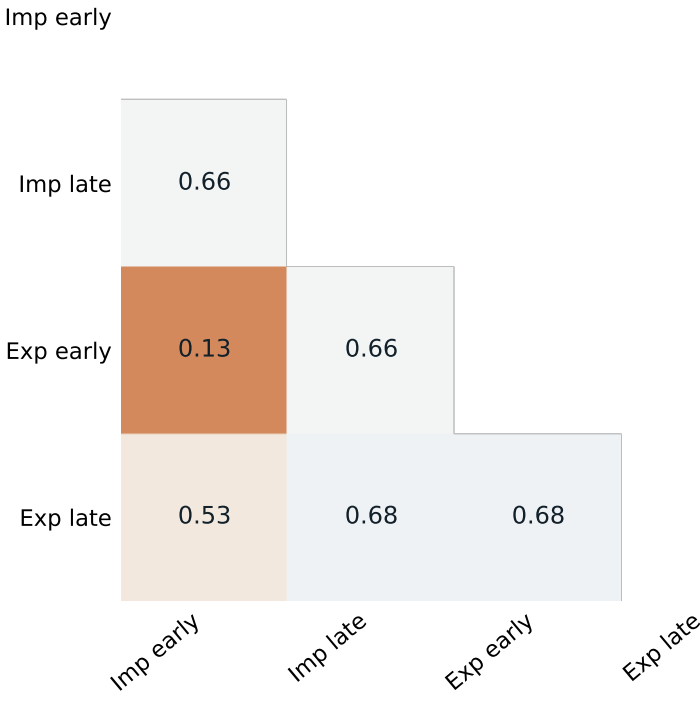
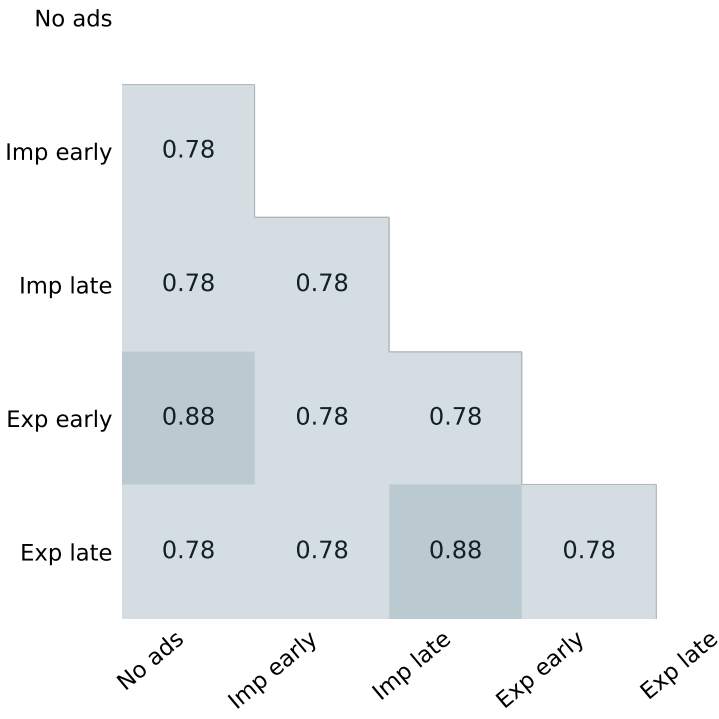
Dataset B, raw post-pre



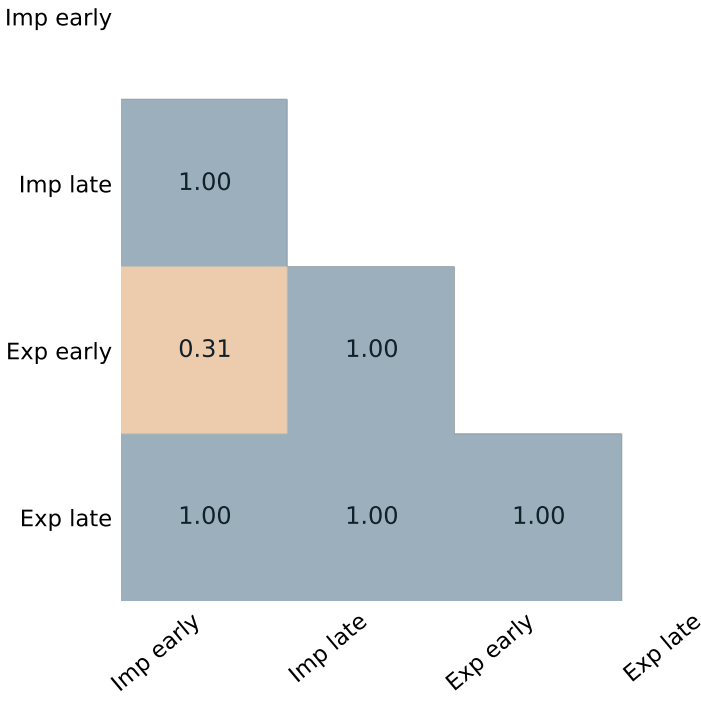
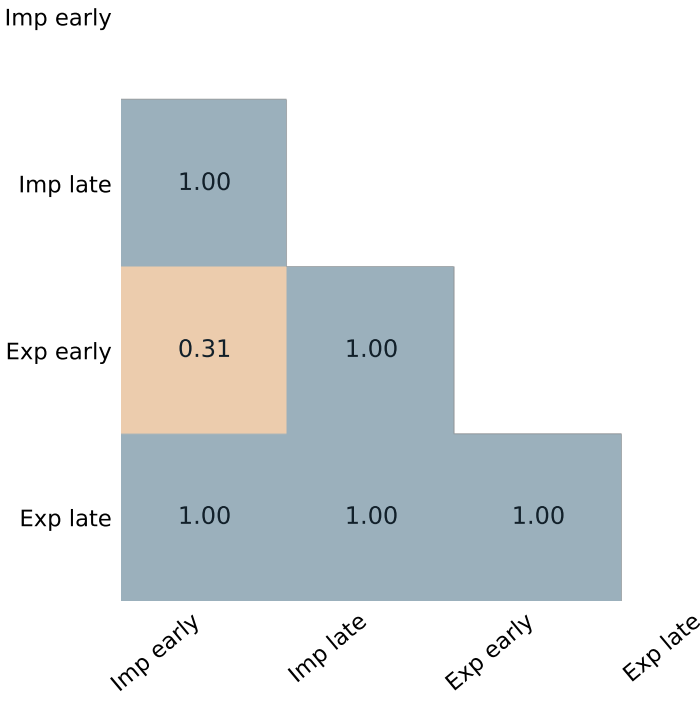
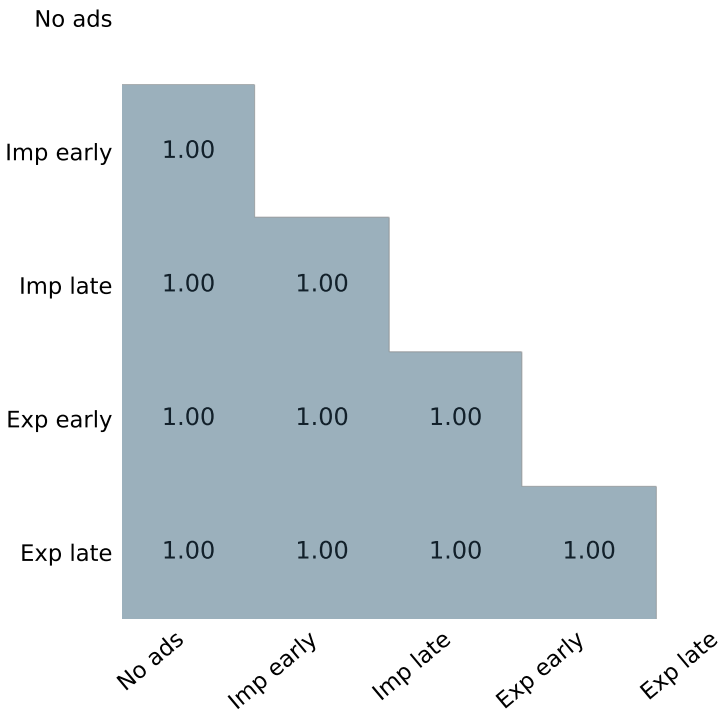
Holm p



BH q



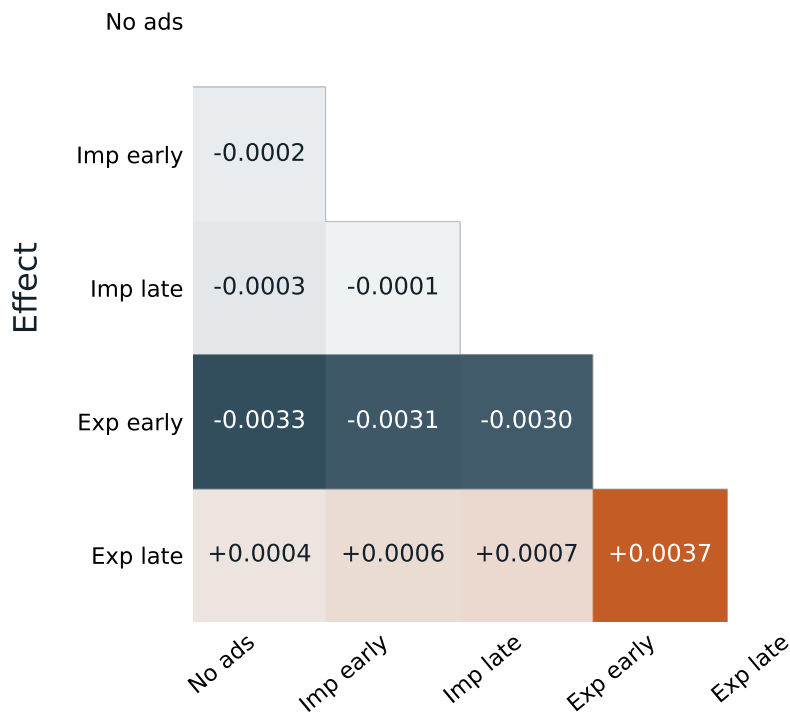
BY q



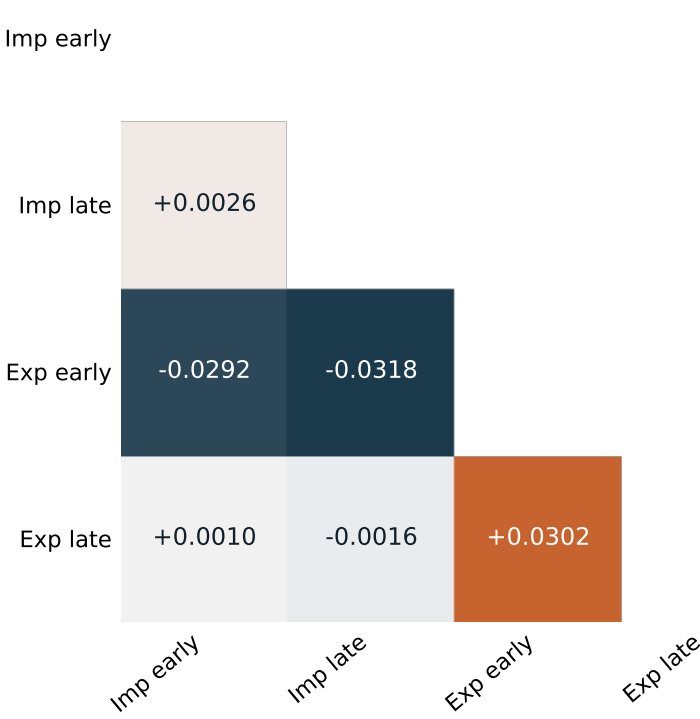
Rel. theta

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

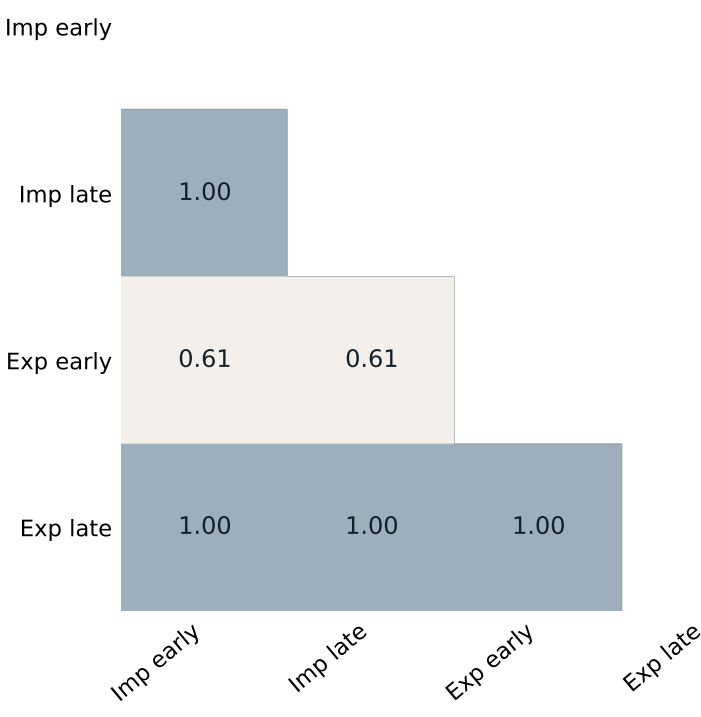
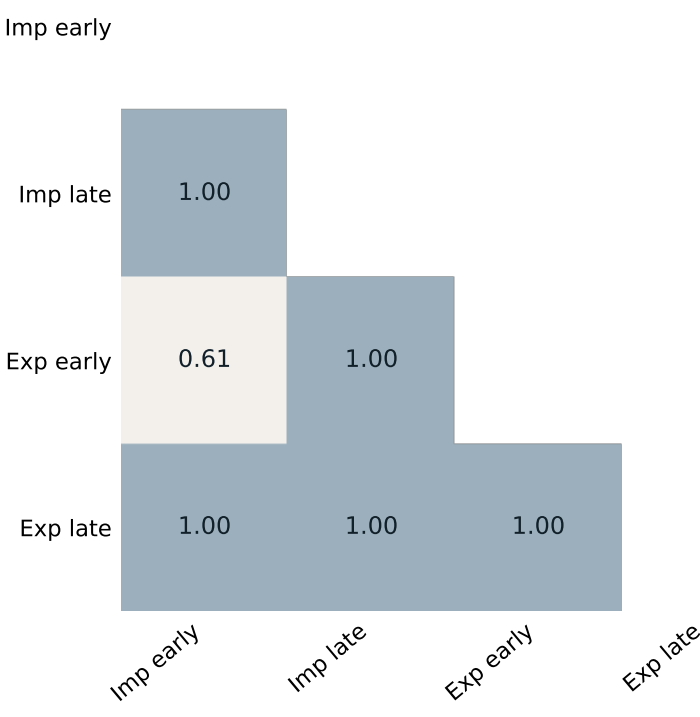
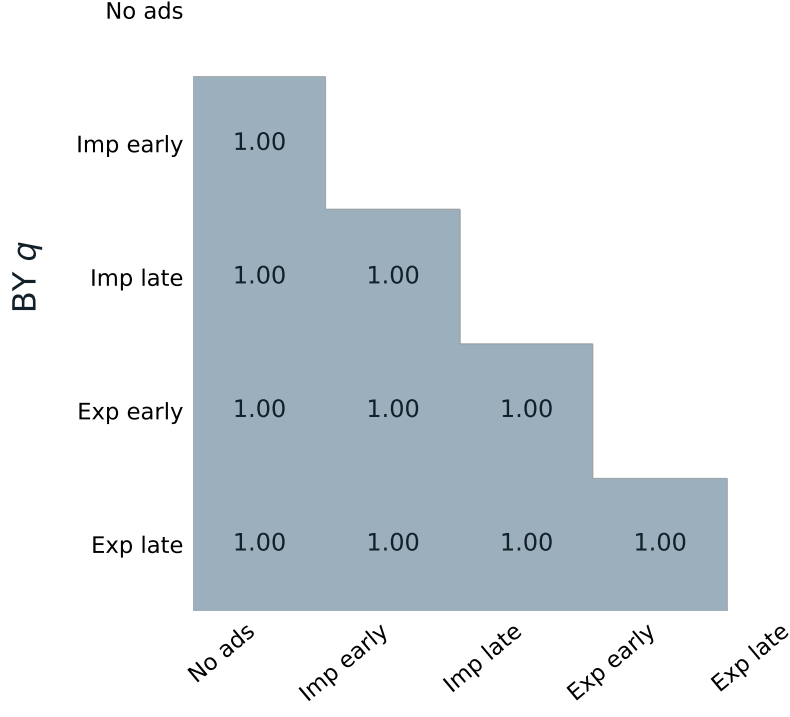
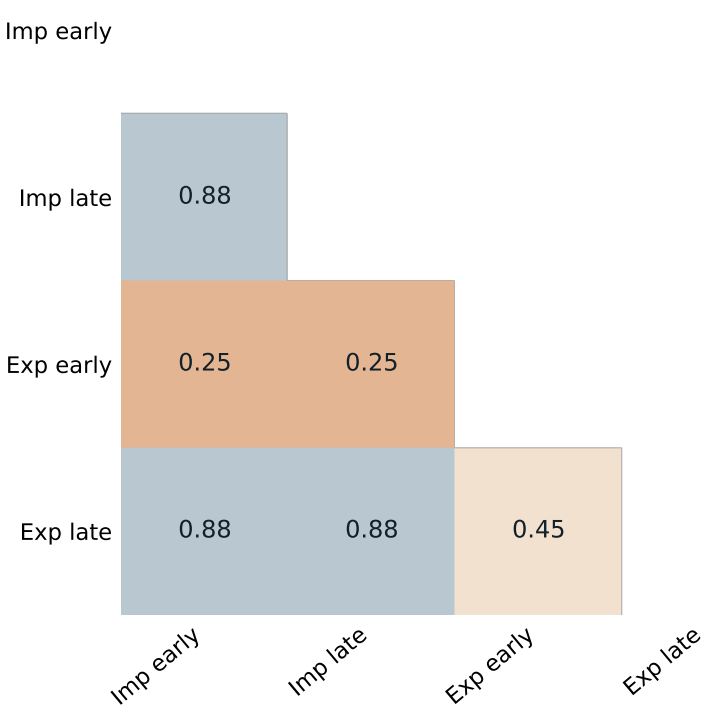
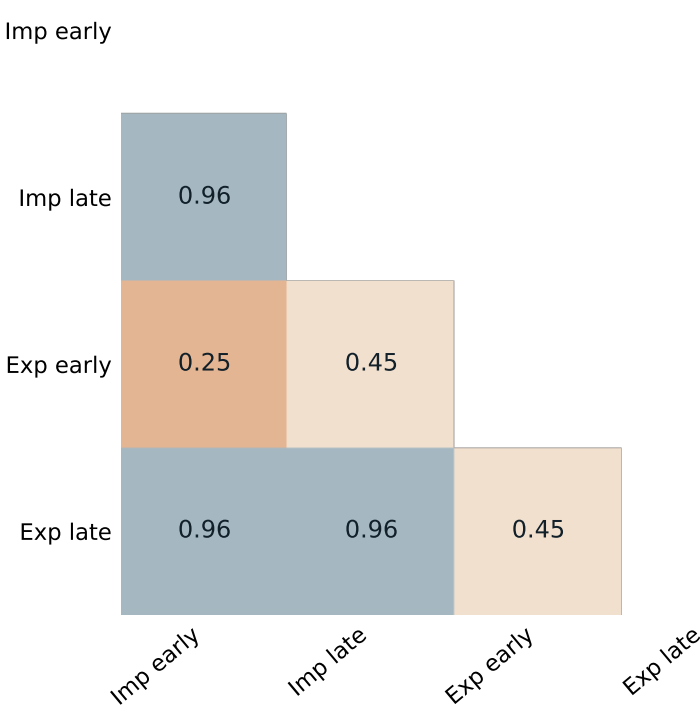
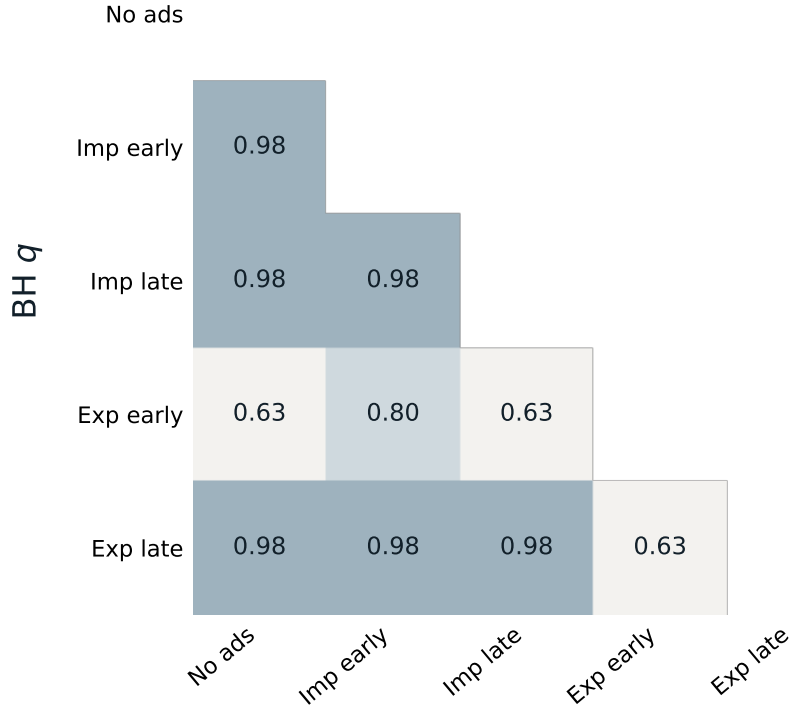
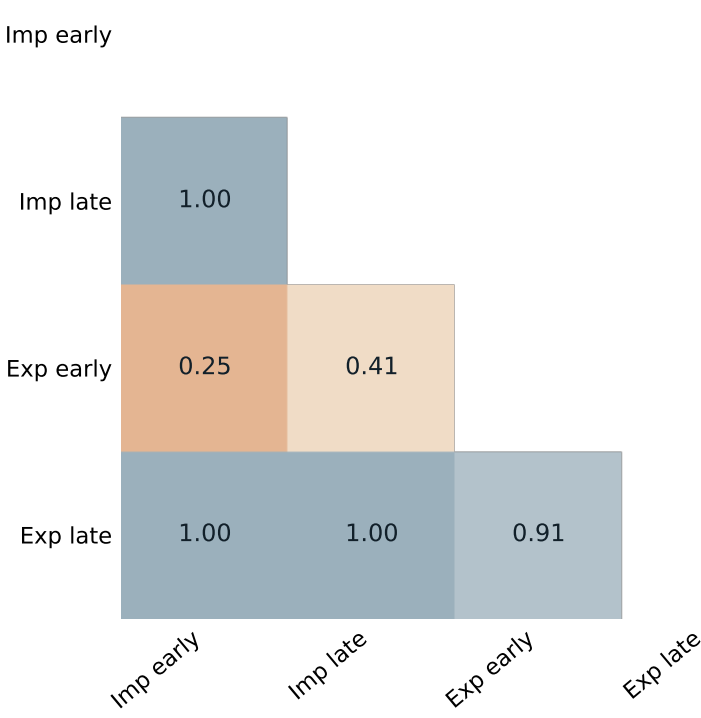
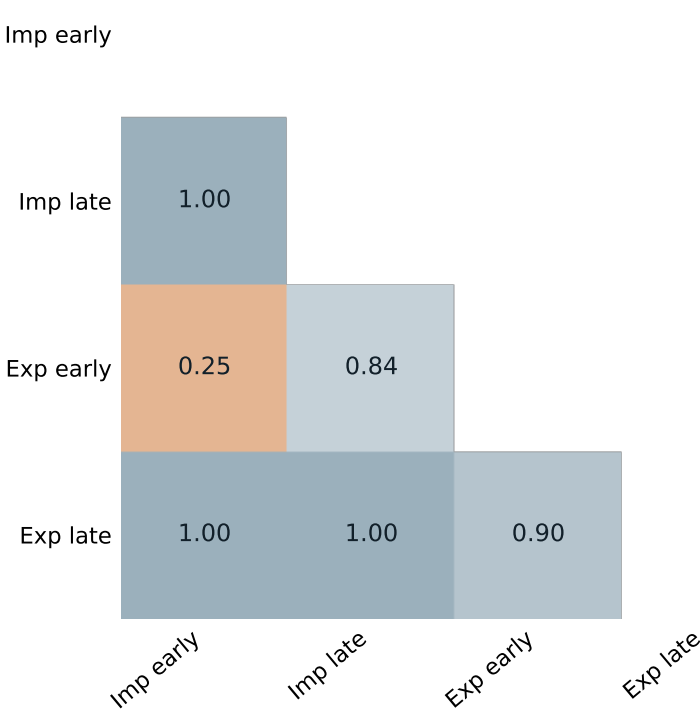
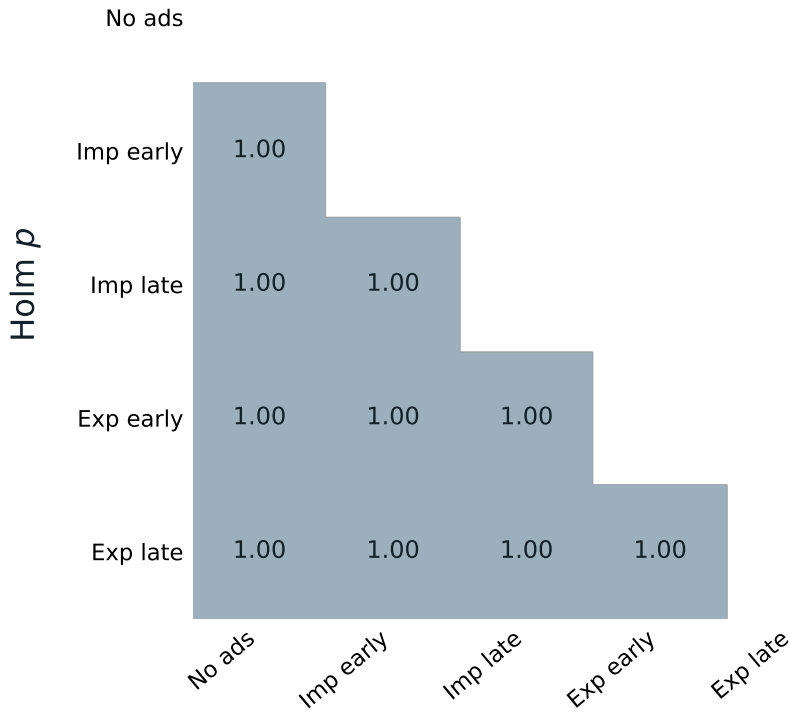
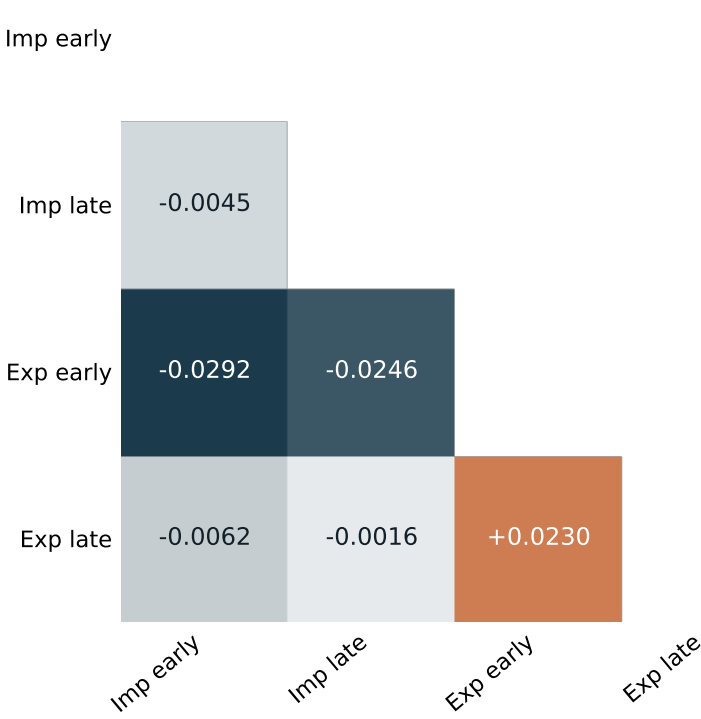
Dataset A (condition medians)



Dataset B, Δ matched control



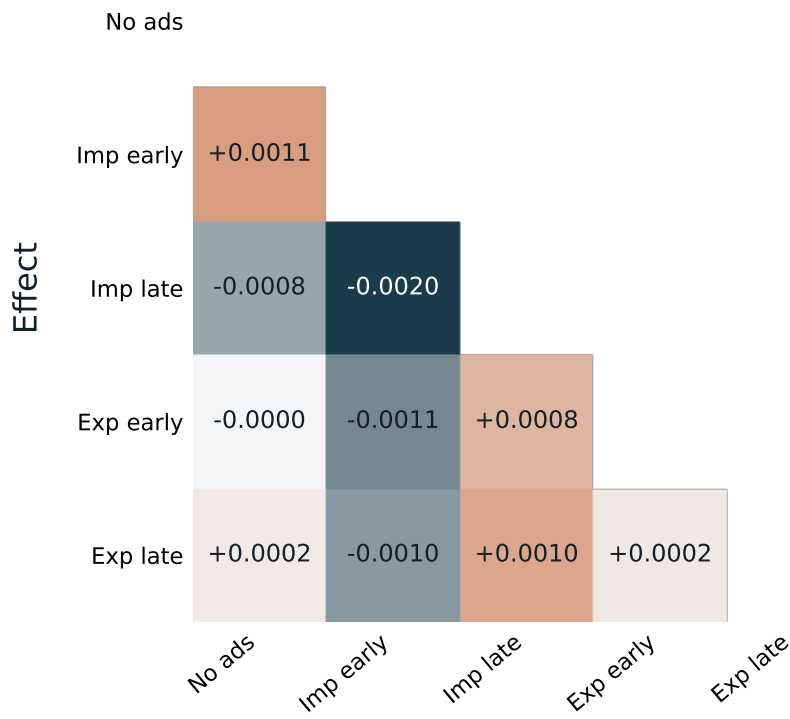
Dataset B, raw post–pre



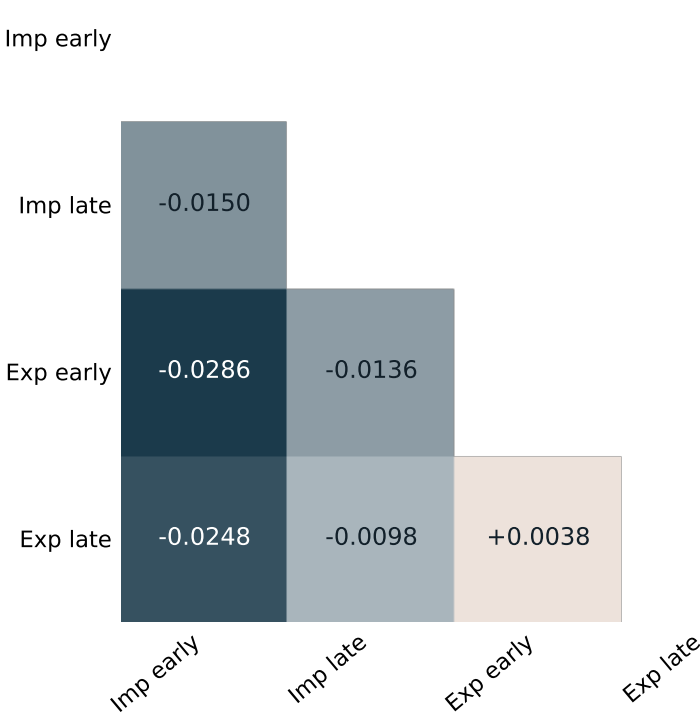
Rel. alpha

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

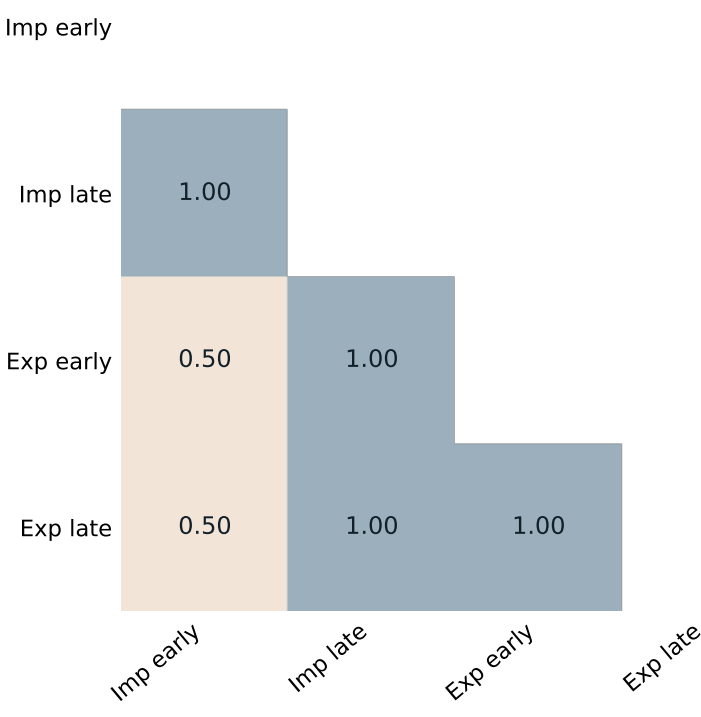
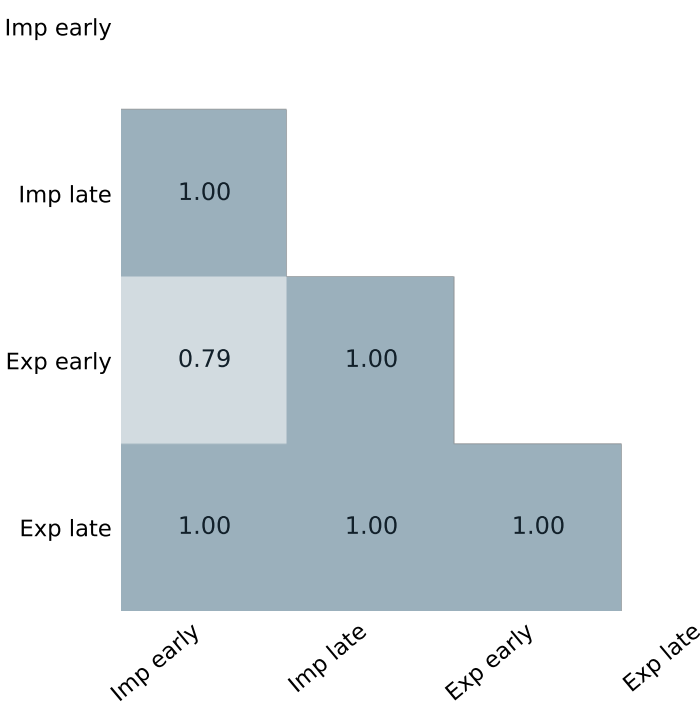
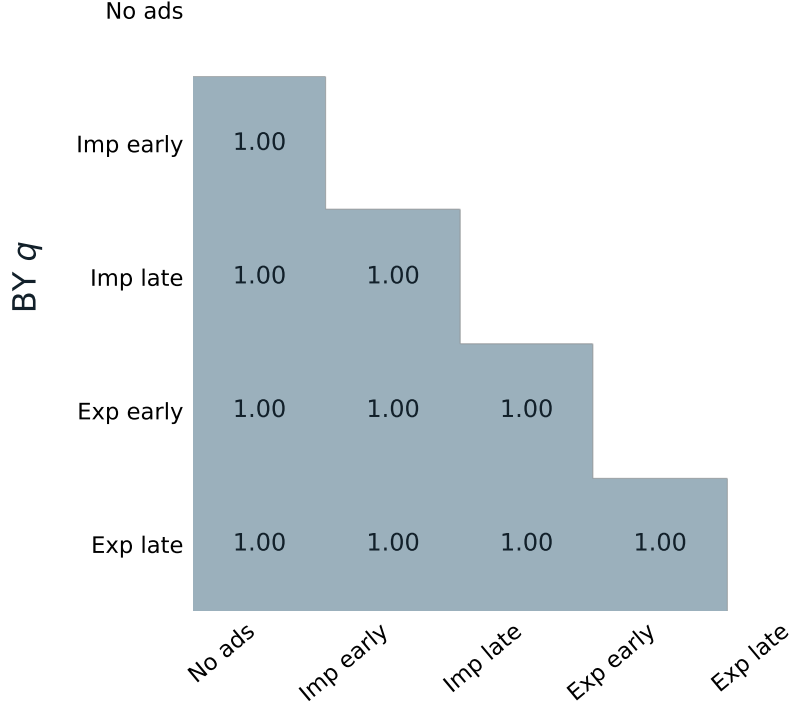
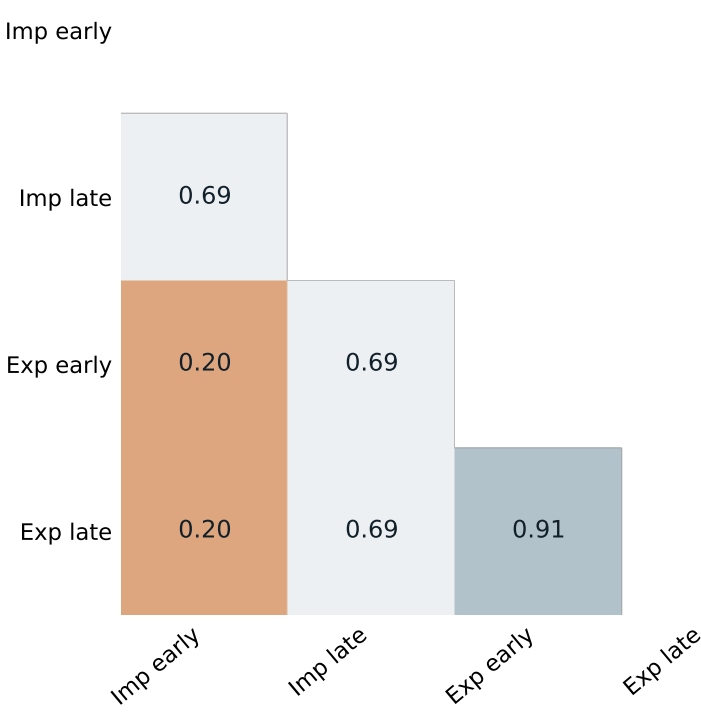
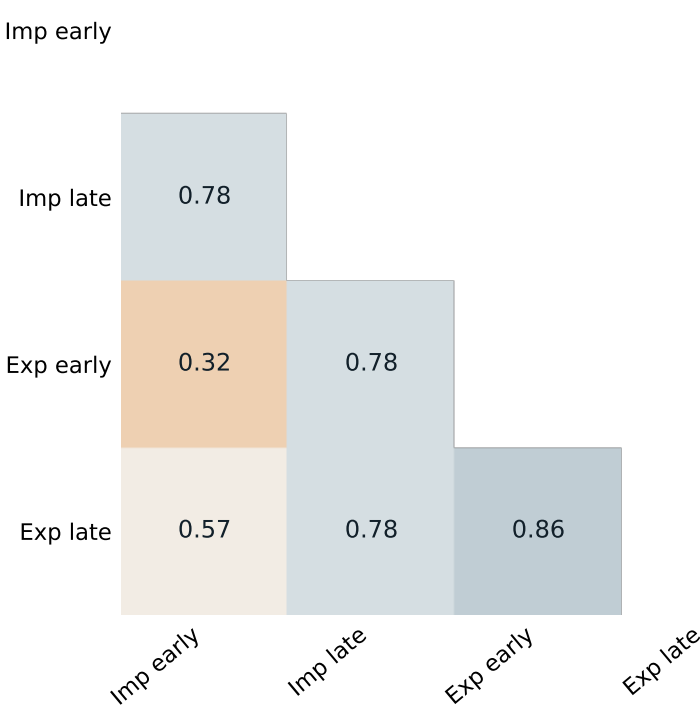
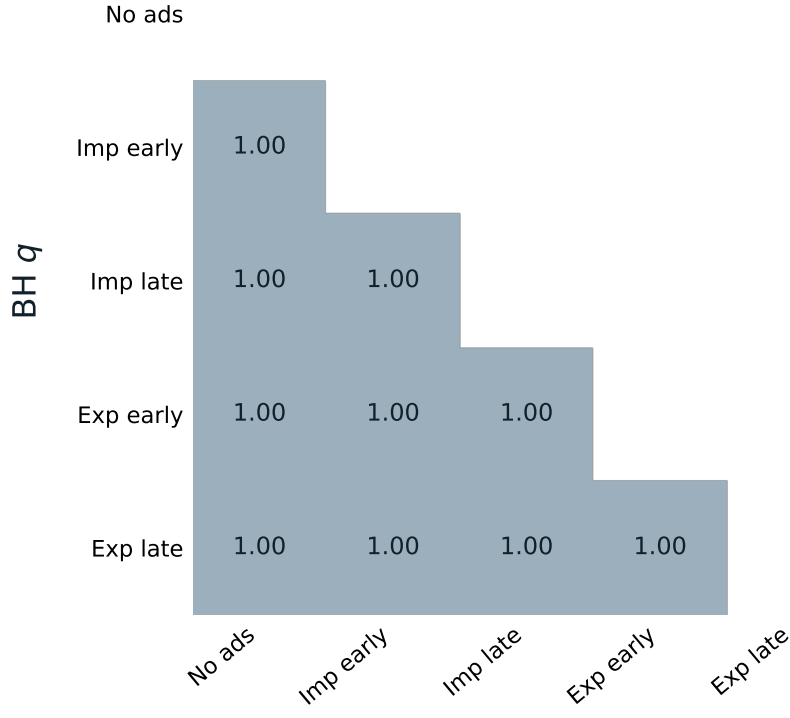
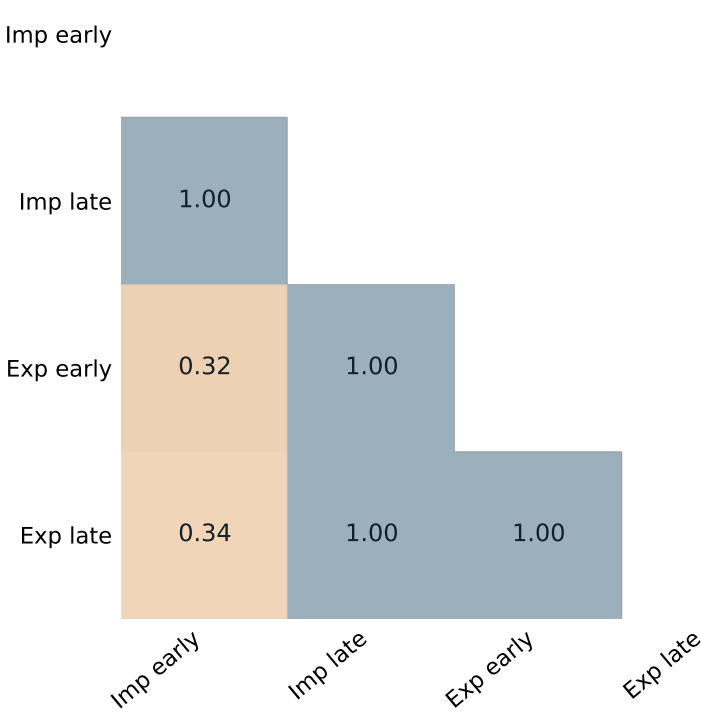
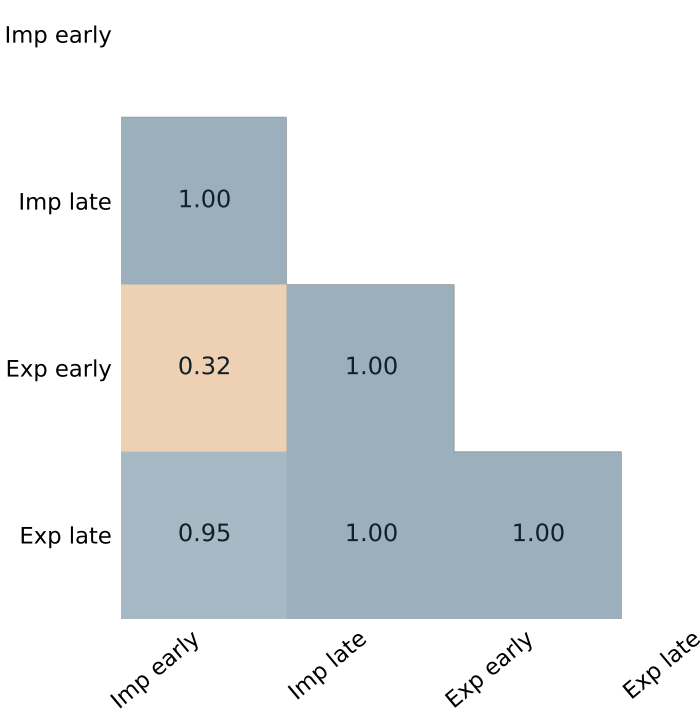
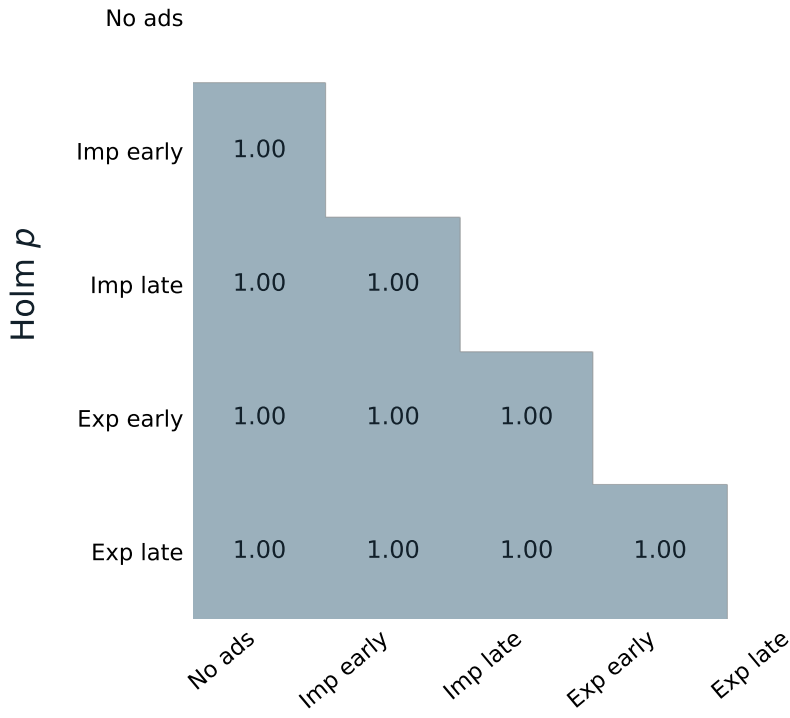
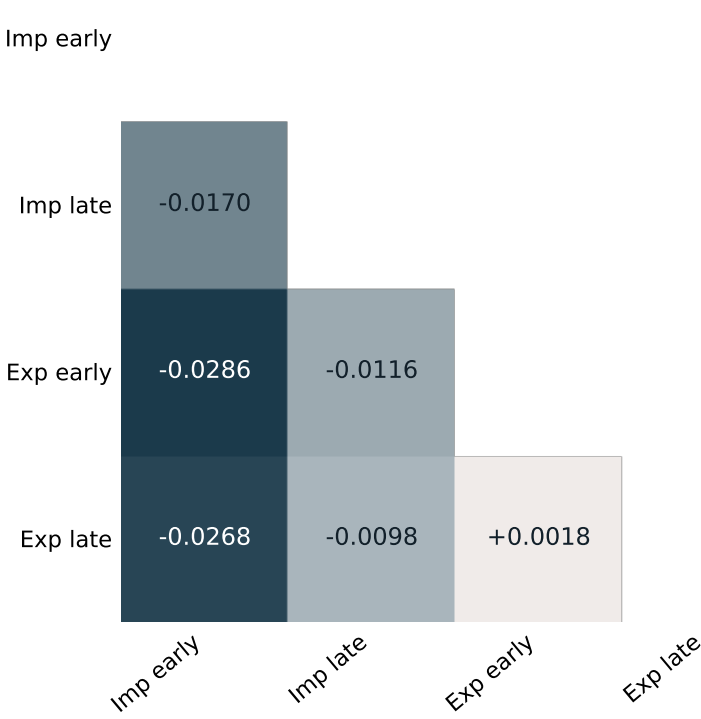
Dataset A (condition medians)



Dataset B, Δ matched control



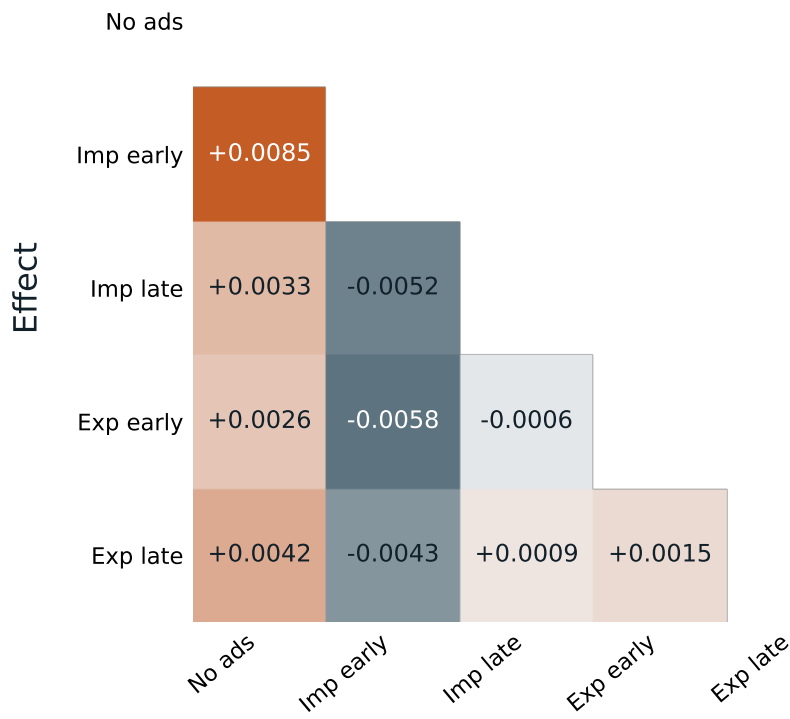
Dataset B, raw post–pre



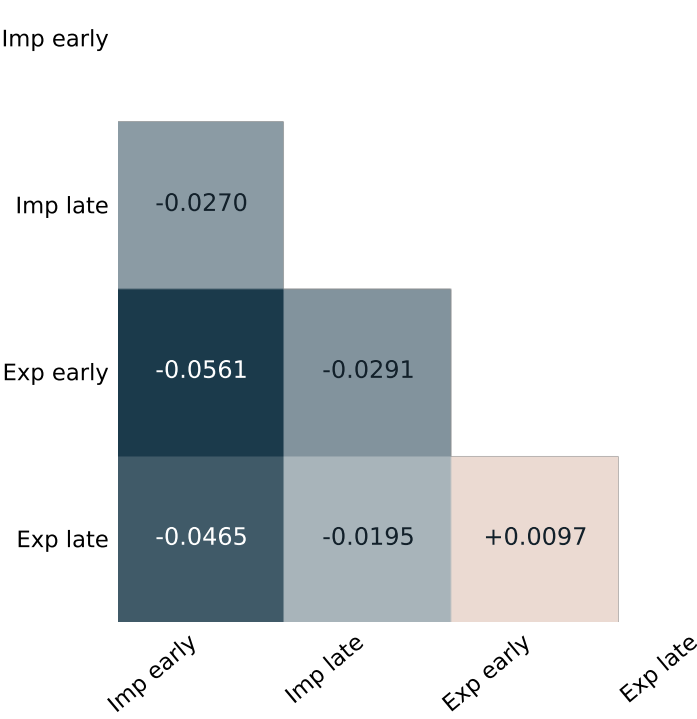
Rel. beta

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

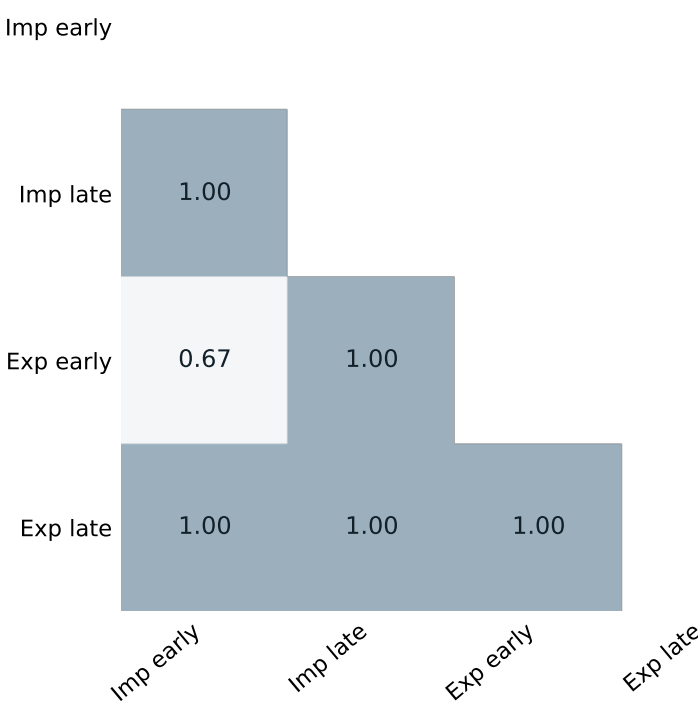
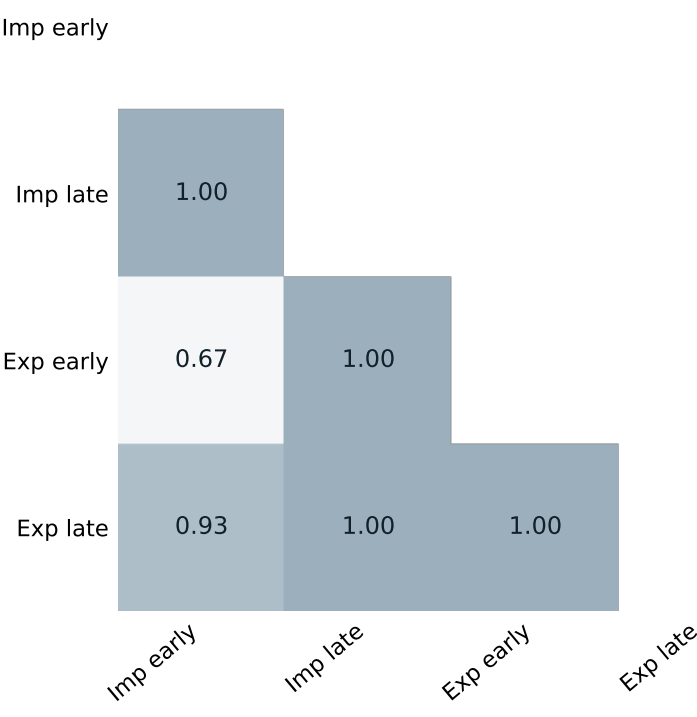
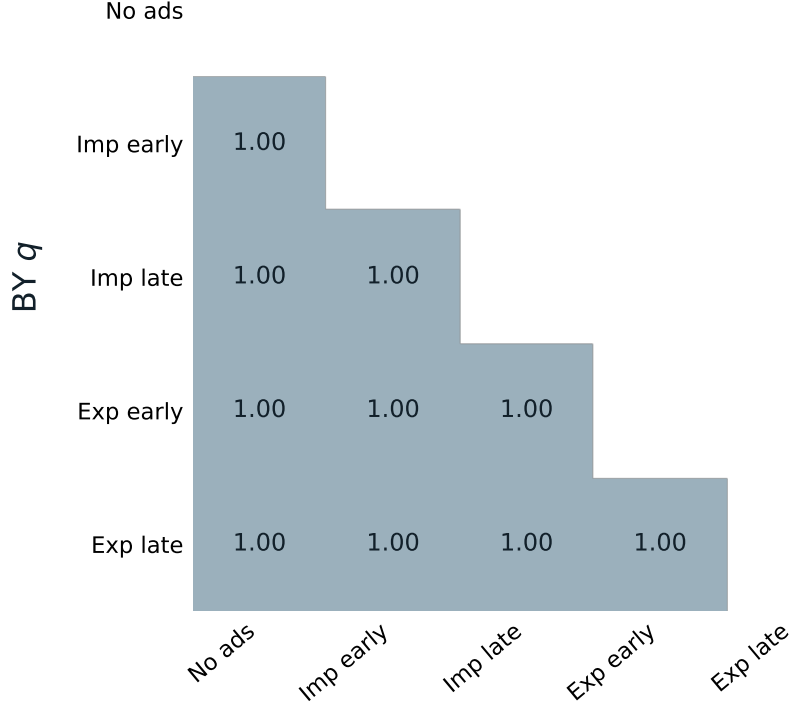
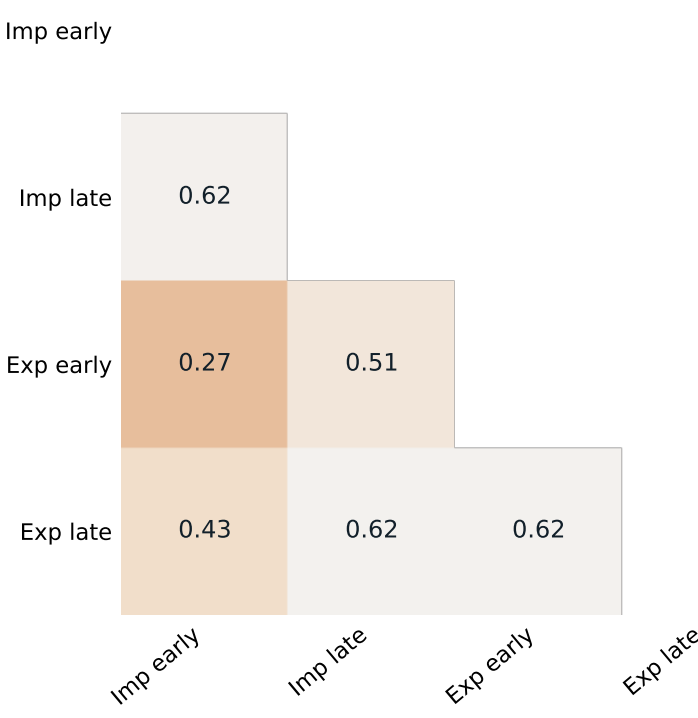
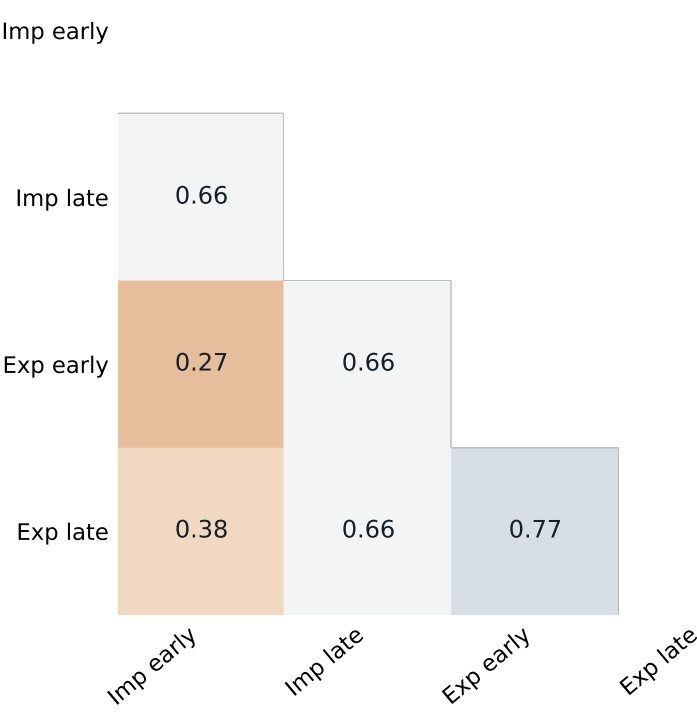
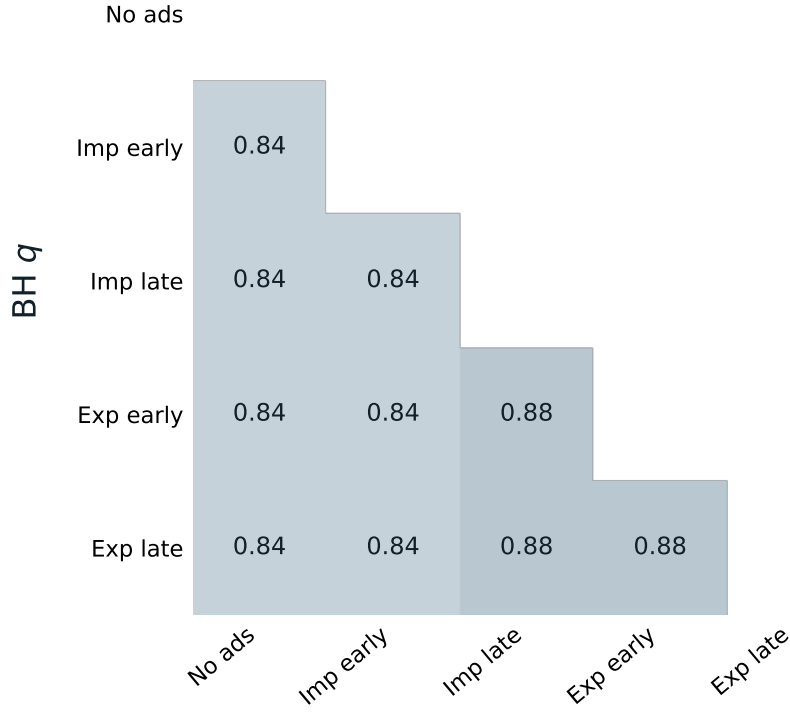
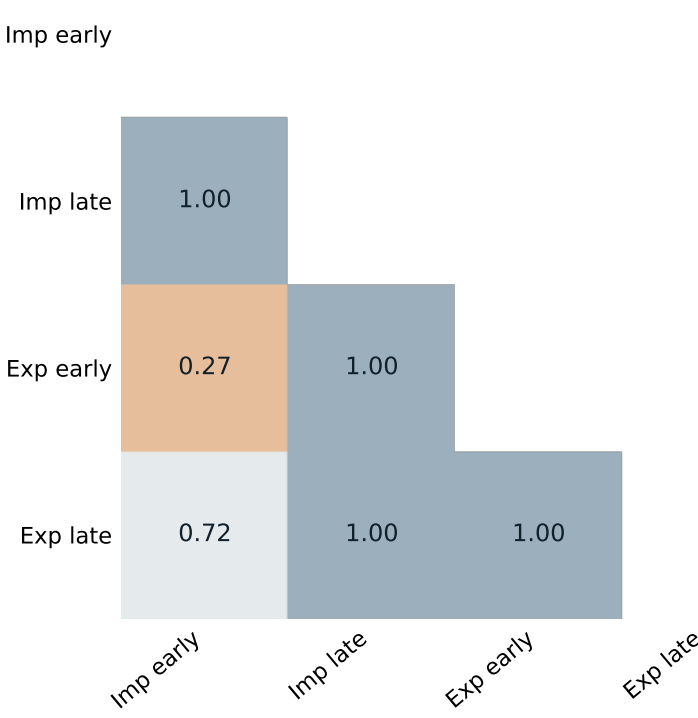
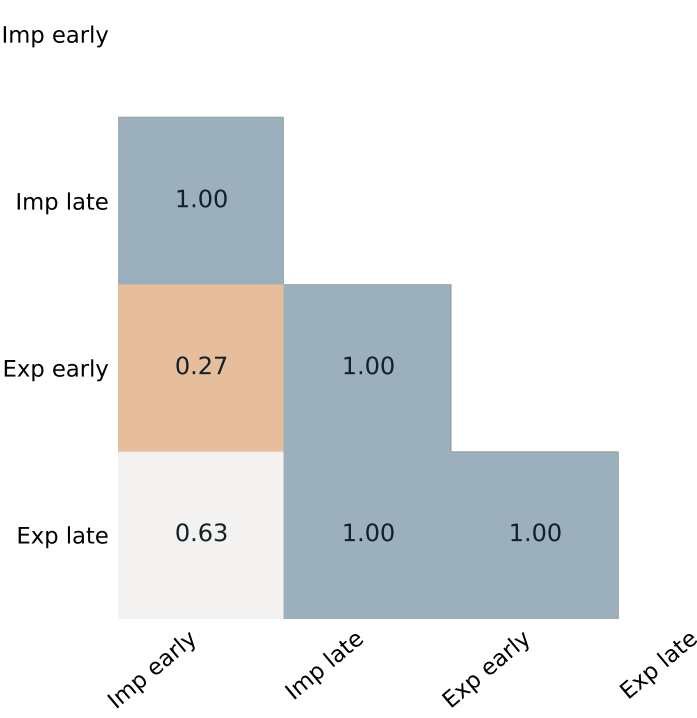
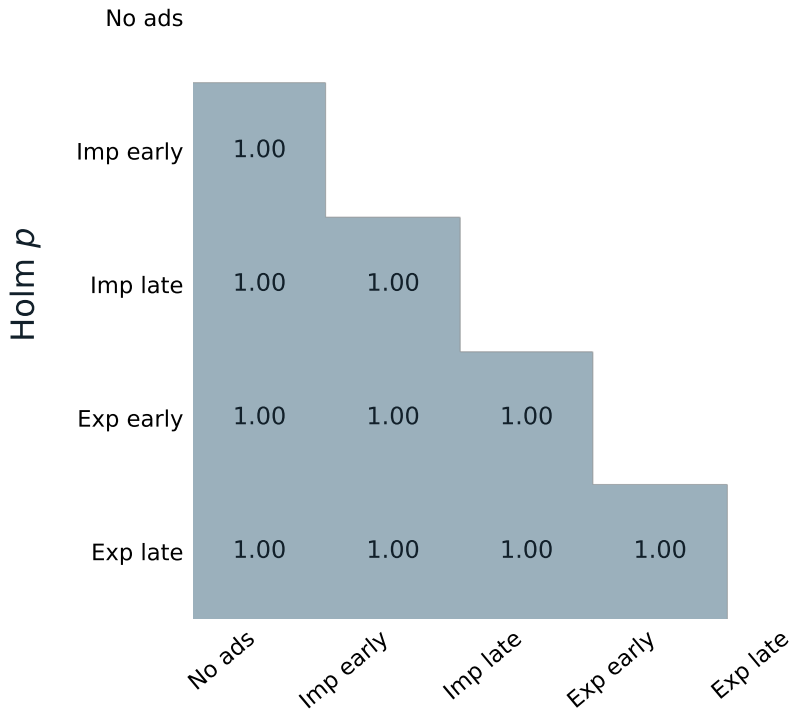
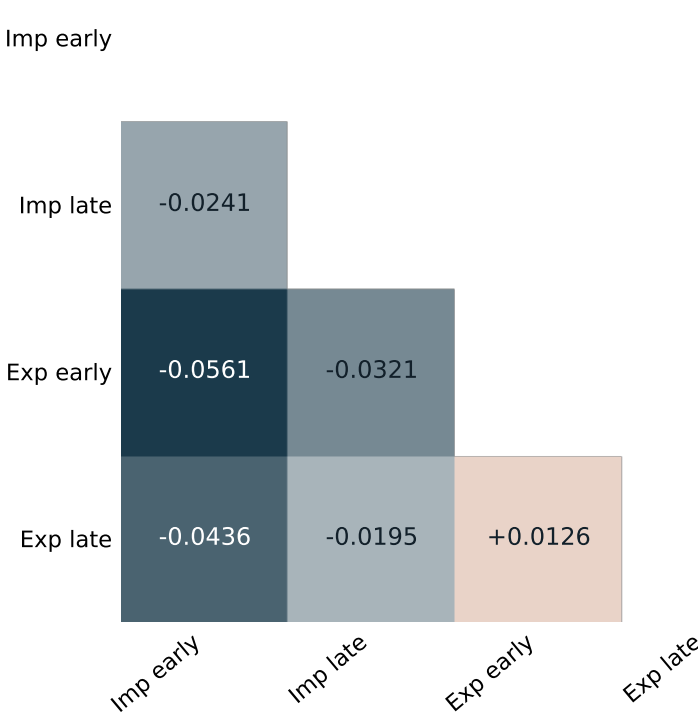
Dataset A (condition medians)



Dataset B, Δ matched control



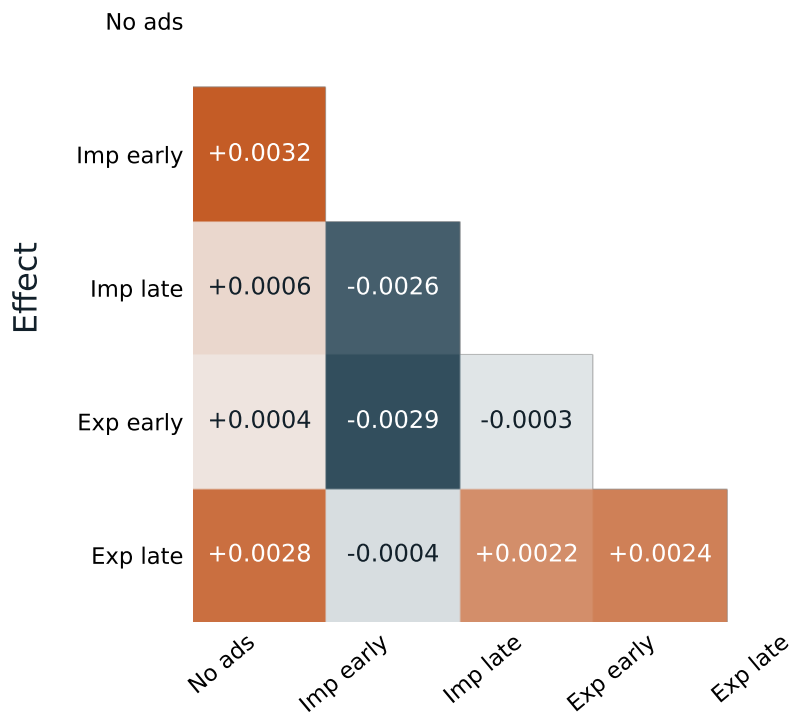
Dataset B, raw post–pre



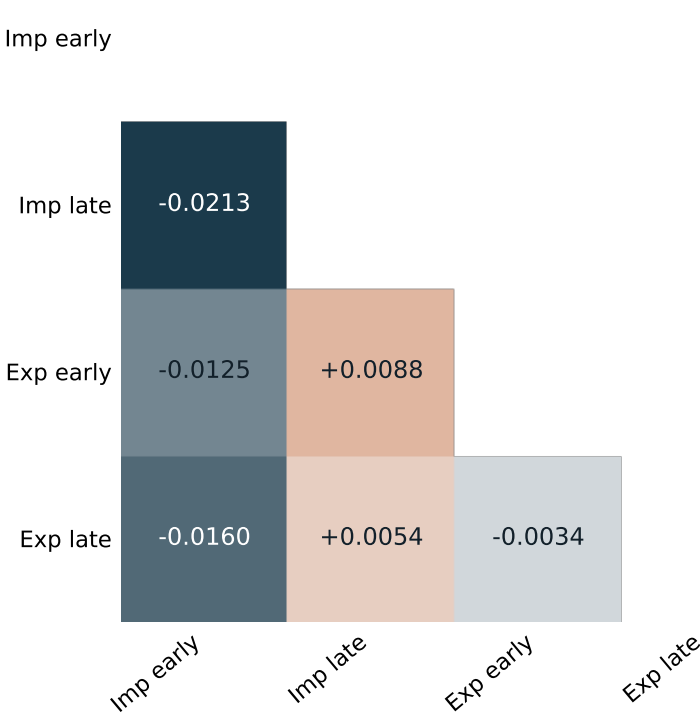
Rel. gamma

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

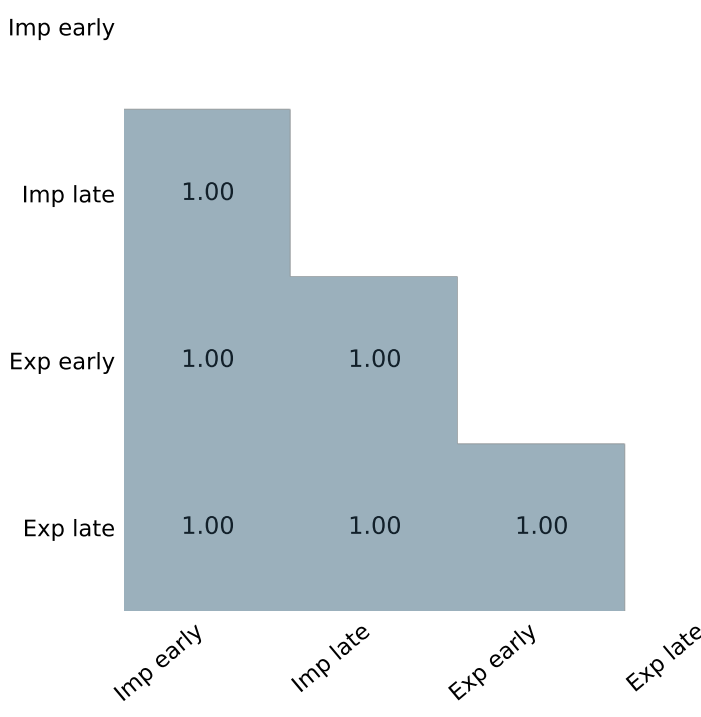
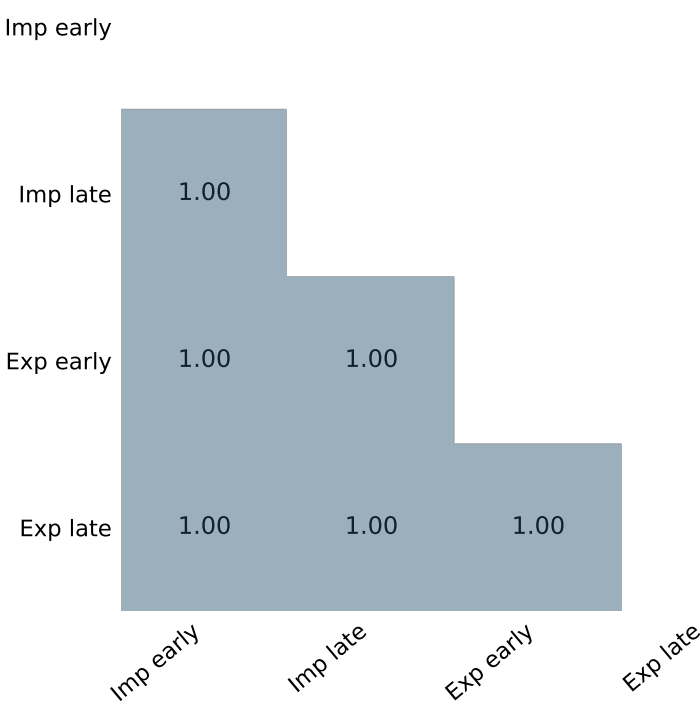
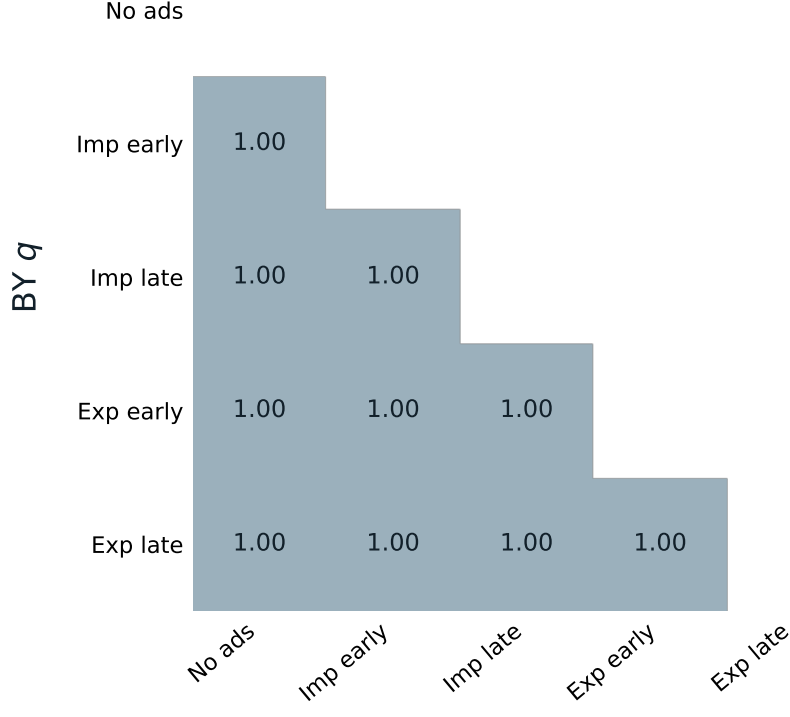
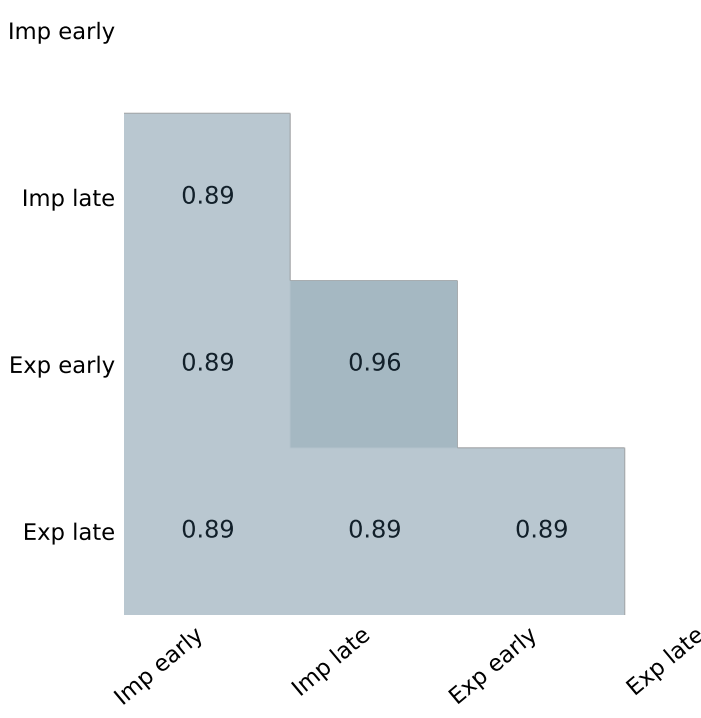
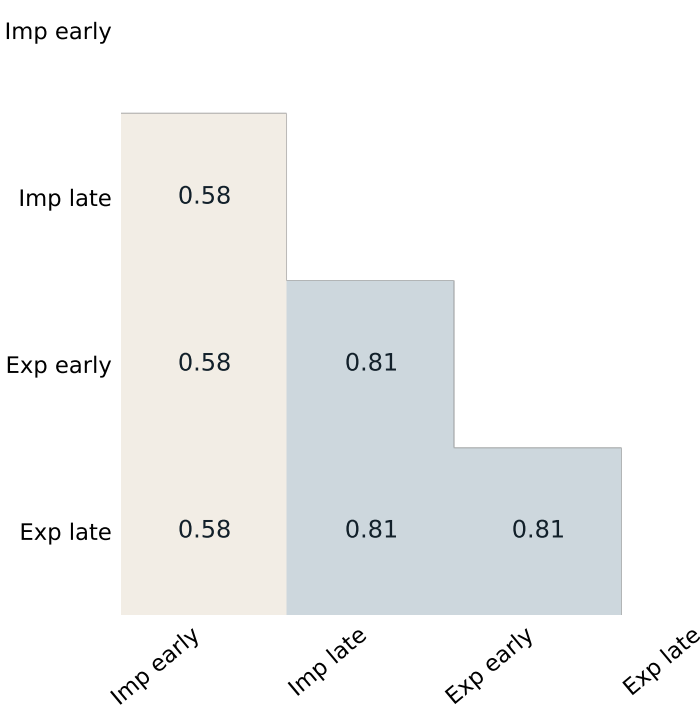
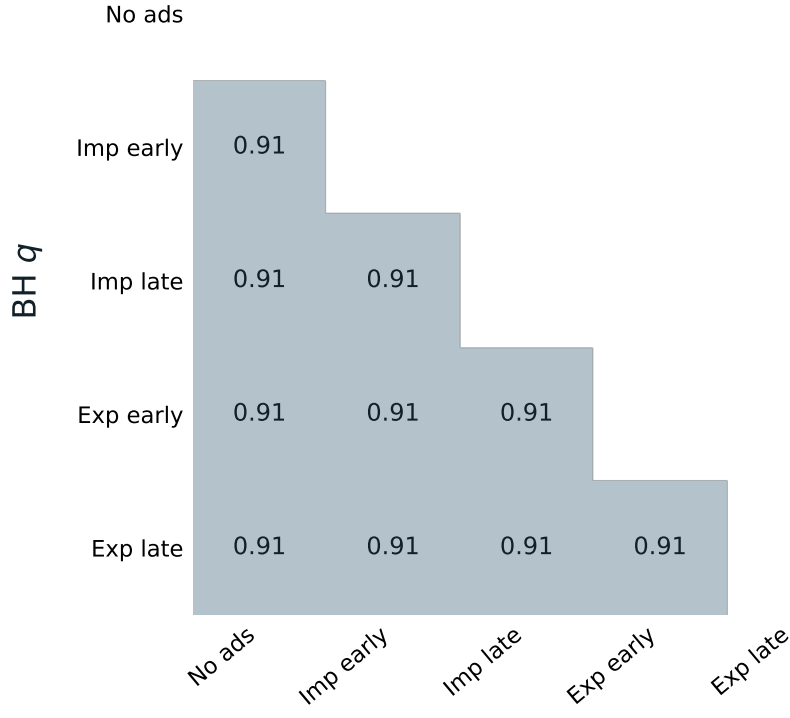
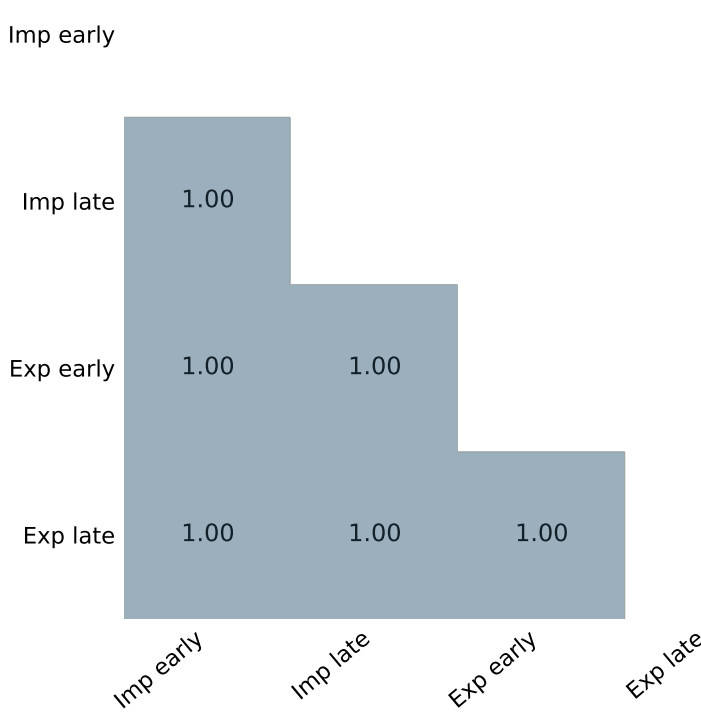
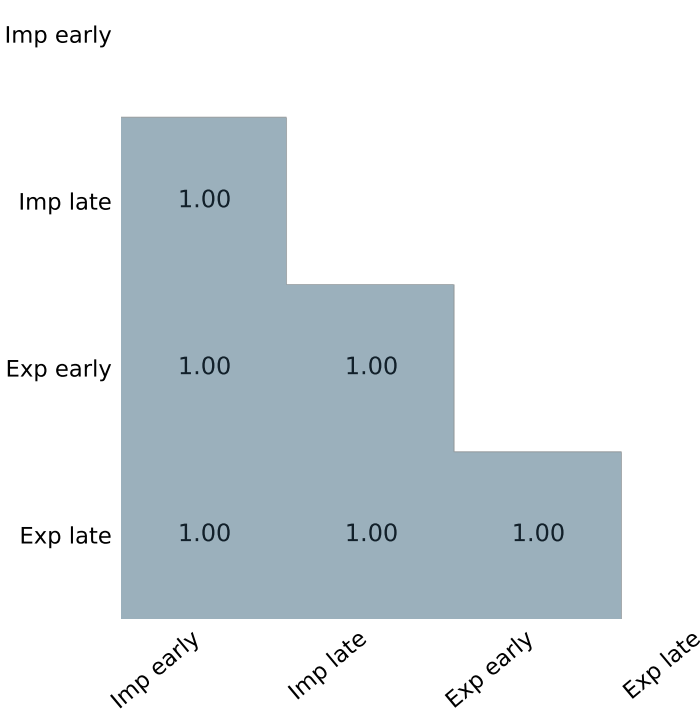
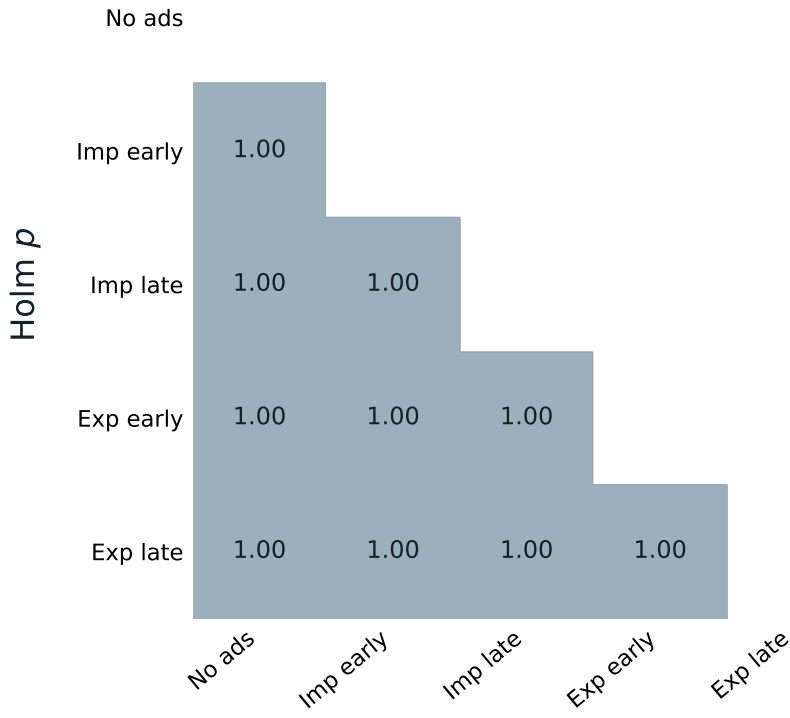
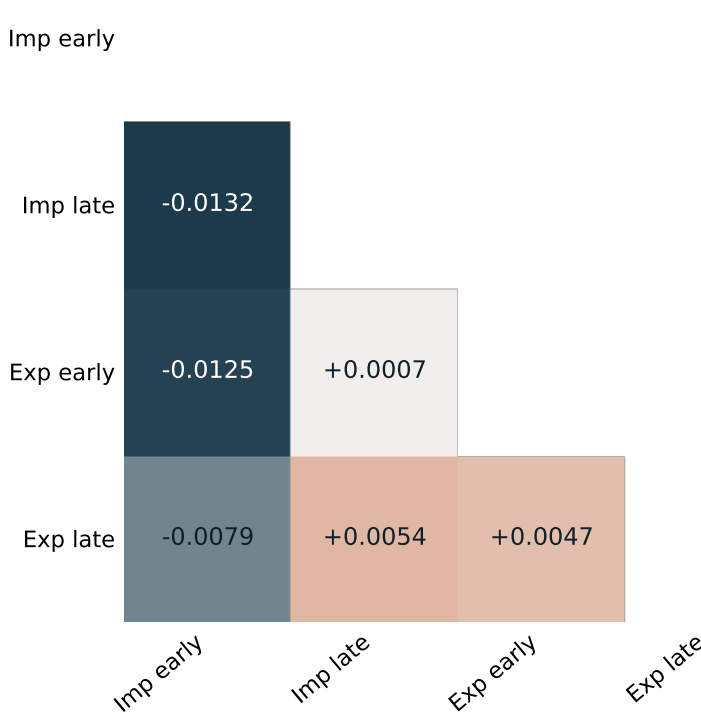
Dataset A (condition medians)



Dataset B, Δ matched control



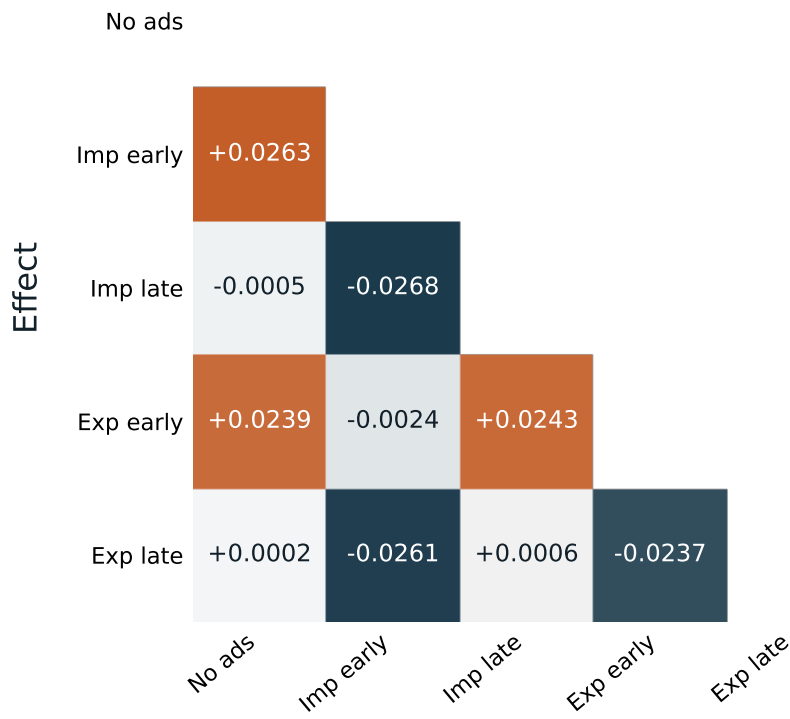
Dataset B, raw post–pre



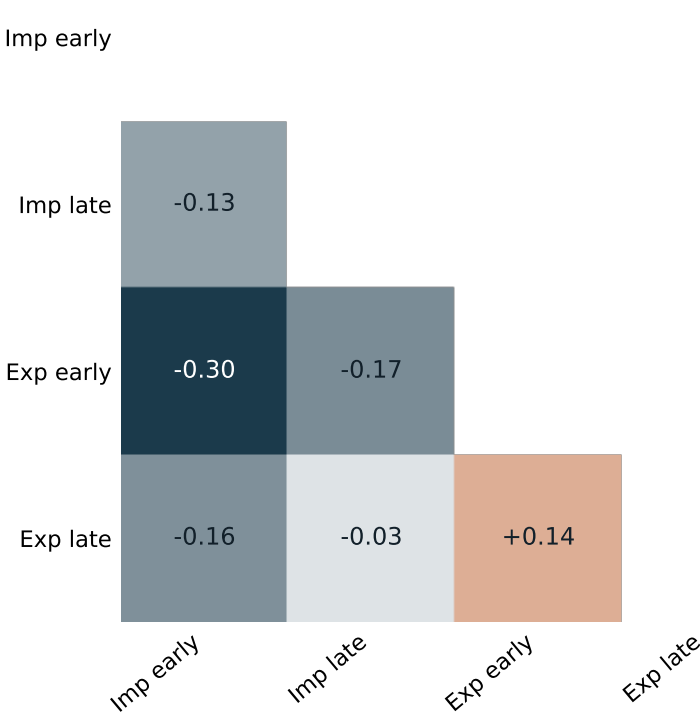
Pope engagement

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

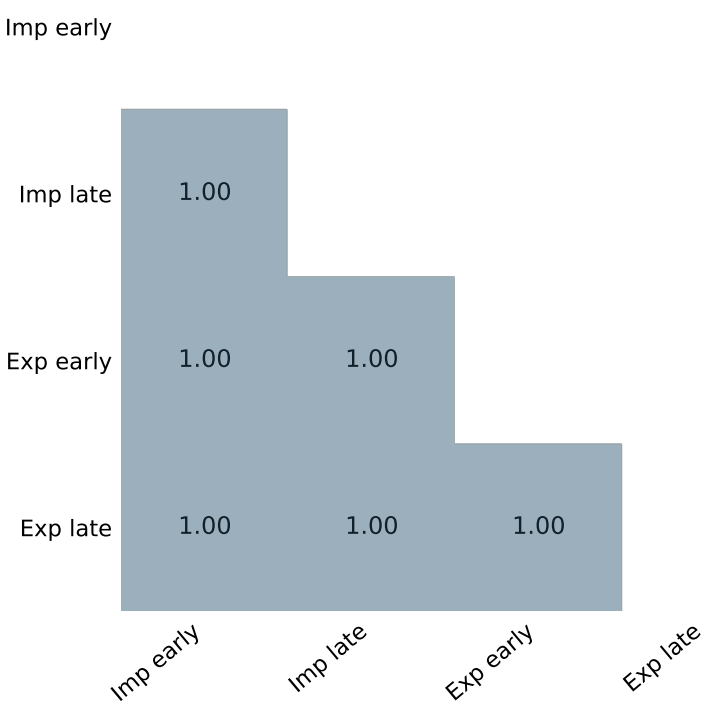
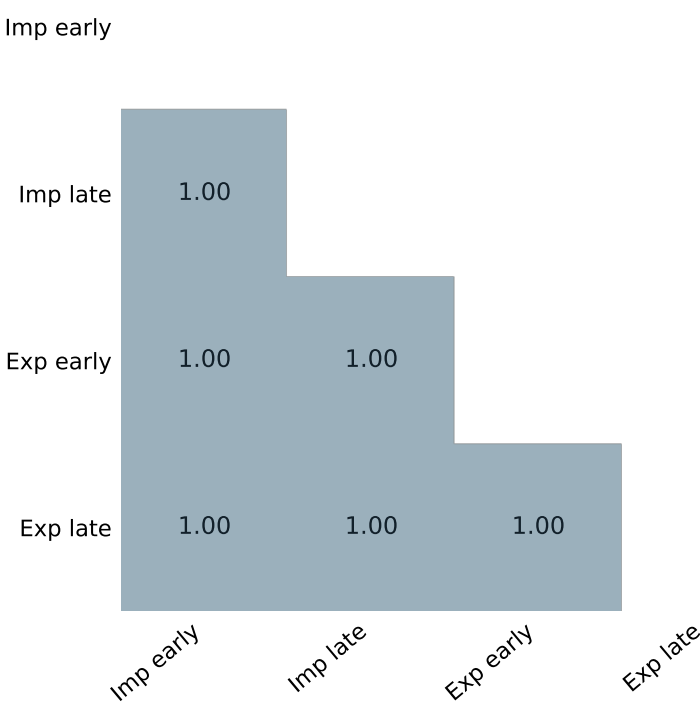
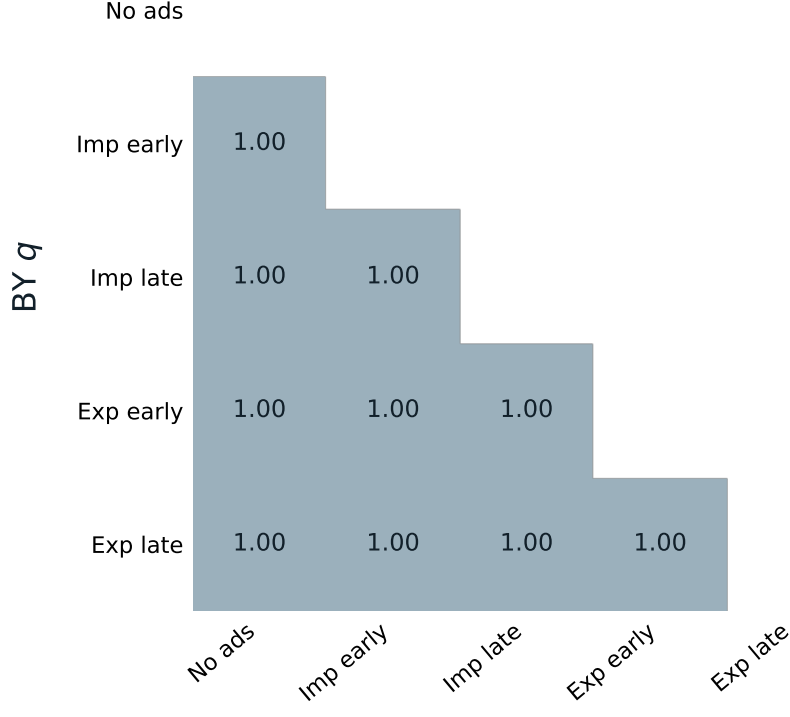
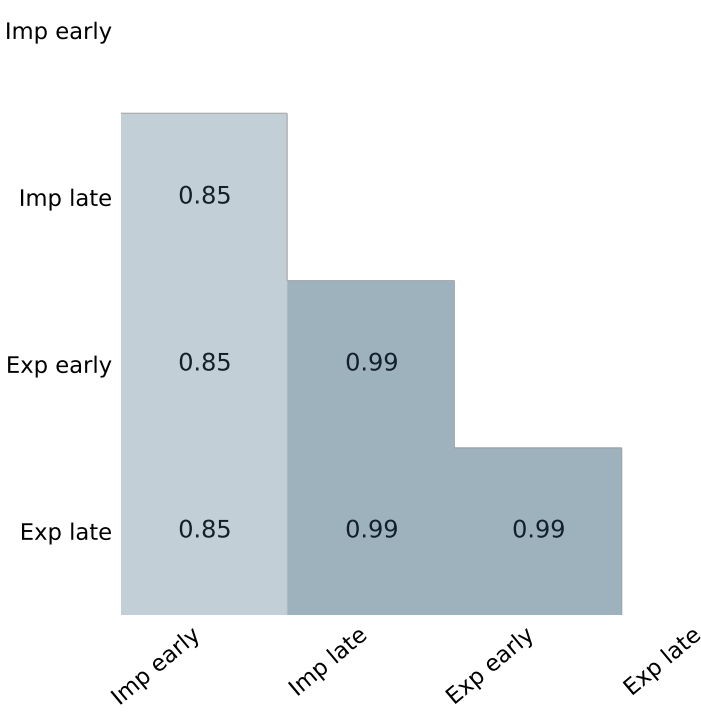
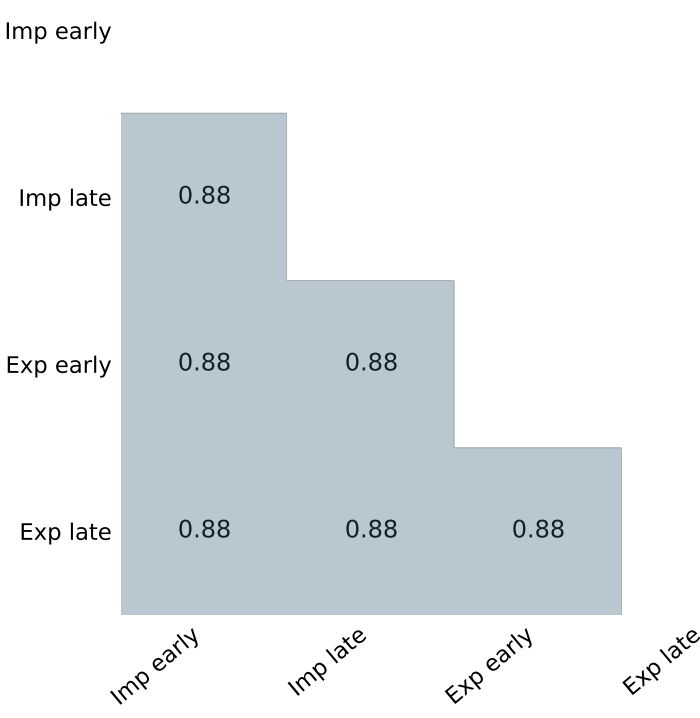
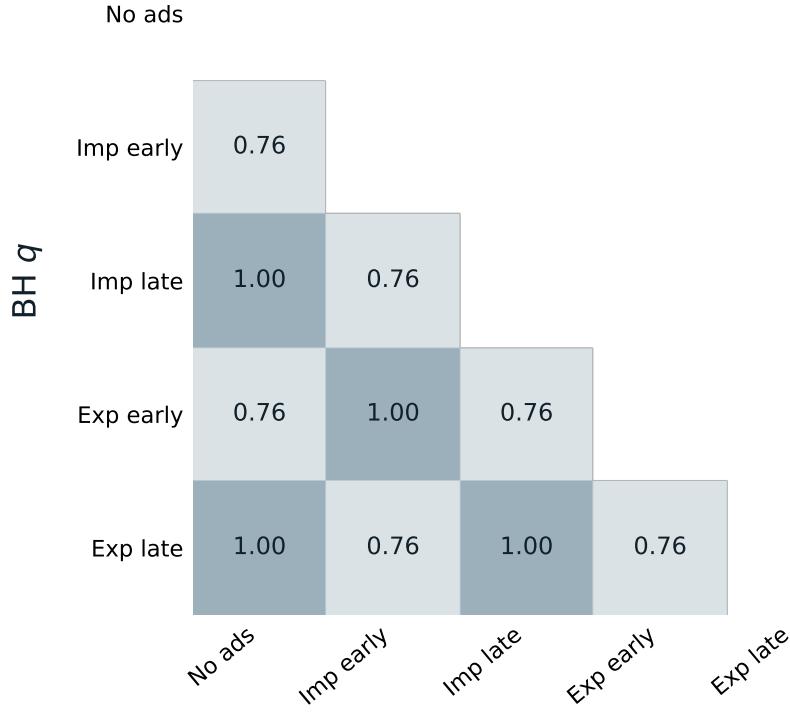
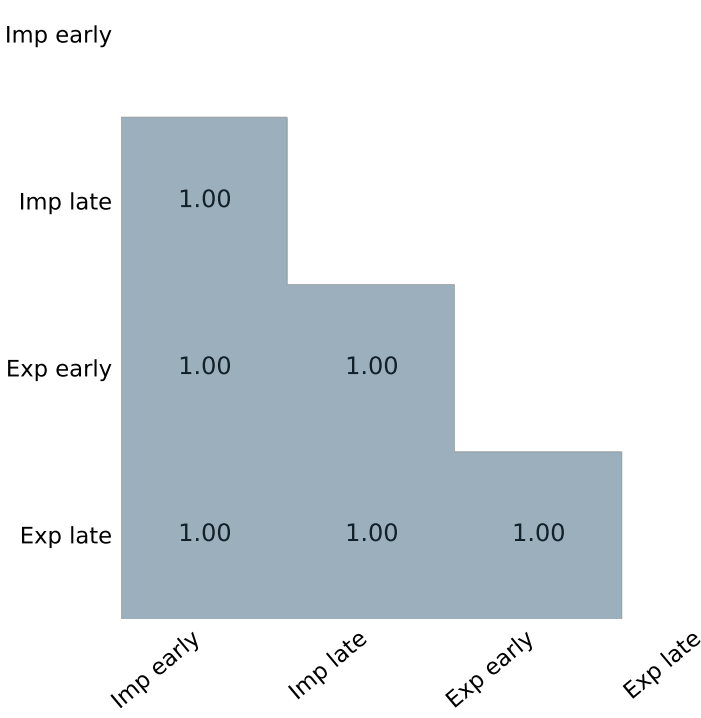
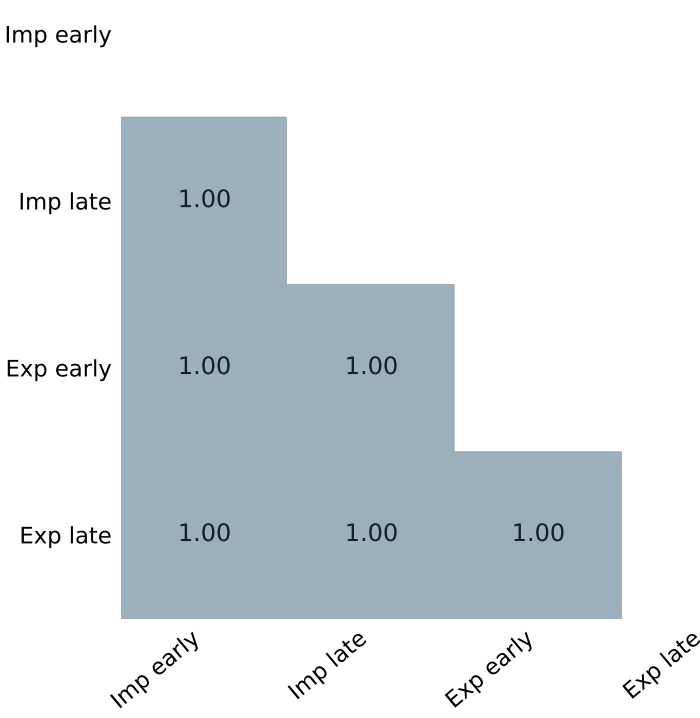
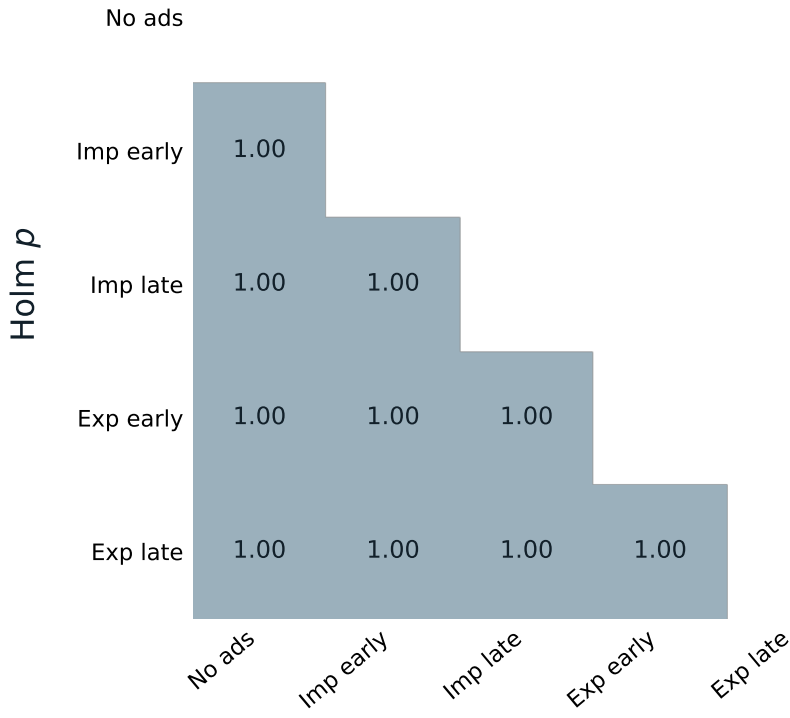
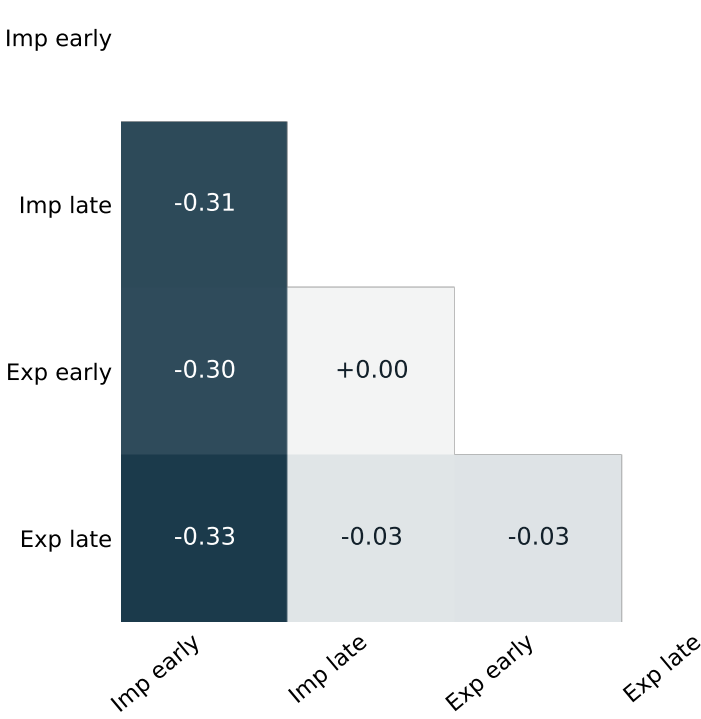
Dataset A (condition medians)



Dataset B, Δ matched control



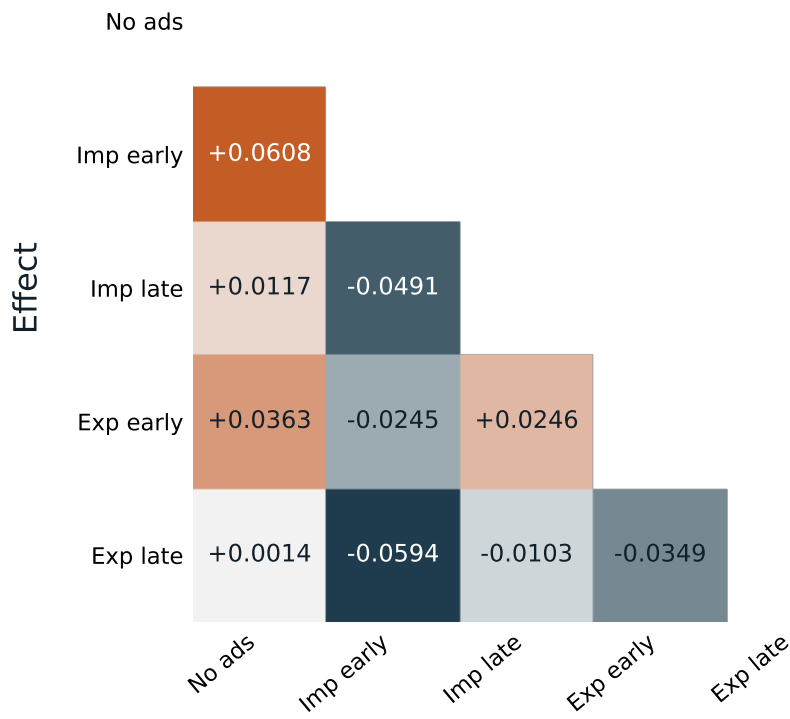
Dataset B, raw post–pre



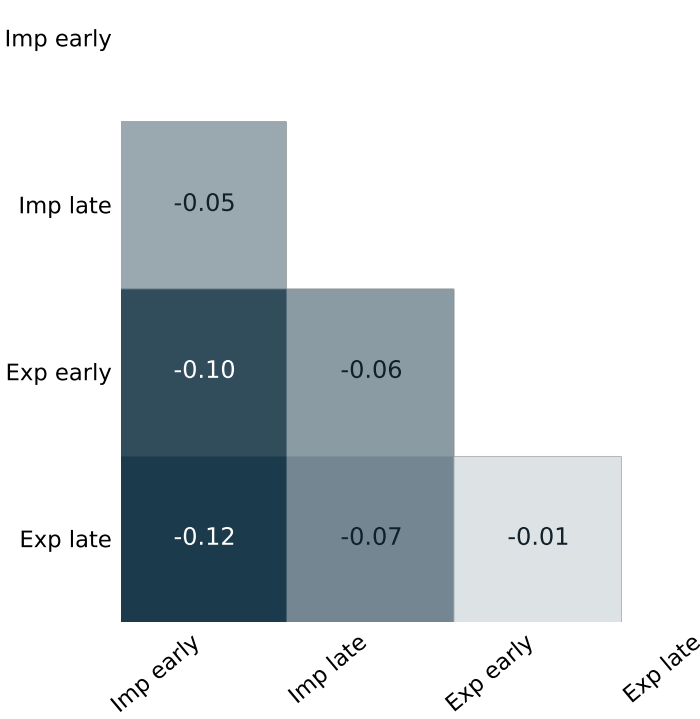
Pope frontocentral

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

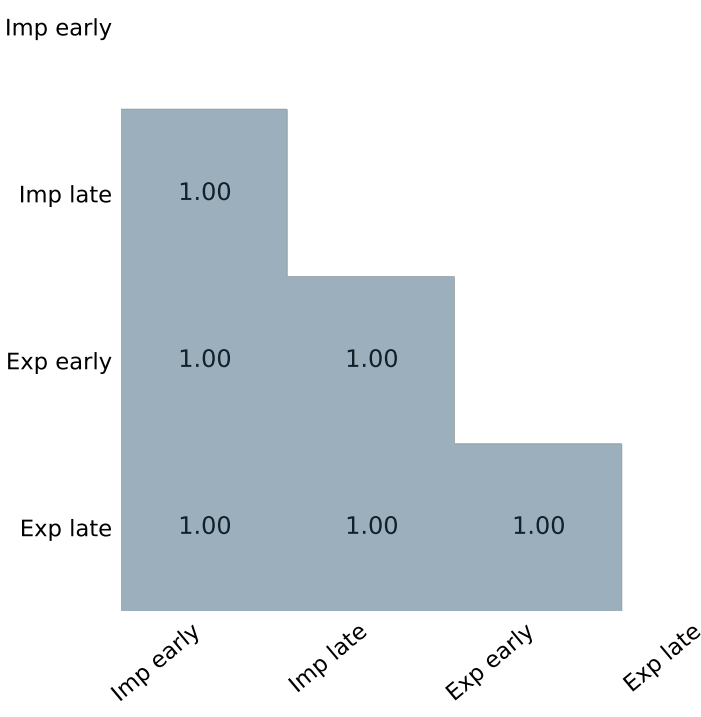
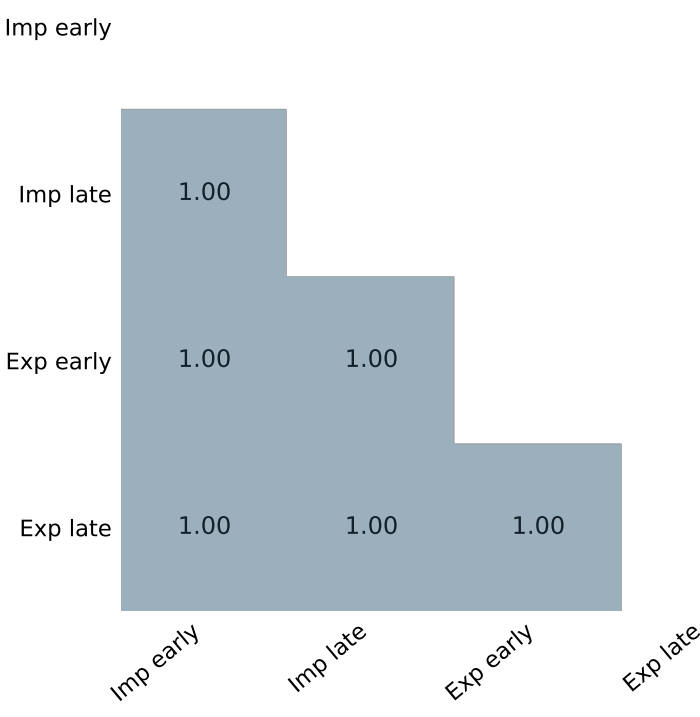
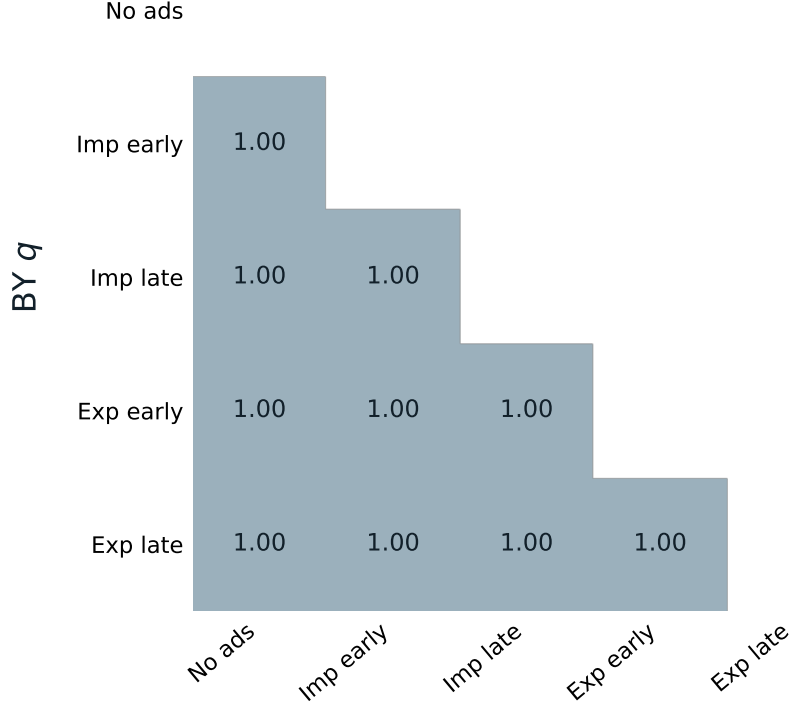
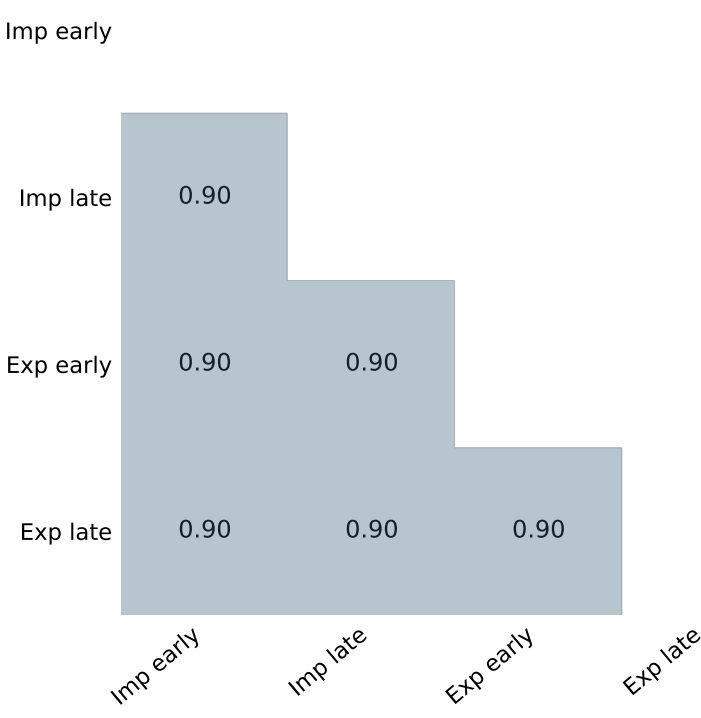
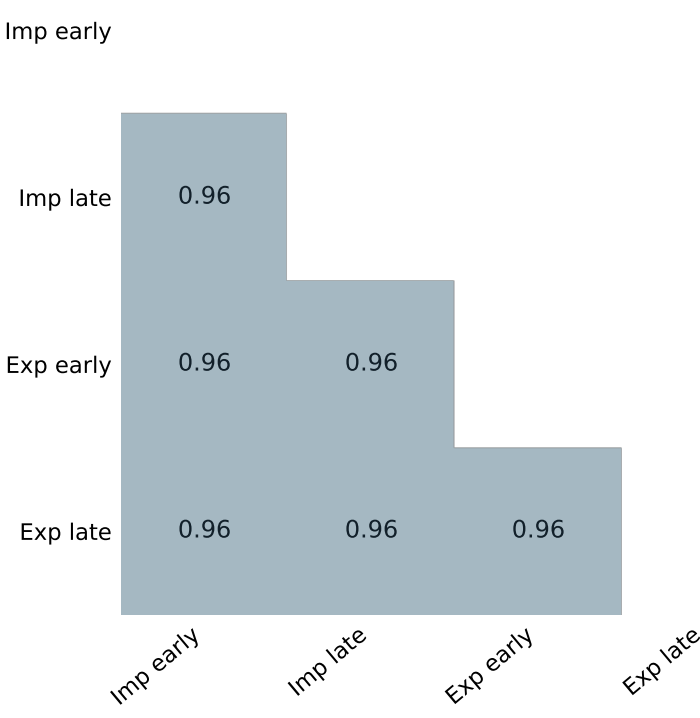
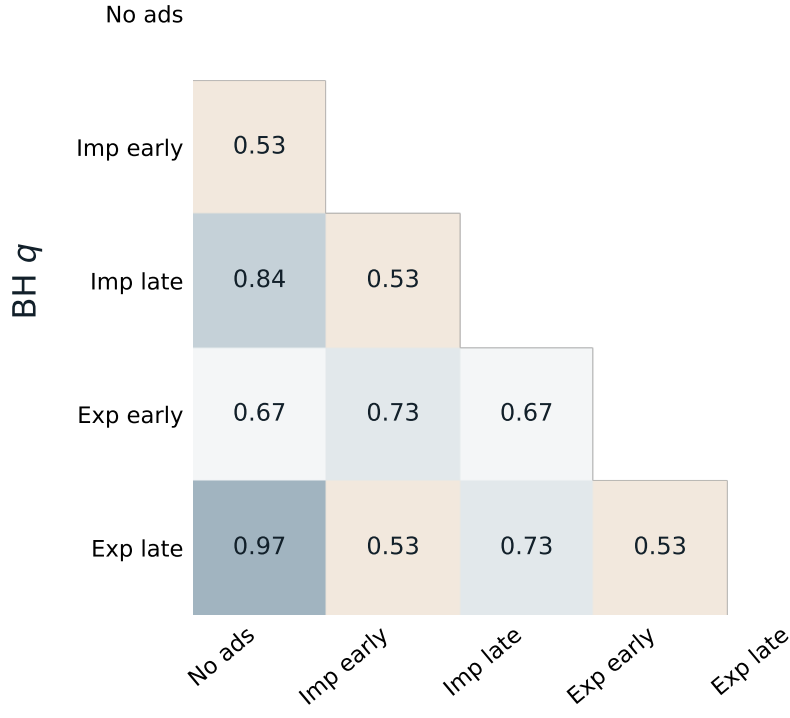
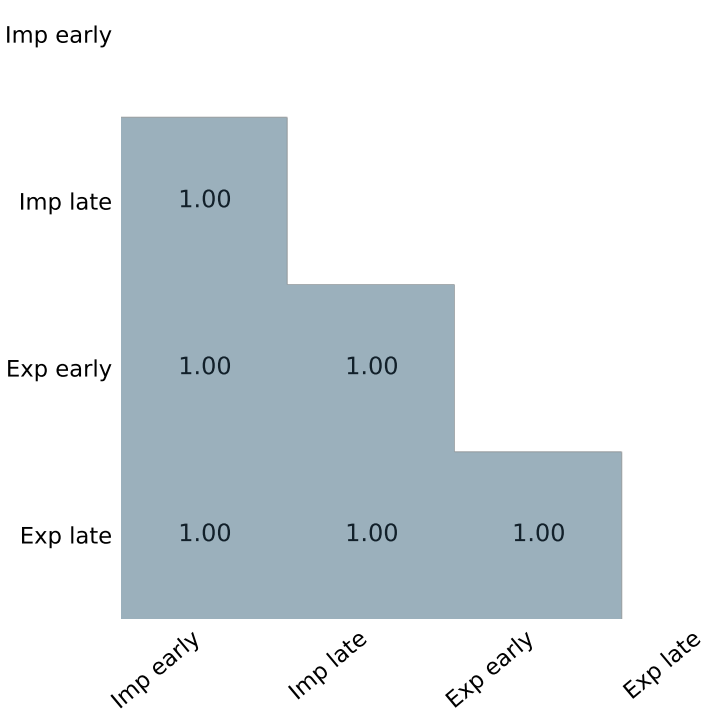
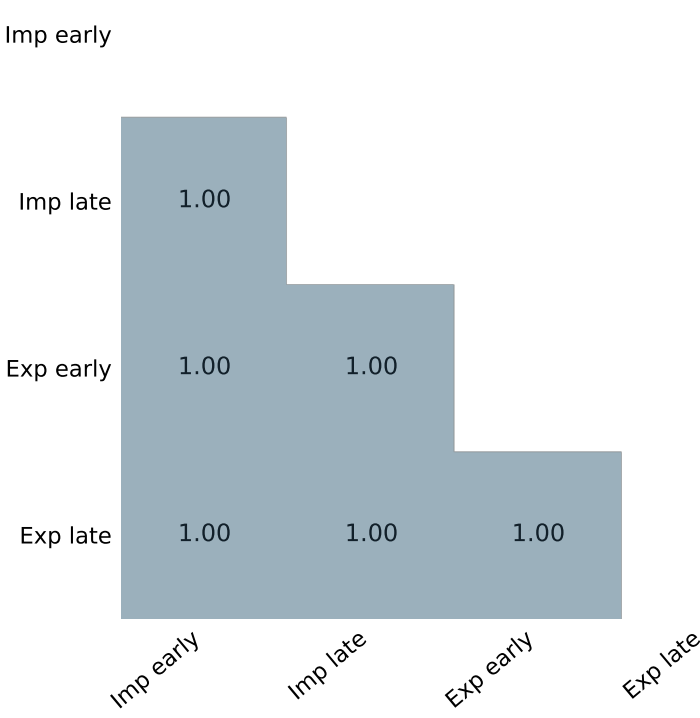
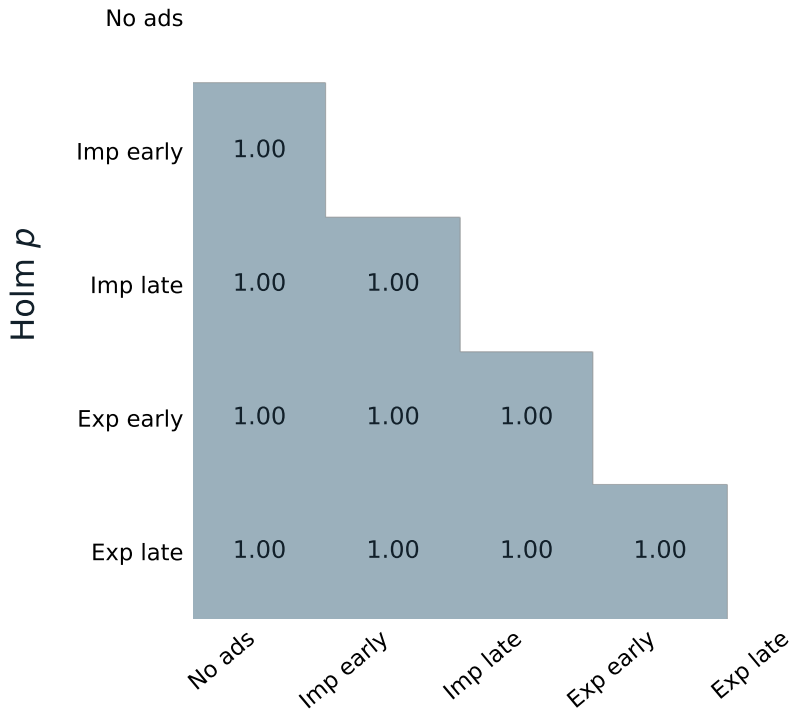
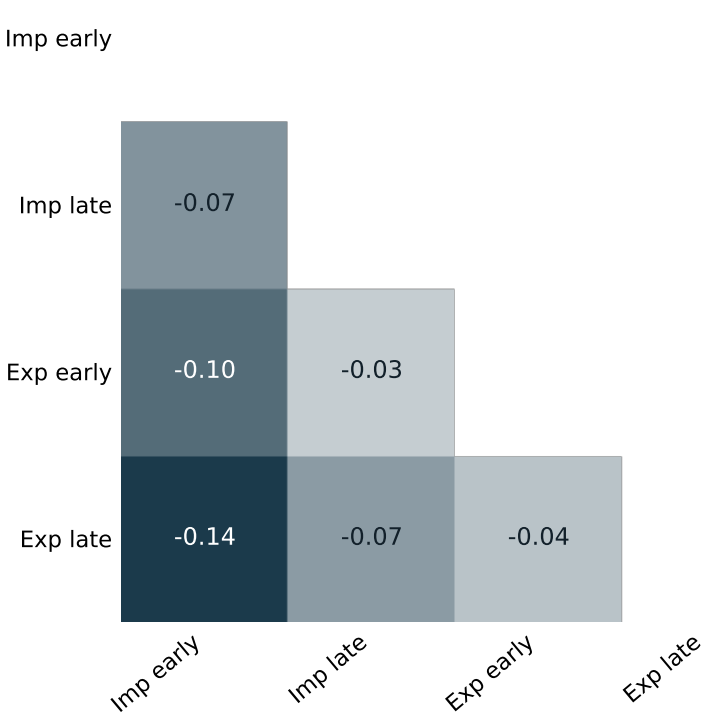
Dataset A (condition medians)



Dataset B, Δ matched control



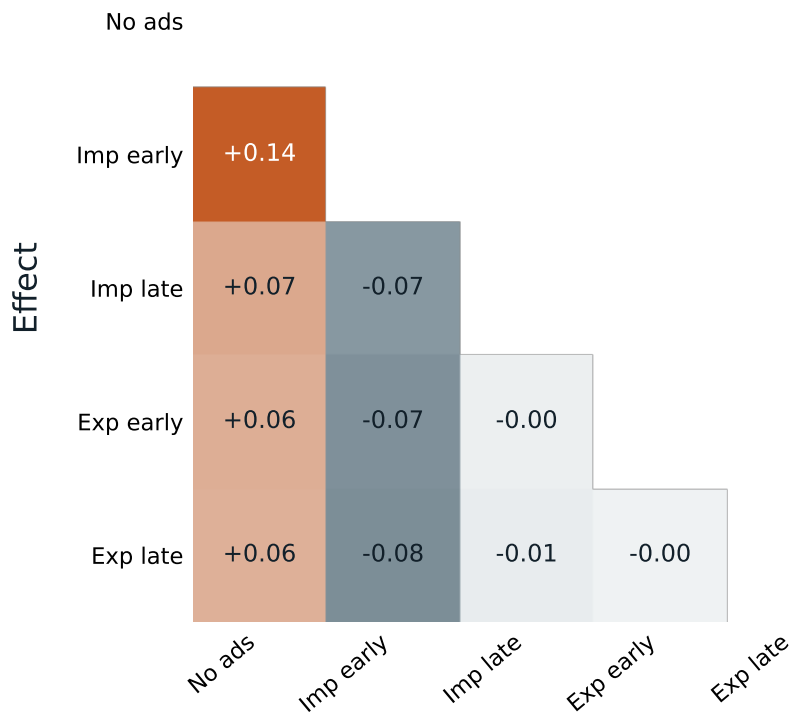
Dataset B, raw post–pre



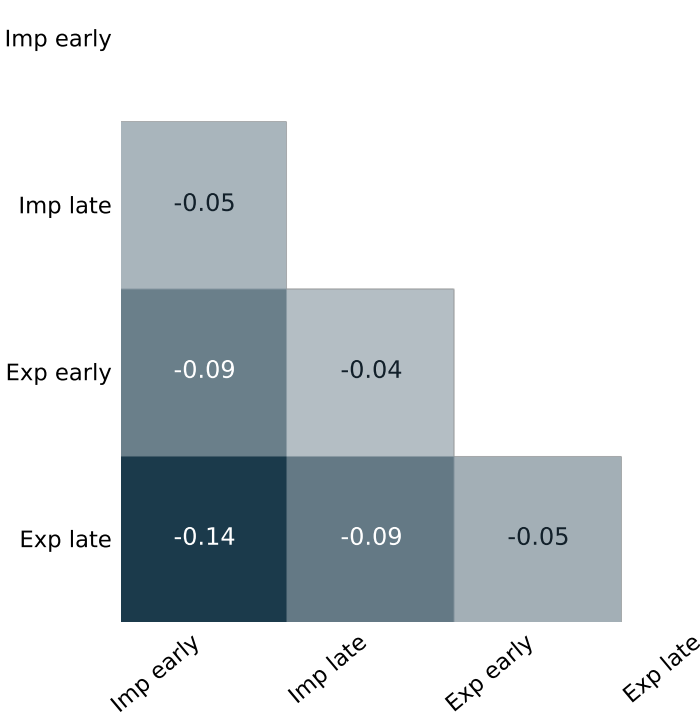
Kislov ratio

Cell = row minus column. n=18, 4 s median ICA. All three corrections within measure within dataset. Exploratory.

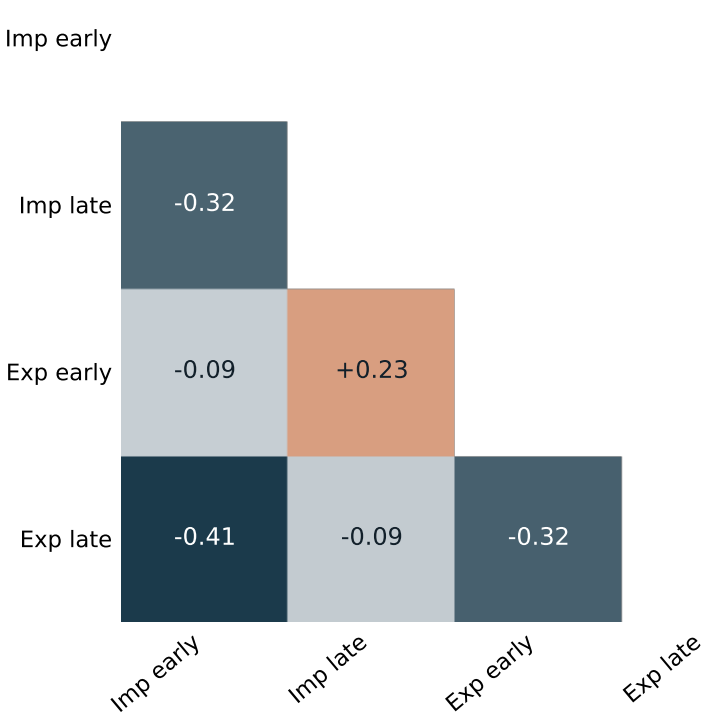
Dataset A (condition medians)



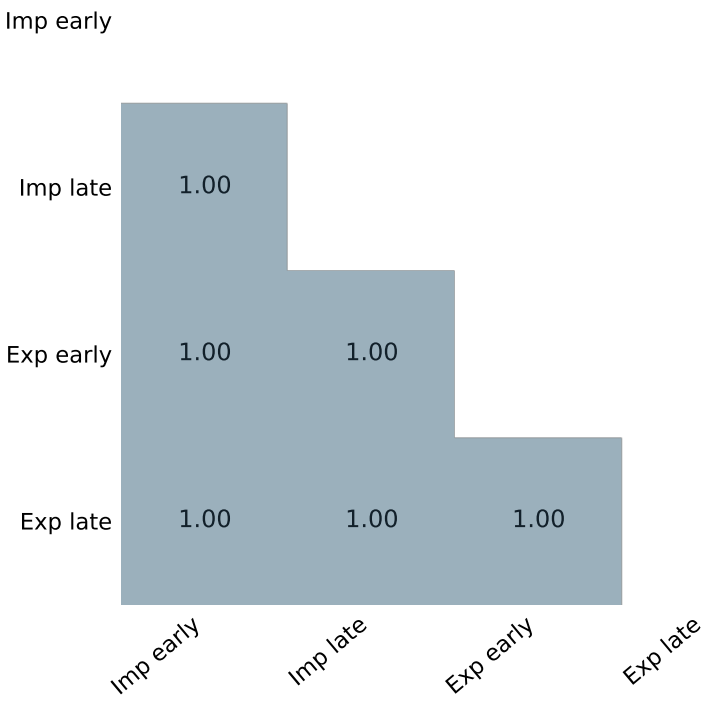
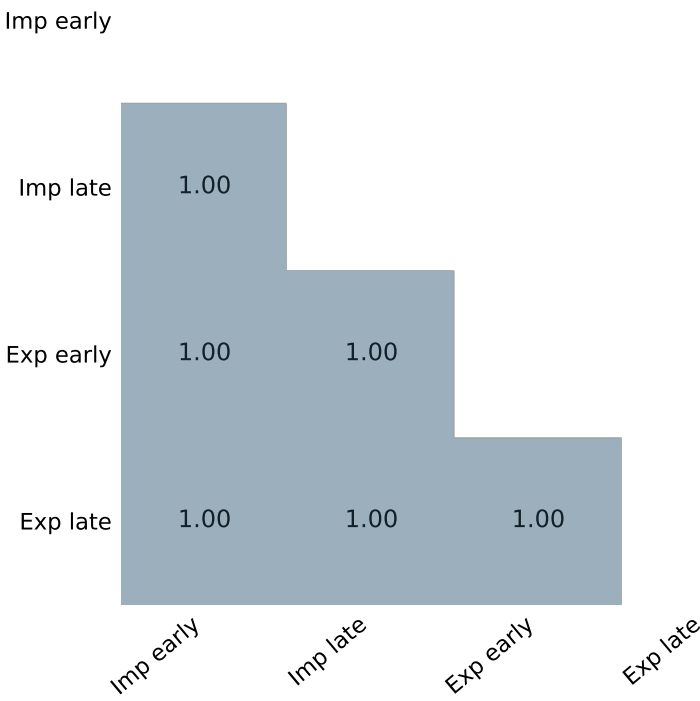
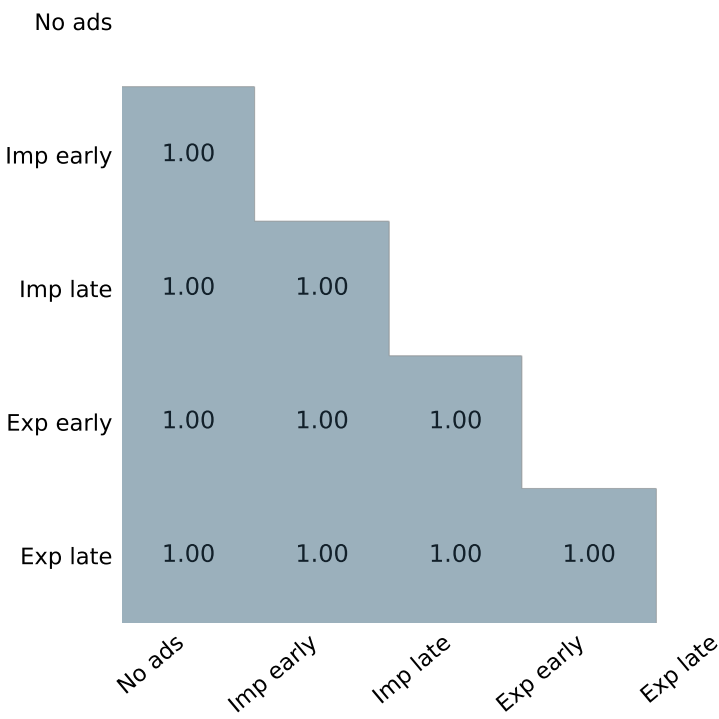
Dataset B, Δ matched control



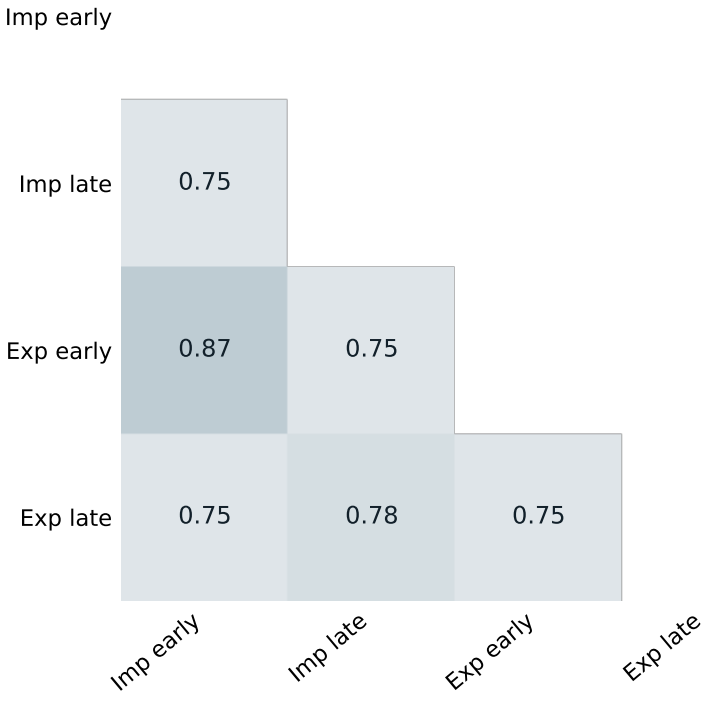
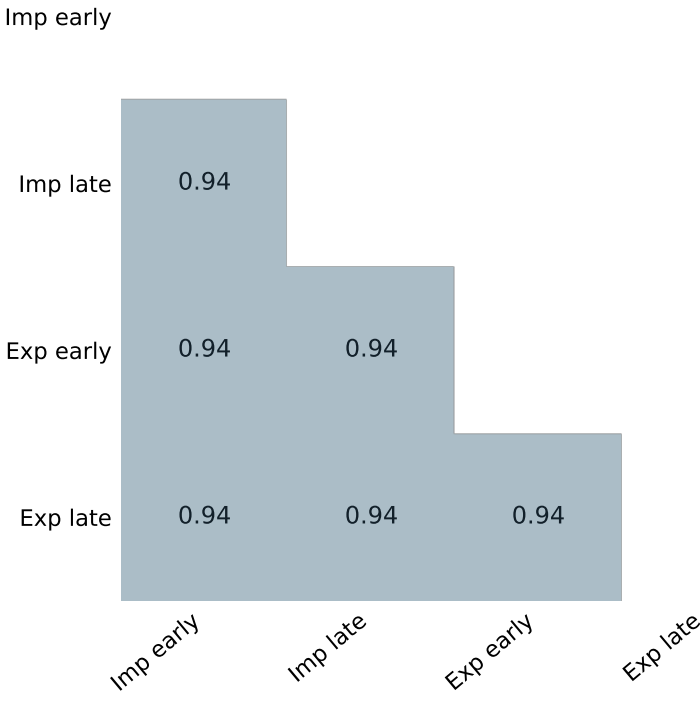
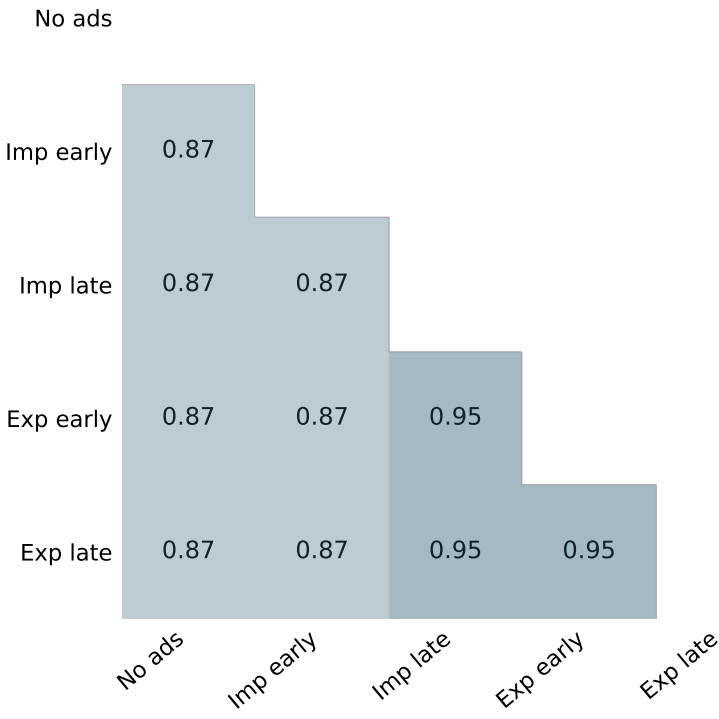
Dataset B, raw post–pre



Holm p



BH q



BY q

